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Narayanan et al.

(54) **MODULAR COFTA-BASED NETWORKING DEVICE SYSTEM**

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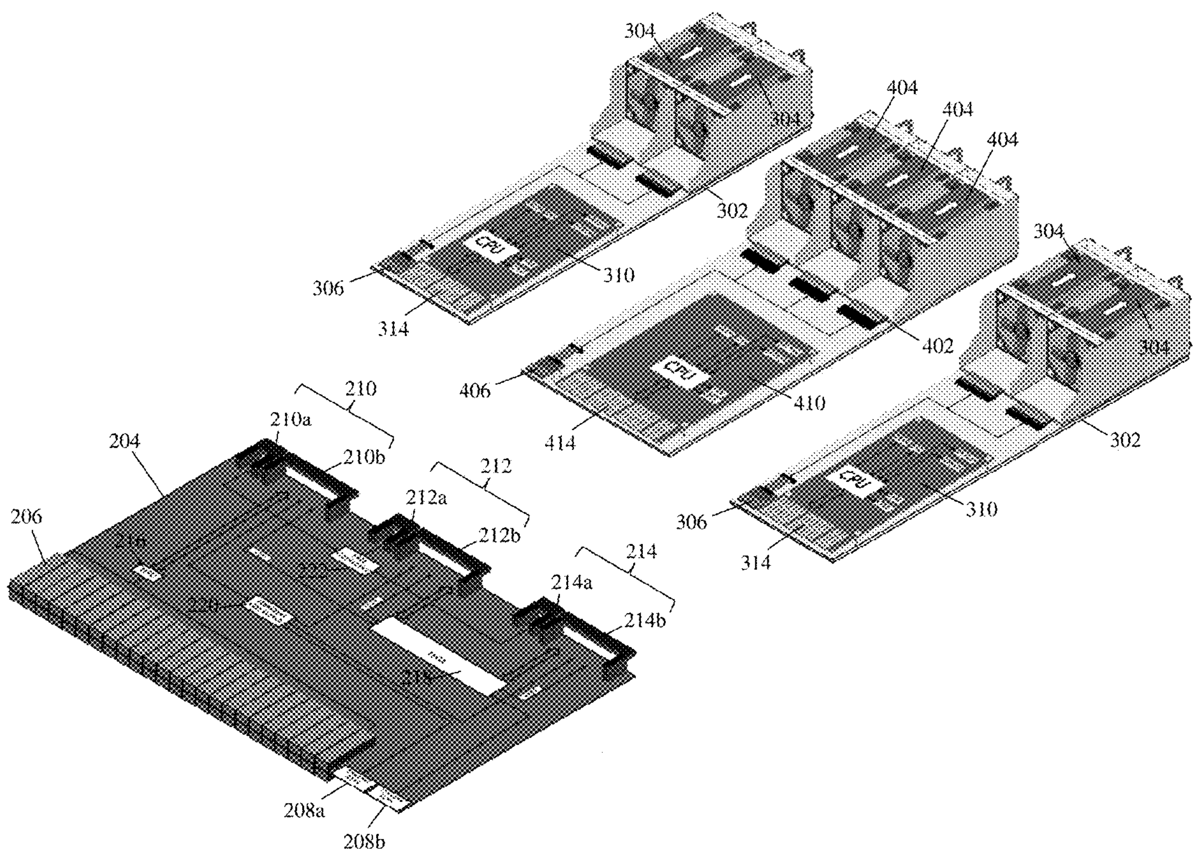
G06F 1/18	(2006.01)
G06F 1/20	(2006.01)
H05K 7/14	(2006.01)

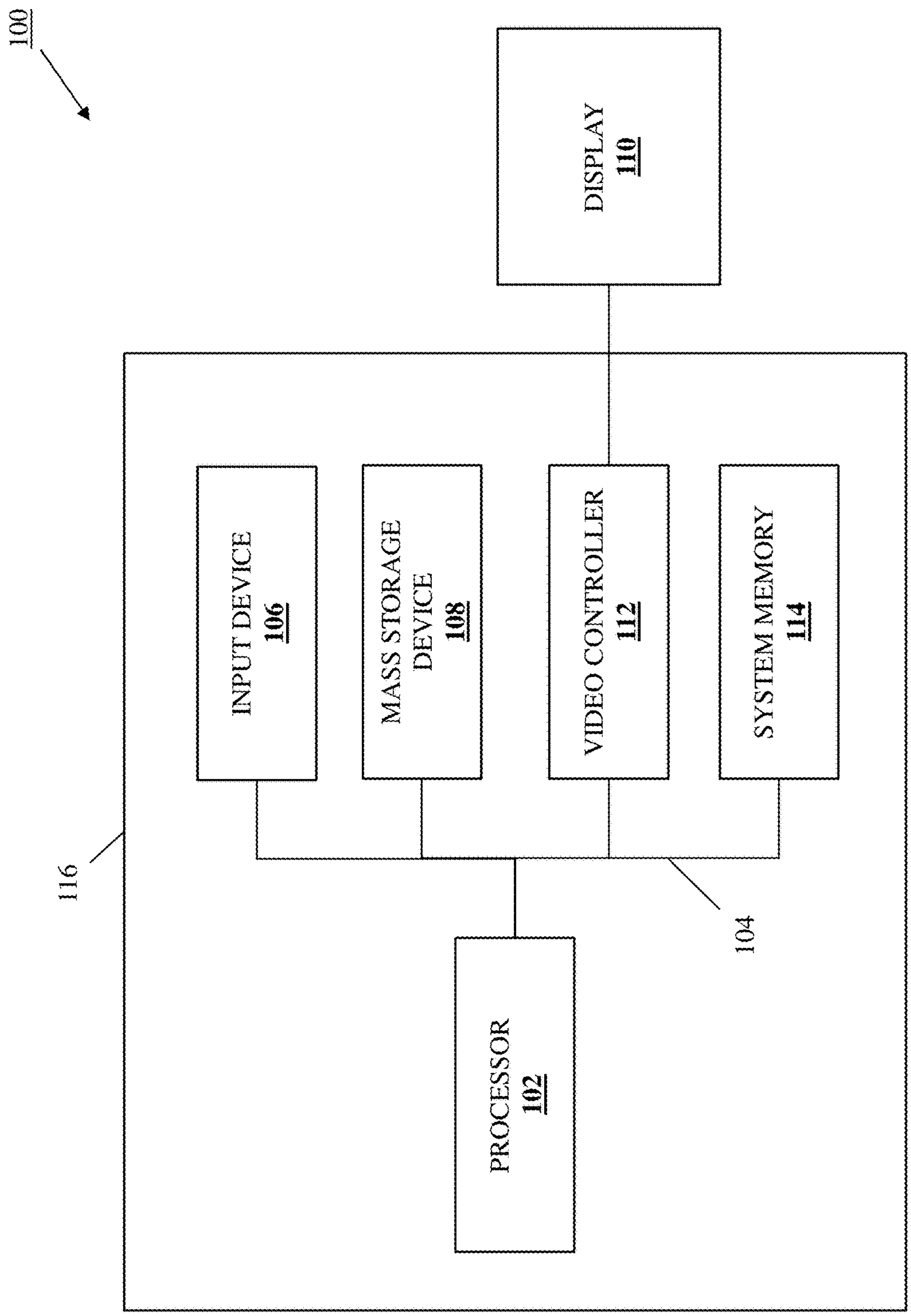
(52) **U.S. Cl.**

CPC ***H05K 7/20727*** (2013.01); ***G06F 1/183*** (2013.01); ***G06F 1/20*** (2013.01); ***H05K 7/1487*** (2013.01); ***H05K 7/20836*** (2013.01)

(57) **ABSTRACT**

A modular Computer-On-Fan-Tray-Assembly (COFTA) networking device system includes a networking device chassis housing a circuit board coupled to a plurality of non-networking components and including: a networking processing system coupled to networking ports that are accessible on the networking device chassis, a networking processing system management connector coupled to the networking processing system, a non-networking component management connector coupled to the non-networking components, and a networking software management connector. At least two COFTA systems are each connected to a different subset of the networking processing system management connector, the non-networking component management connector, and the networking software management connector. Each COFTA system provides a respective operating system that manages a different subset of the networking processing system via the networking processing system management connector, the plurality of non-networking components via the non-networking component management connector, and networking software via the networking software management connector.

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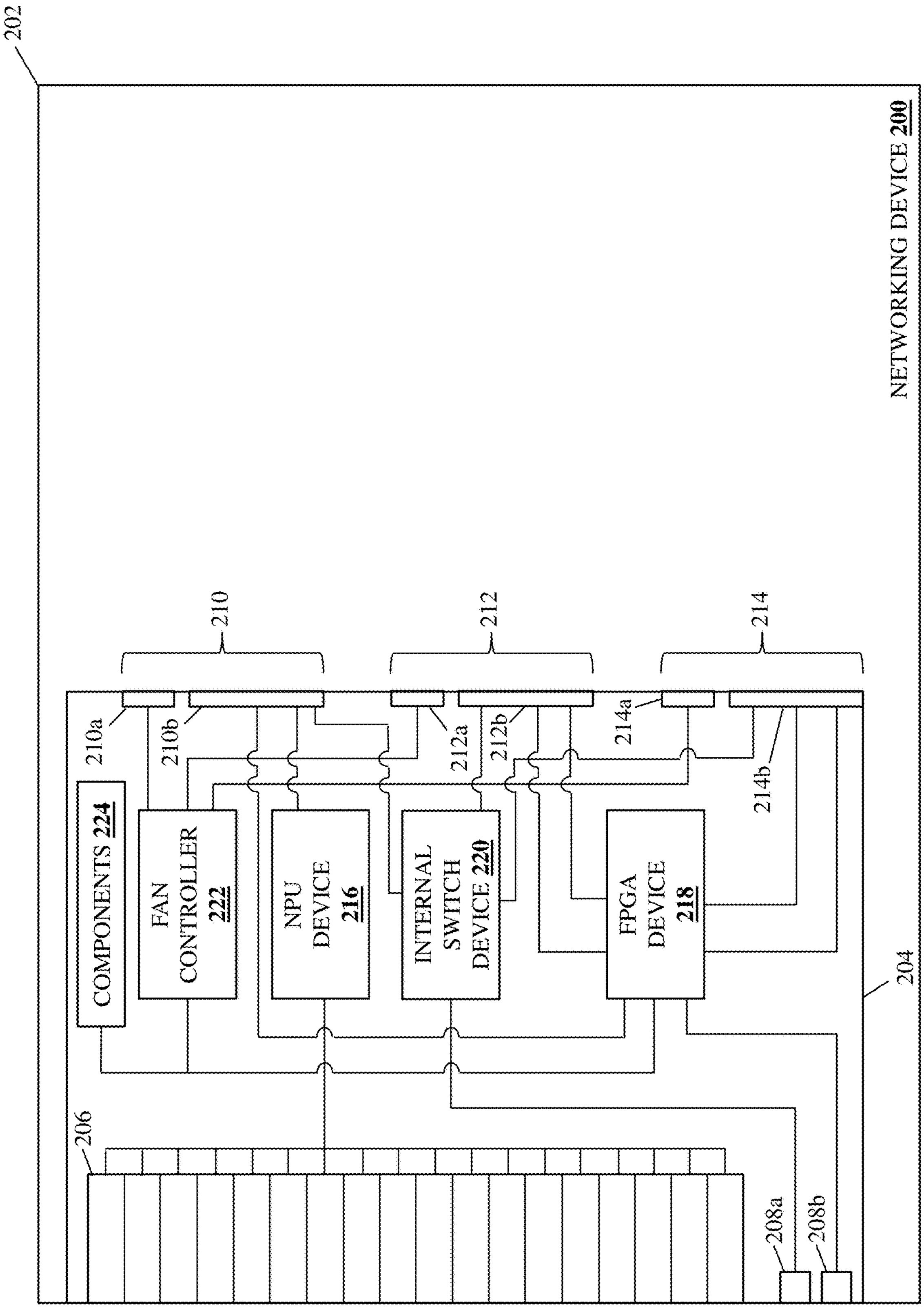
INPUT DEVICE 106
MASS STORAGE DEVICE 108
VIDEO CONTROLLER 112
SYSTEM MEMORY 114
PROCESSOR 102
116
104

FIG. 1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204
206
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210b
212
212a
212b
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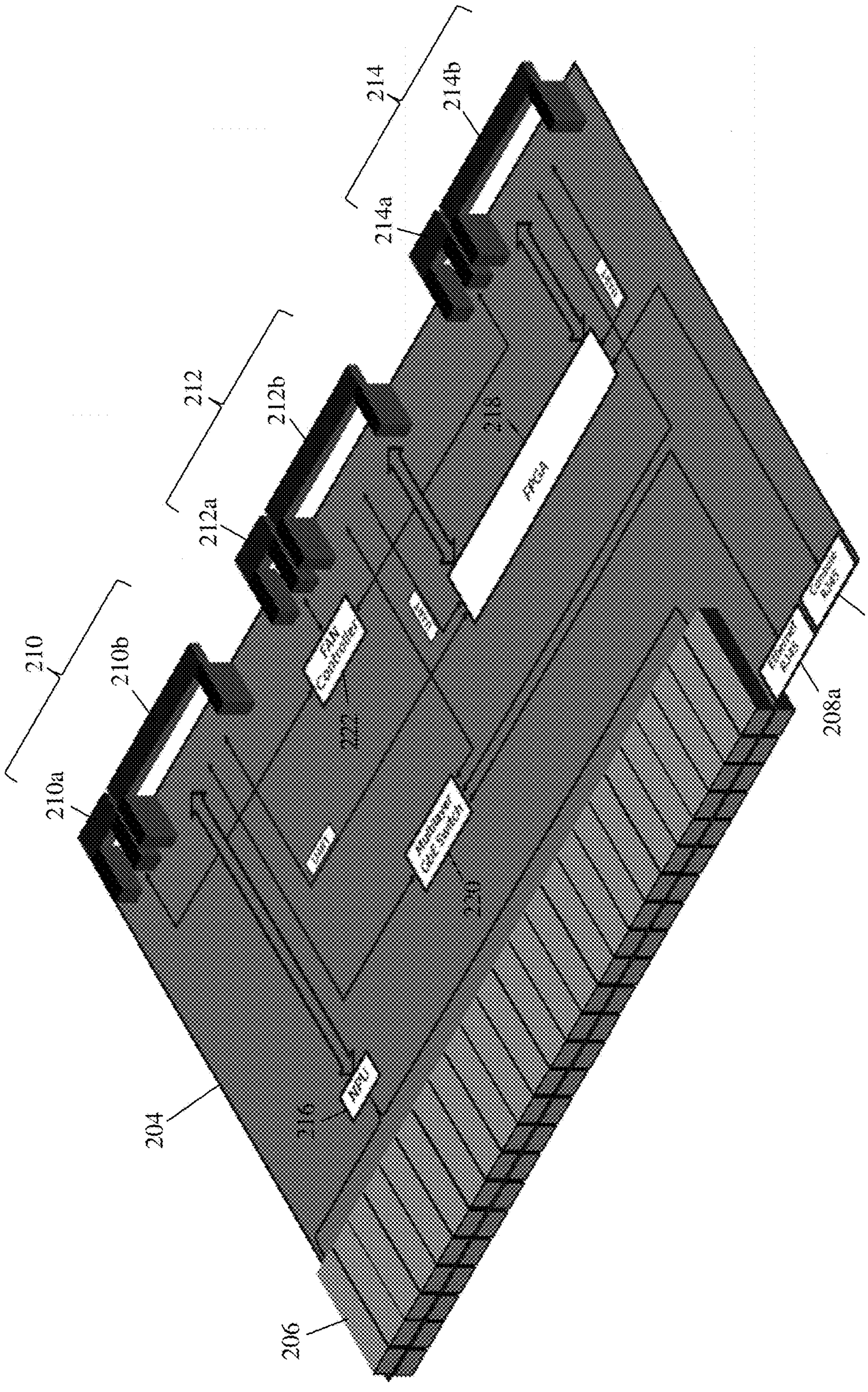
214a
214b
COMPONENTS 224
FAN CONTROLLER 222
NPU DEVICE 216
INTERNAL SWITCH DEVICE 220
FPGA DEVICE 218

FIG. 2A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214a
212
212b
212a
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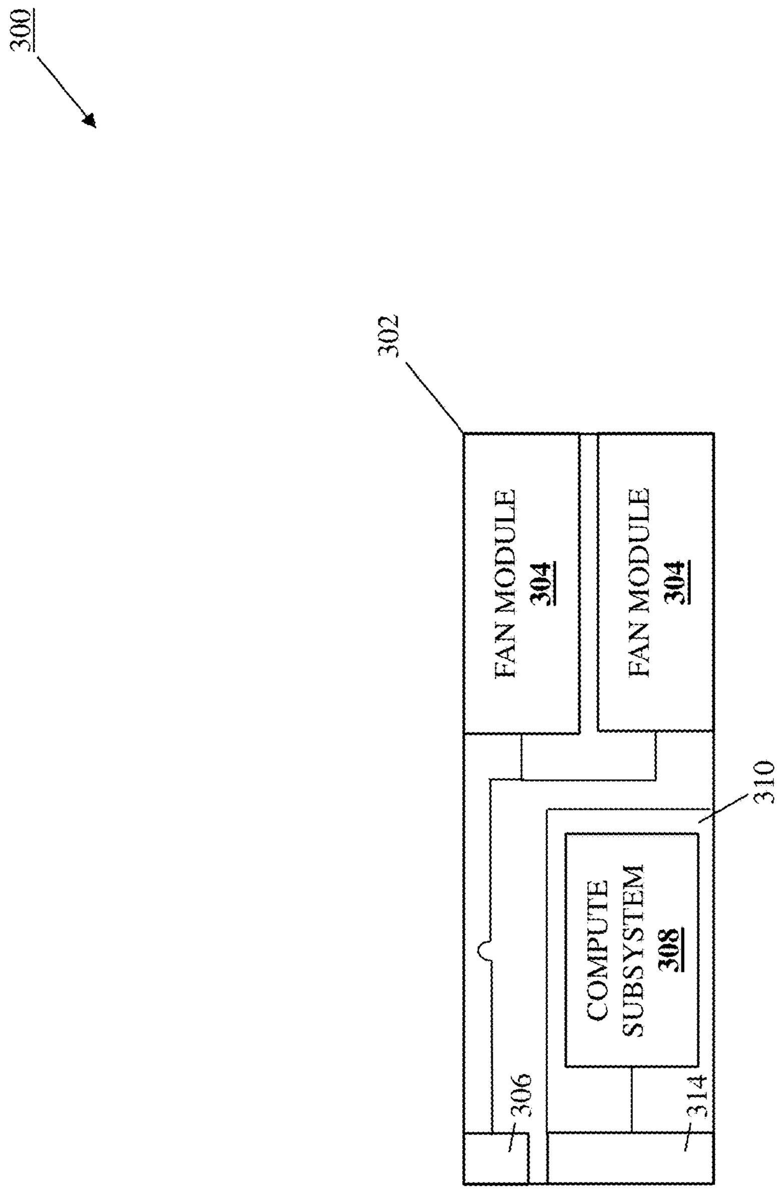
FPGA
210
210b
210a
FAN Controller
222
220
NPU
216
204
206
208a
208b

FIG. 2B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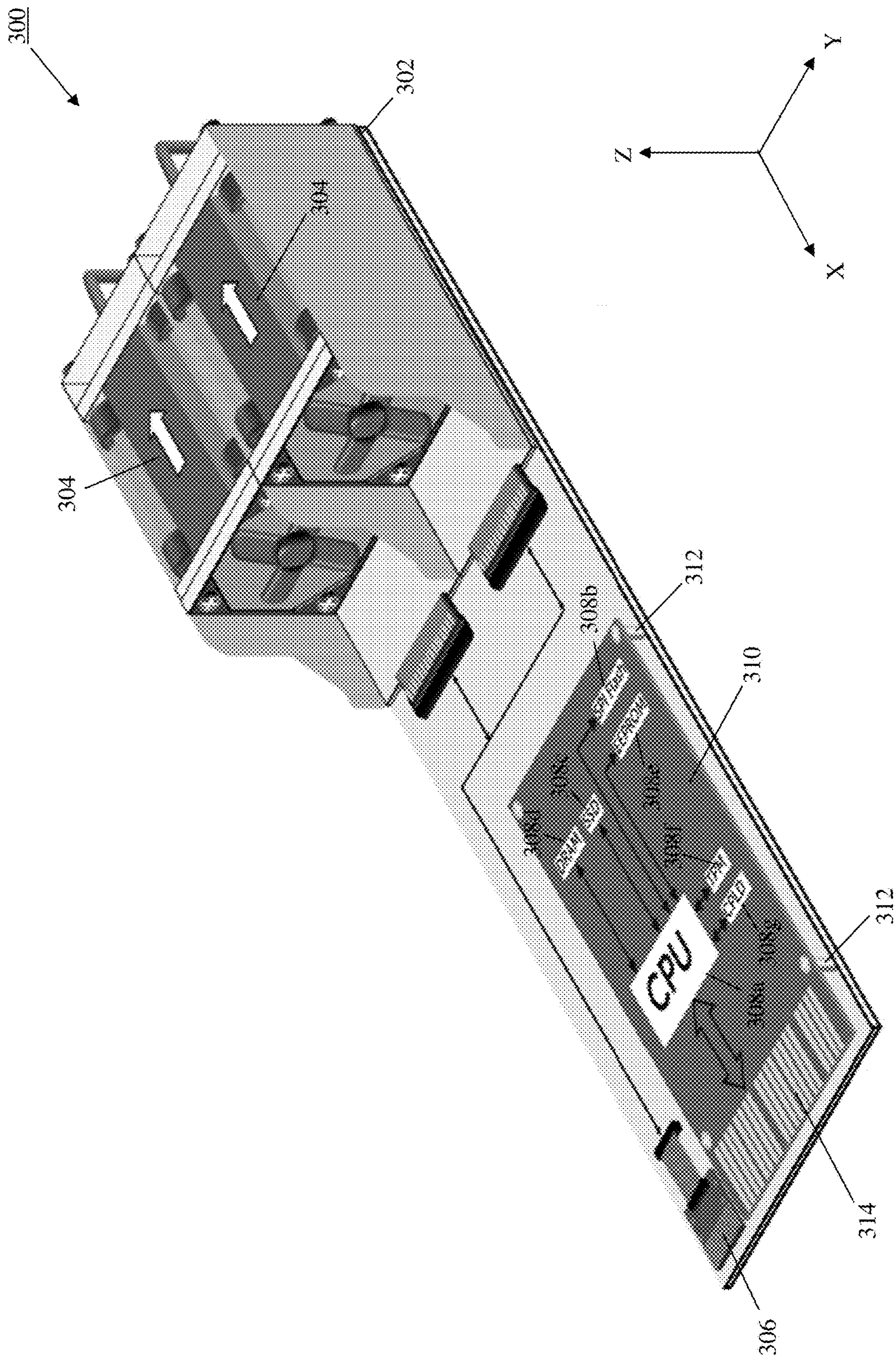
FAN MODULE 304
FAN MODULE 304
310
COMPUTE SUBSYSTEM 308
306
314

FIG. 3A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302
304
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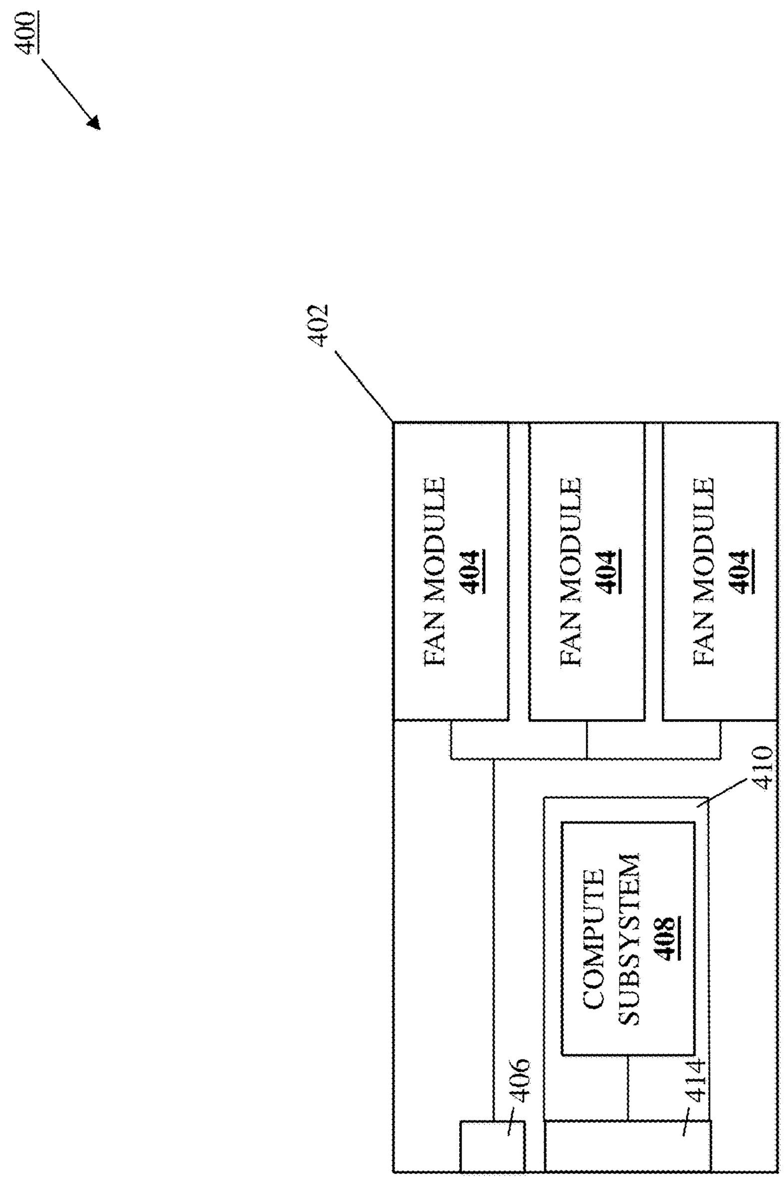
308a
308b
308c
308d
308e
308f
308g
310
312
312
314
CPU
X
Y
Z

FIG. 3B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FAN MODULE
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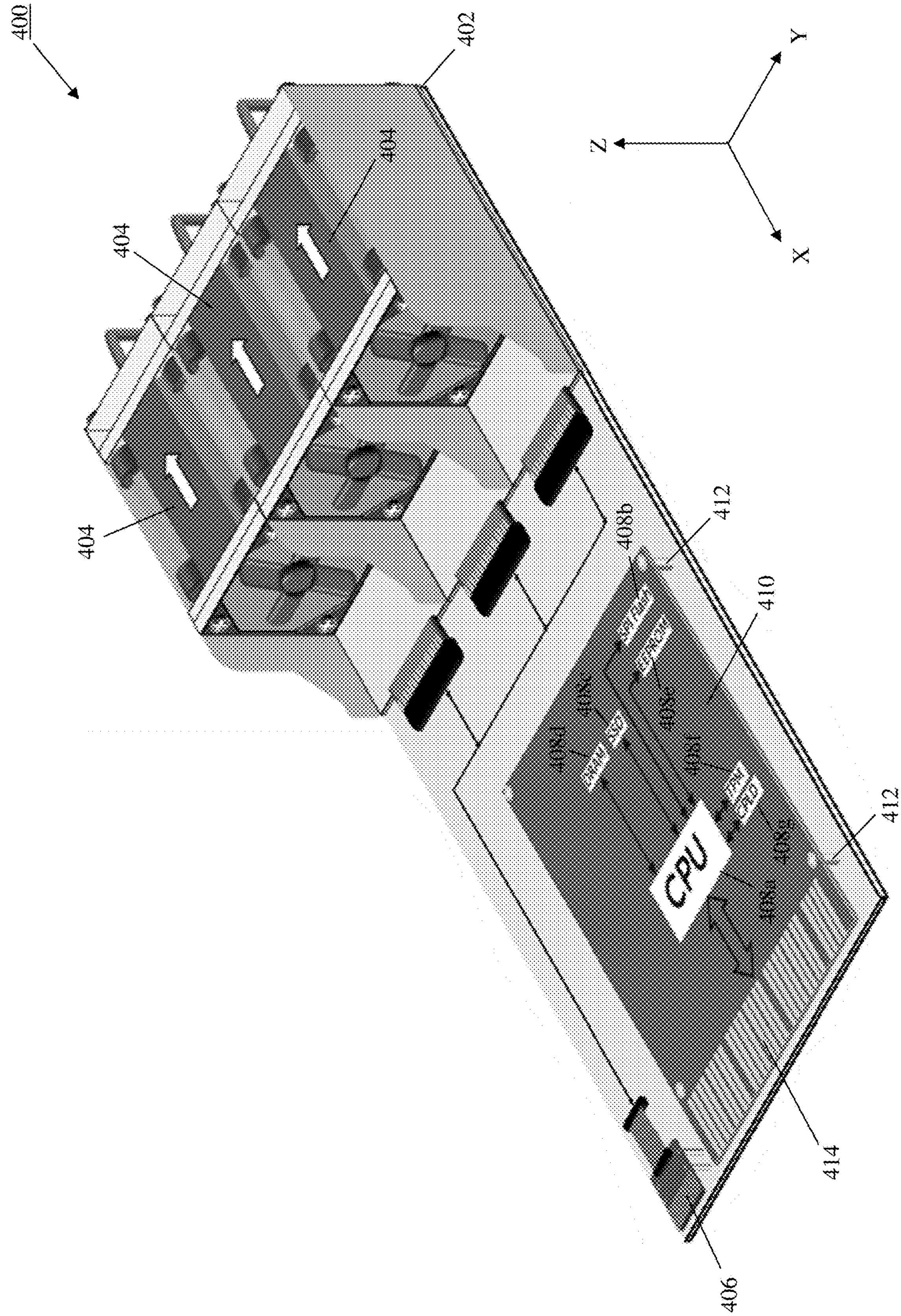
FAN MODULE
404
410
COMPUTE
SUBSYSTEM
408
406
414

FIG. 4A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404
404
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Y
X
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412
410
408c
408e
408d
408f
CPU
408g
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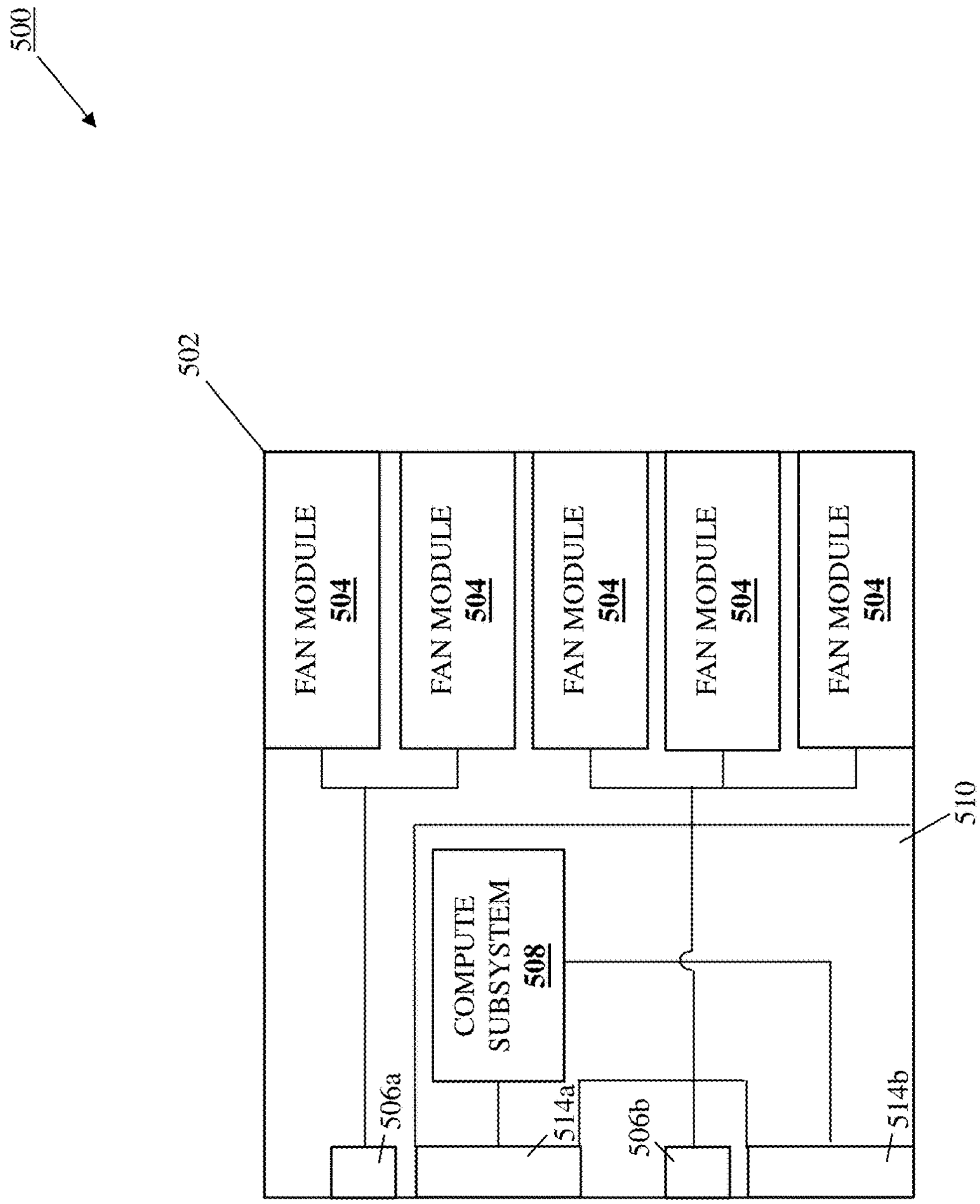
412
414
406

FIG. 4B



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502
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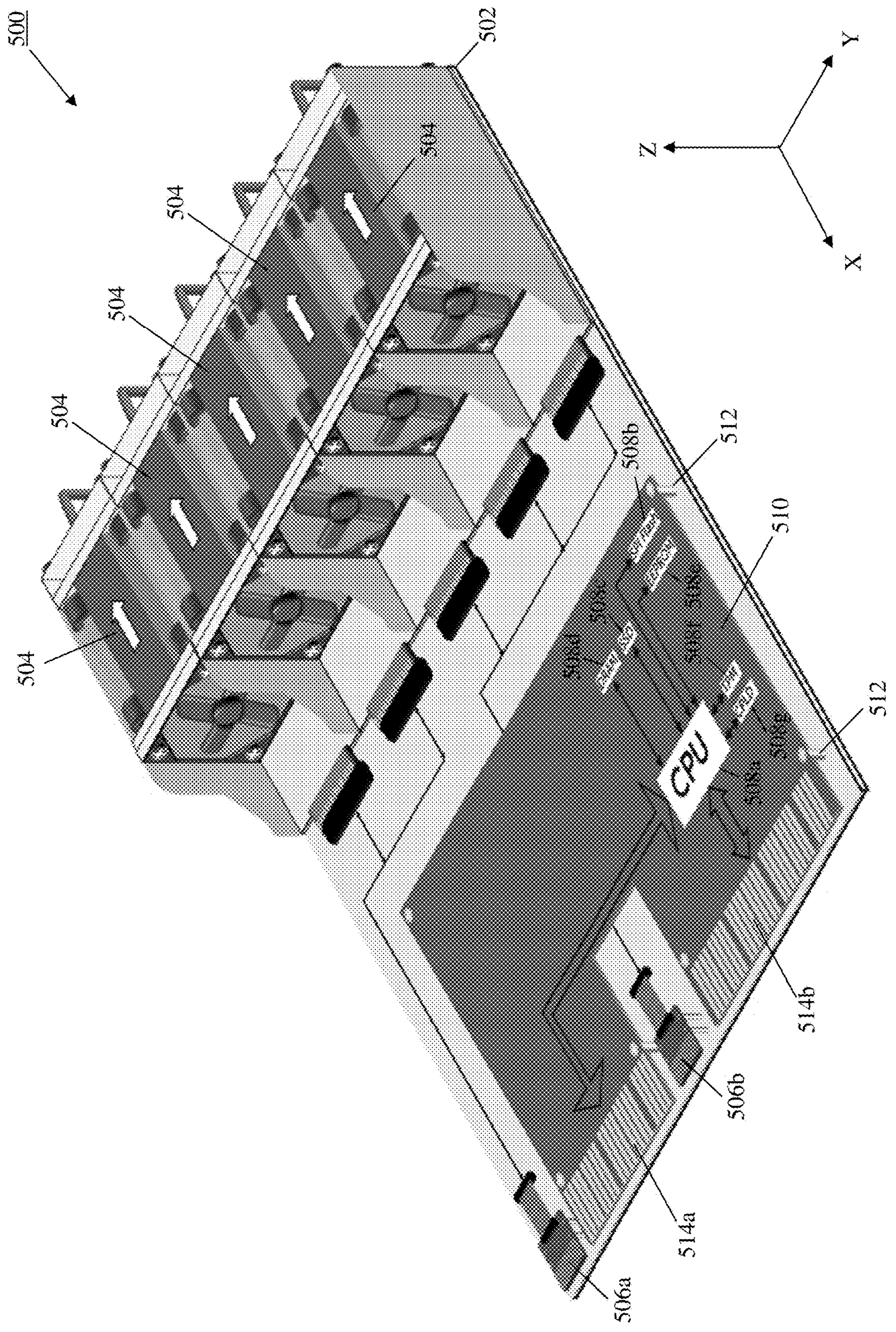
FAN MODULE 504
FAN MODULE 504
FAN MODULE 504
FAN MODULE 504
510
COMPUTE SUBSYSTEM 508
506a
514a
506b
514b

FIG. 5A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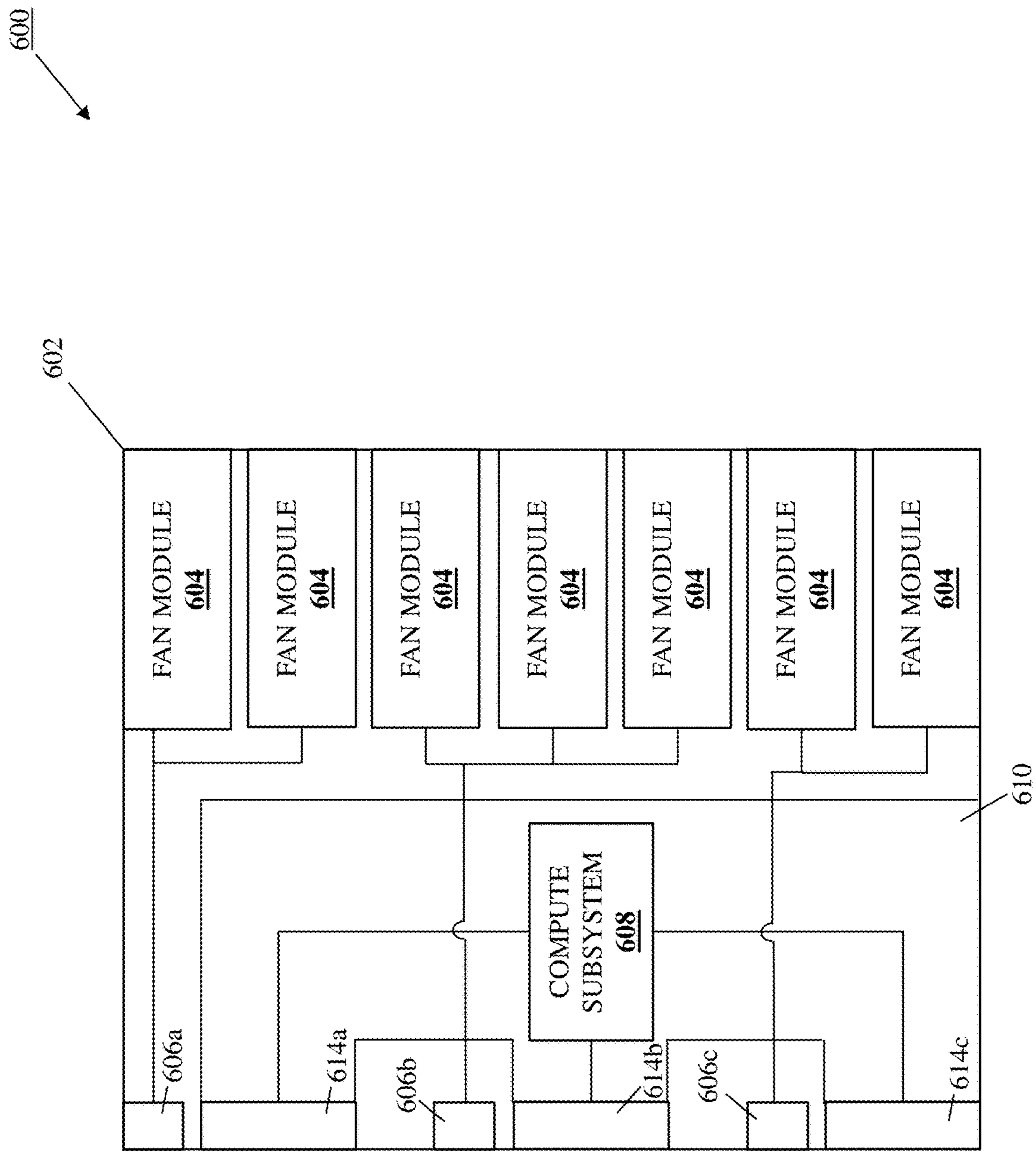
508b
508c
508d
508e
508f
508g
510
512
512
514a
514b
CPU
X
Y
Z

FIG. 5B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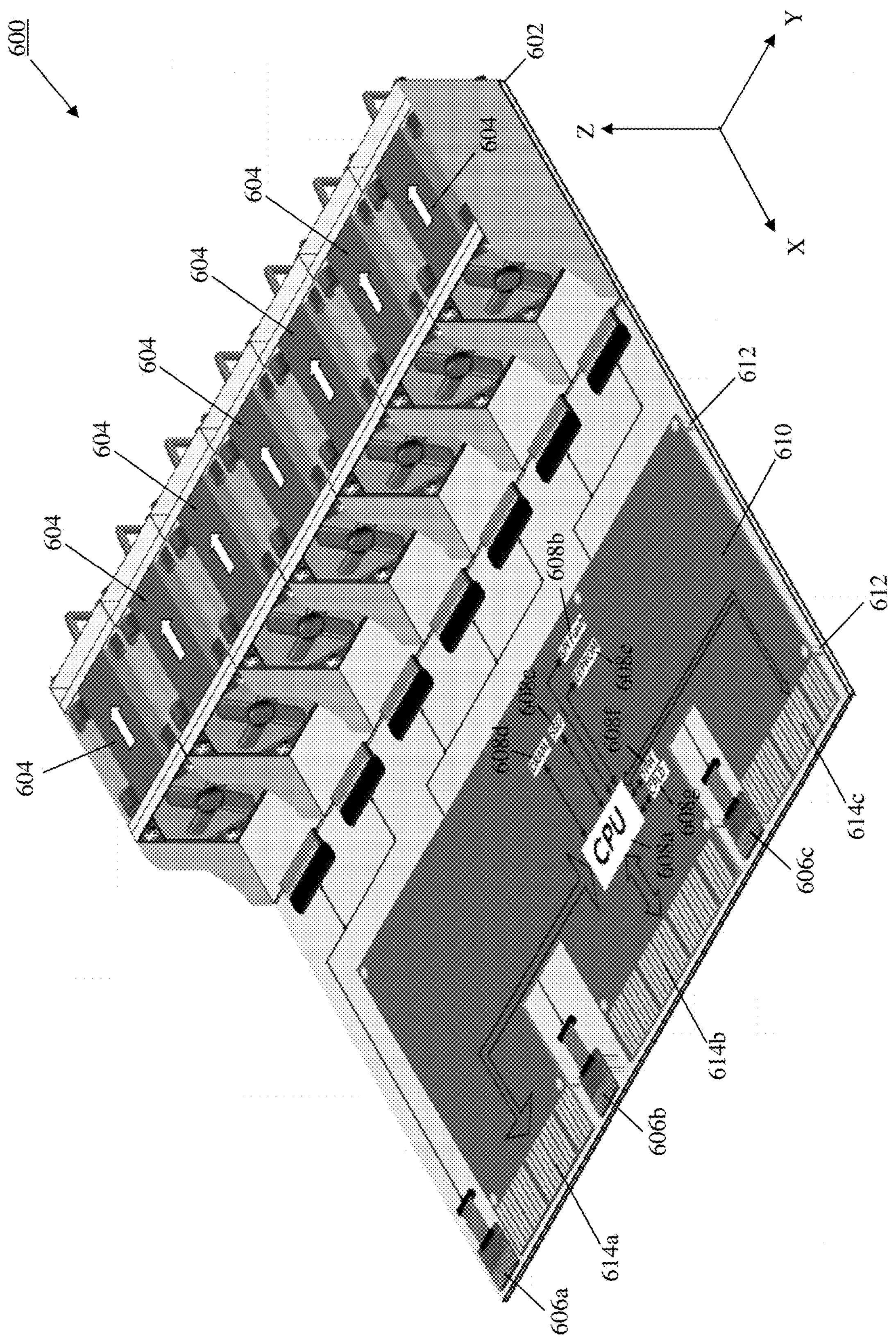
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
610
COMPUTE SUBSYSTEM 608
606a
614a
606b
614b
606c
614c

FIG. 6A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CPU
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614c
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606b
614a
606a
X
Y
Z

FIG. 6B

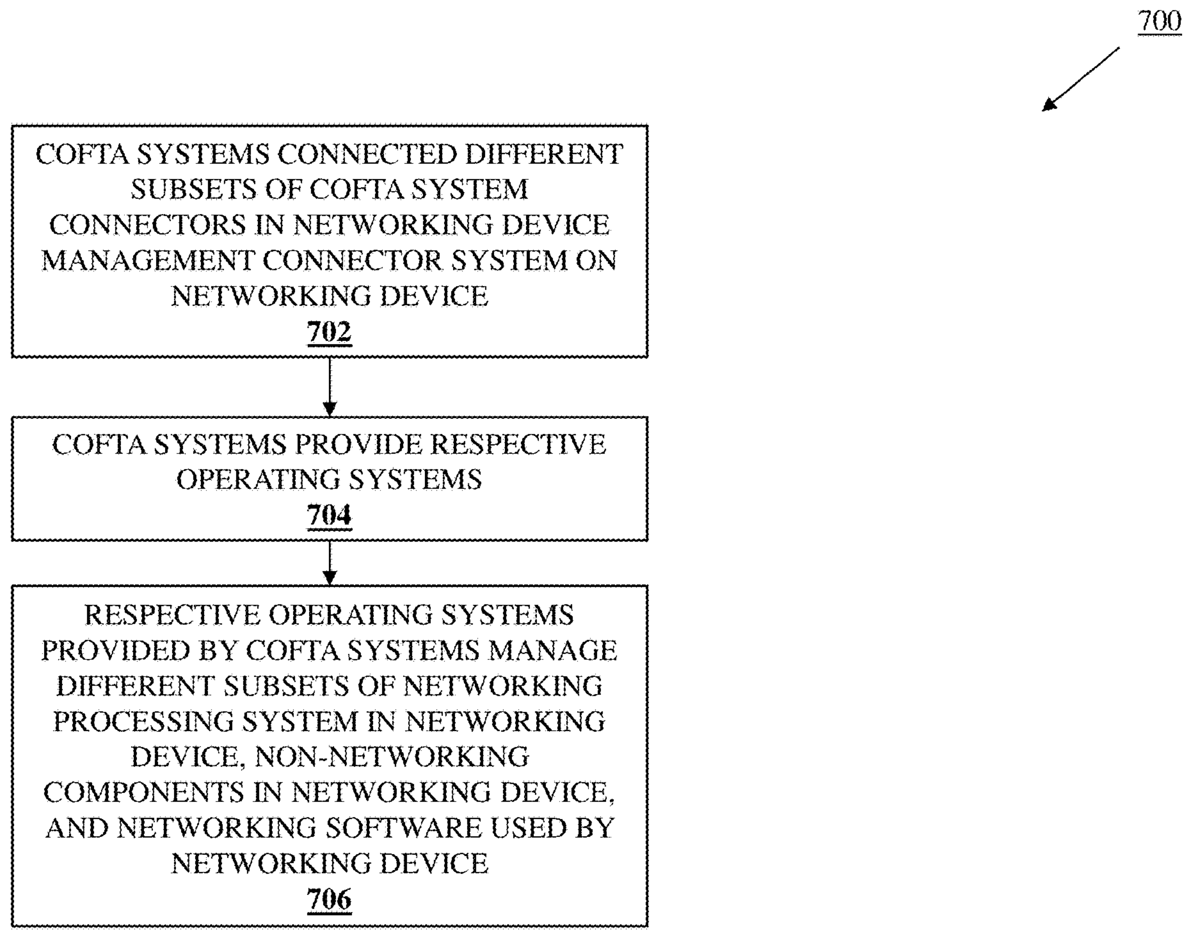
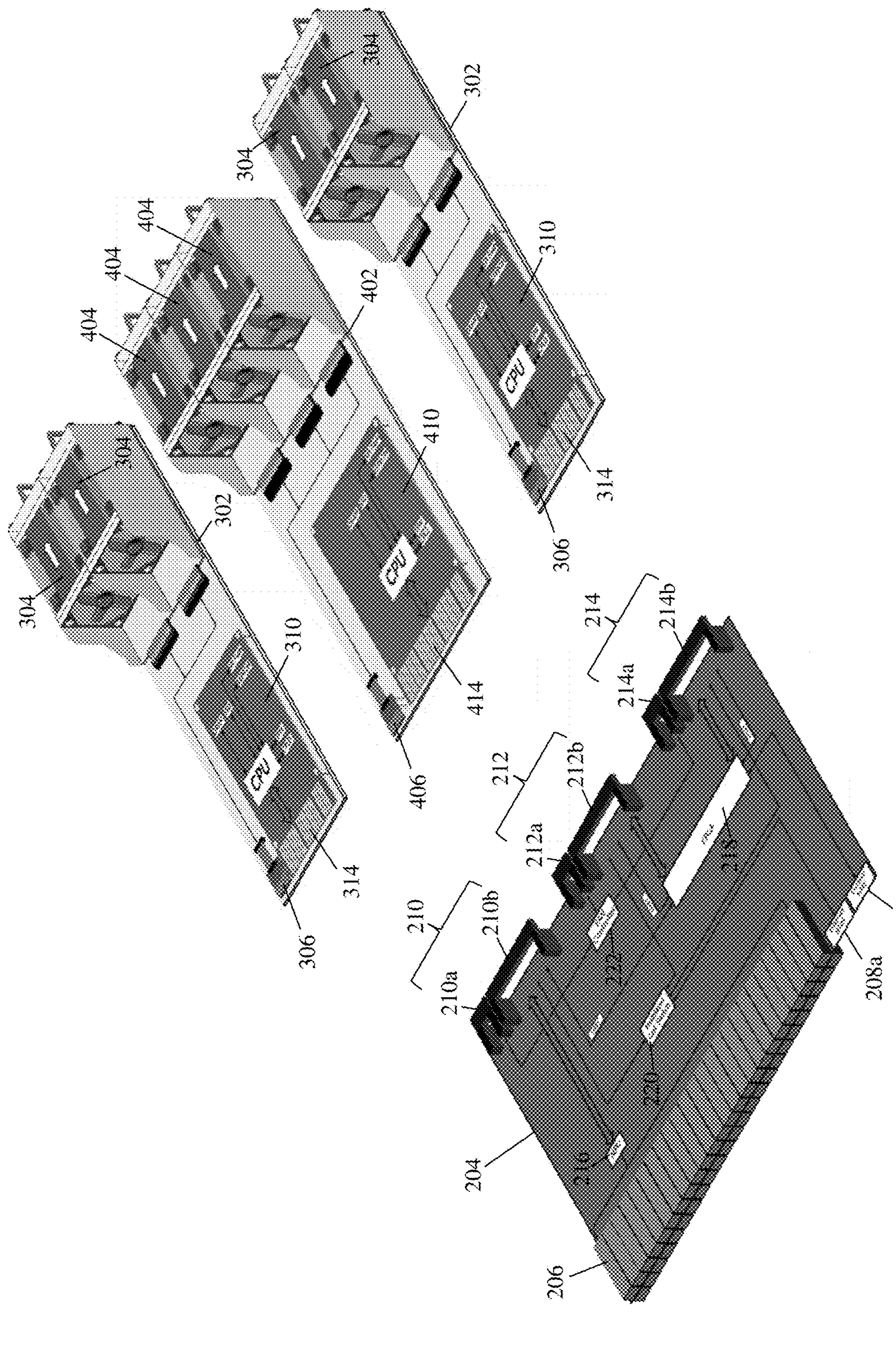

700
COFTA SYSTEMS CONNECTED DIFFERENT SUBSETS OF COFTA SYSTEM CONNECTORS IN NETWORKING DEVICE MANAGEMENT CONNECTOR SYSTEM ON NETWORKING DEVICE
702
COFTA SYSTEMS PROVIDE RESPECTIVE OPERATING SYSTEMS
704
RESPECTIVE OPERATING SYSTEMS PROVIDED BY COFTA SYSTEMS MANAGE DIFFERENT SUBSETS OF NETWORKING PROCESSING SYSTEM IN NETWORKING DEVICE, NON-NETWORKING COMPONENTS IN NETWORKING DEVICE, AND NETWORKING SOFTWARE USED BY NETWORKING DEVICE
706

FIG. 7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210
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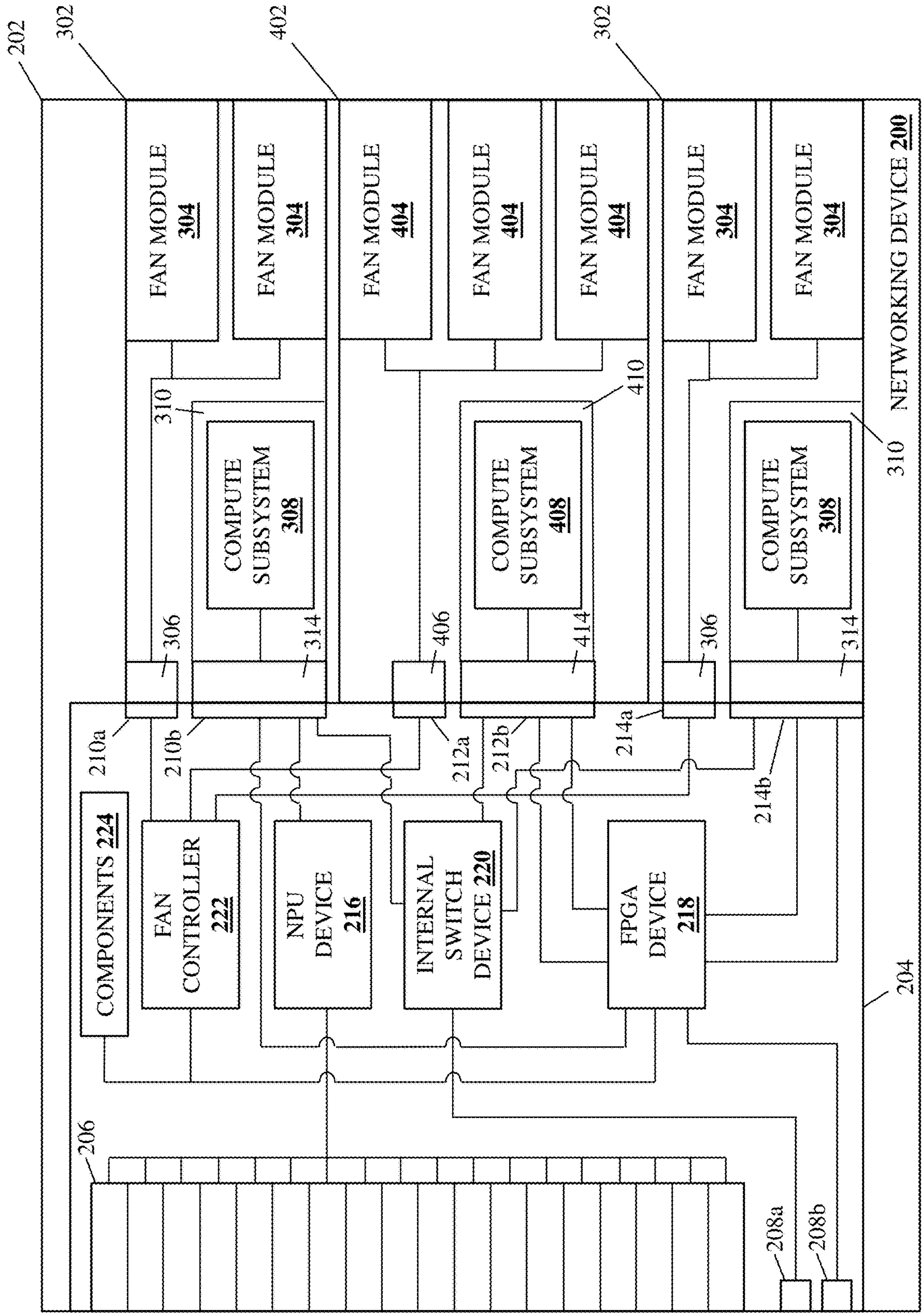
208b

FIG. 8A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COMPUTE SUBSYSTEM 308
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COMPUTE SUBSYSTEM 408
310
COMPUTE SUBSYSTEM 308
NETWORKING DEVICE 200
306
314
406
414
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210a
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212b
214a
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INTERNAL SWITCH DEVICE 220
FPGA DEVICE 218
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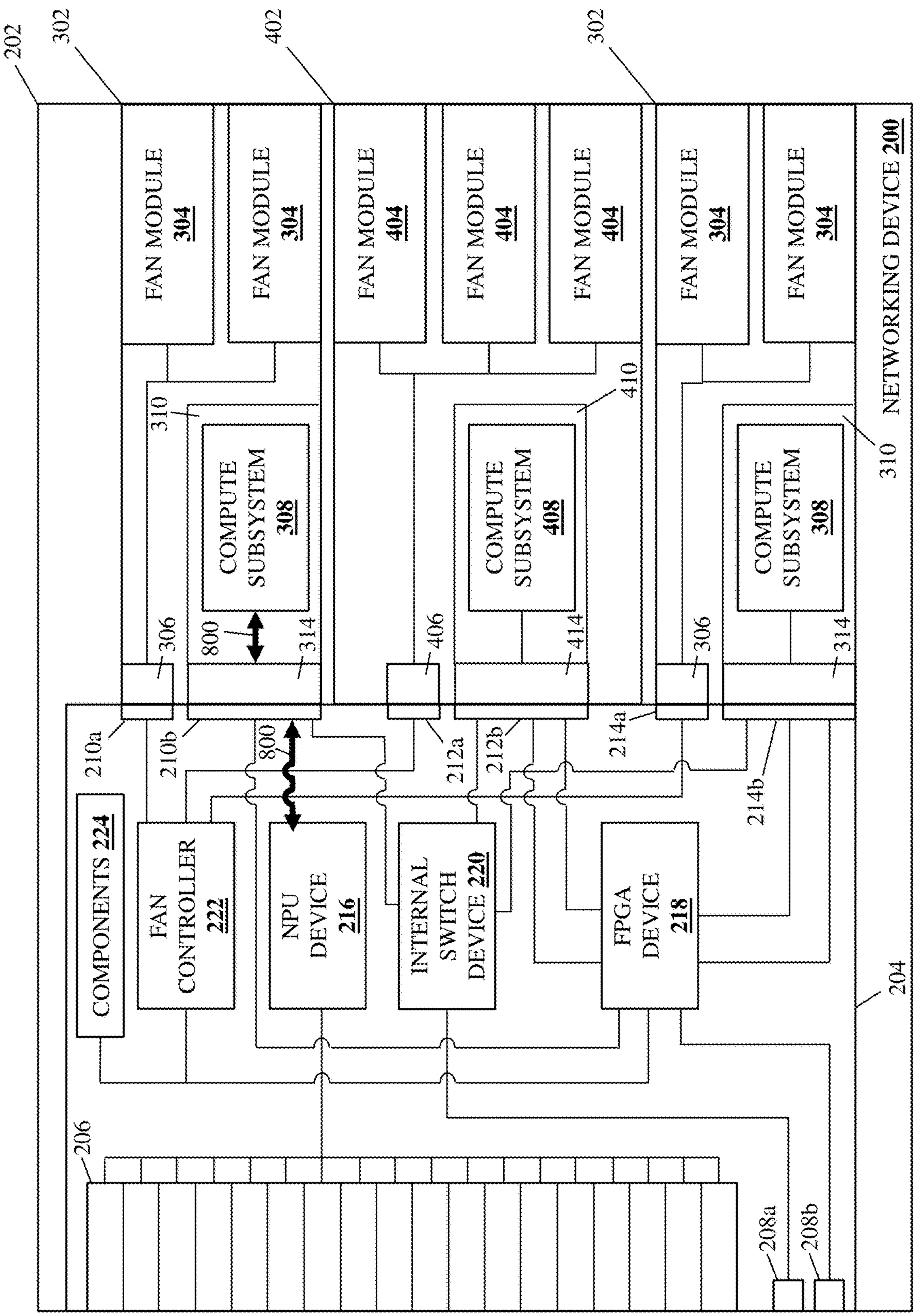
208a
208b

FIG. 8B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COMPUTE SUBSYSTEM 308
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800
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212b
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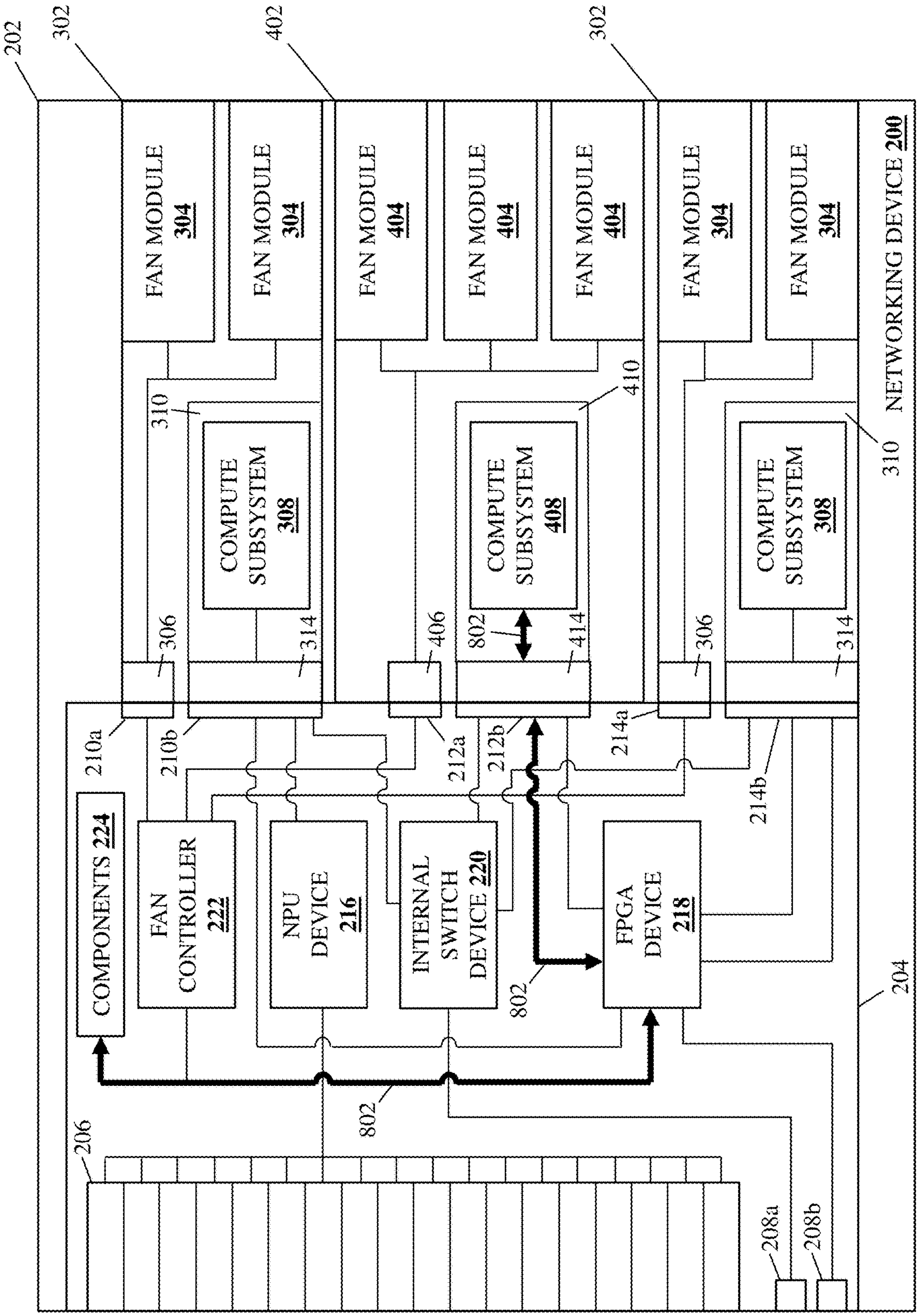
FIG. 8C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COMPUTE SUBSYSTEM 408
COMPUTE SUBSYSTEM 308
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314
406
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306
314
210a
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212b
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214b
COMPONENTS 224
FAN CONTROLLER 222
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INTERNAL SWITCH DEVICE 220
FPGA DEVICE 218
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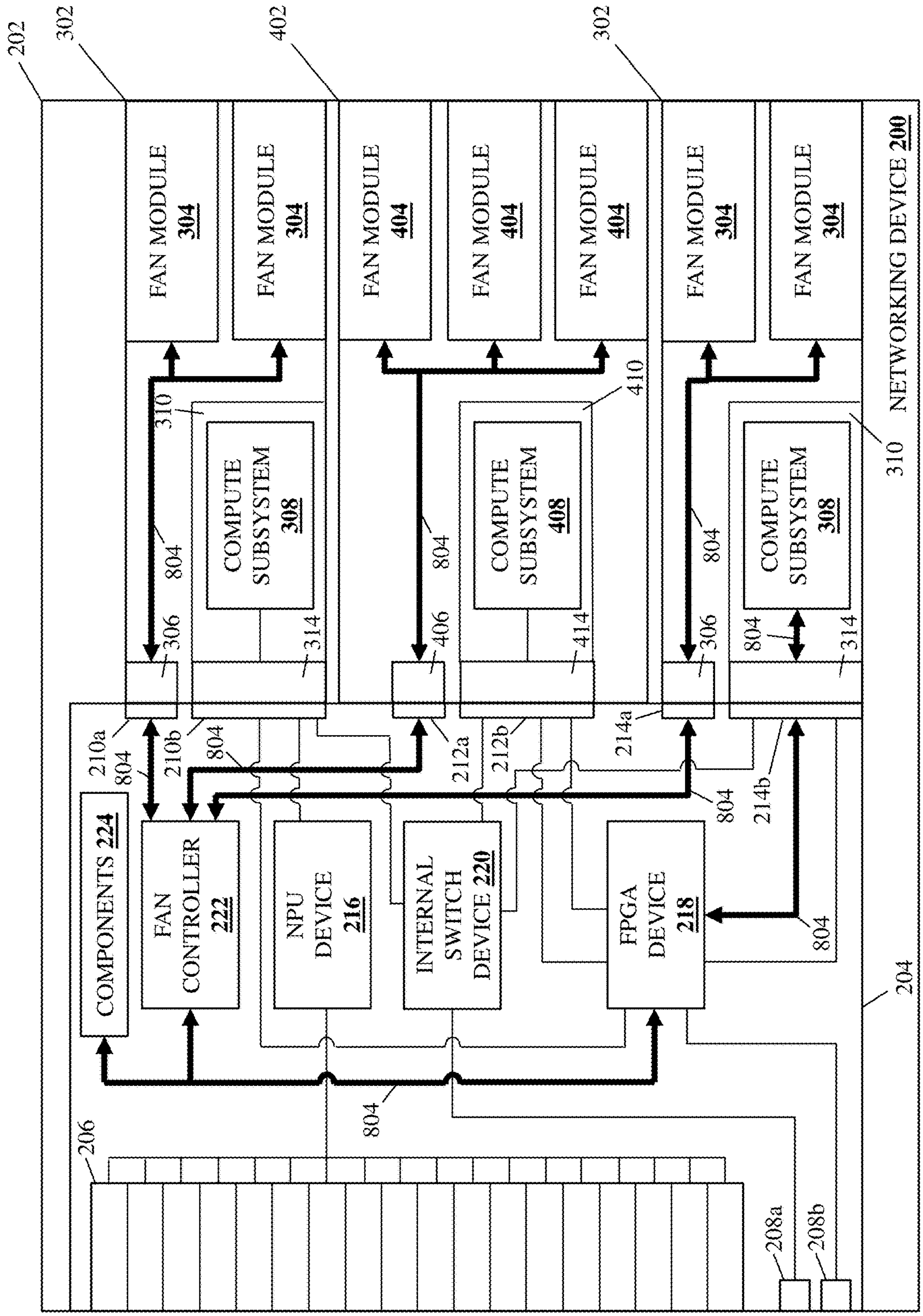
208a
208b
NETWORKING DEVICE 200

FIG. 8D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COMPUTE SUBSYSTEM 308
COMPUTE SUBSYSTEM 408
COMPUTE SUBSYSTEM 308
804
306
314
406
414
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314
210a
210b
212a
212b
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NPU DEVICE 216
INTERNAL SWITCH DEVICE 220
FPGA DEVICE 218
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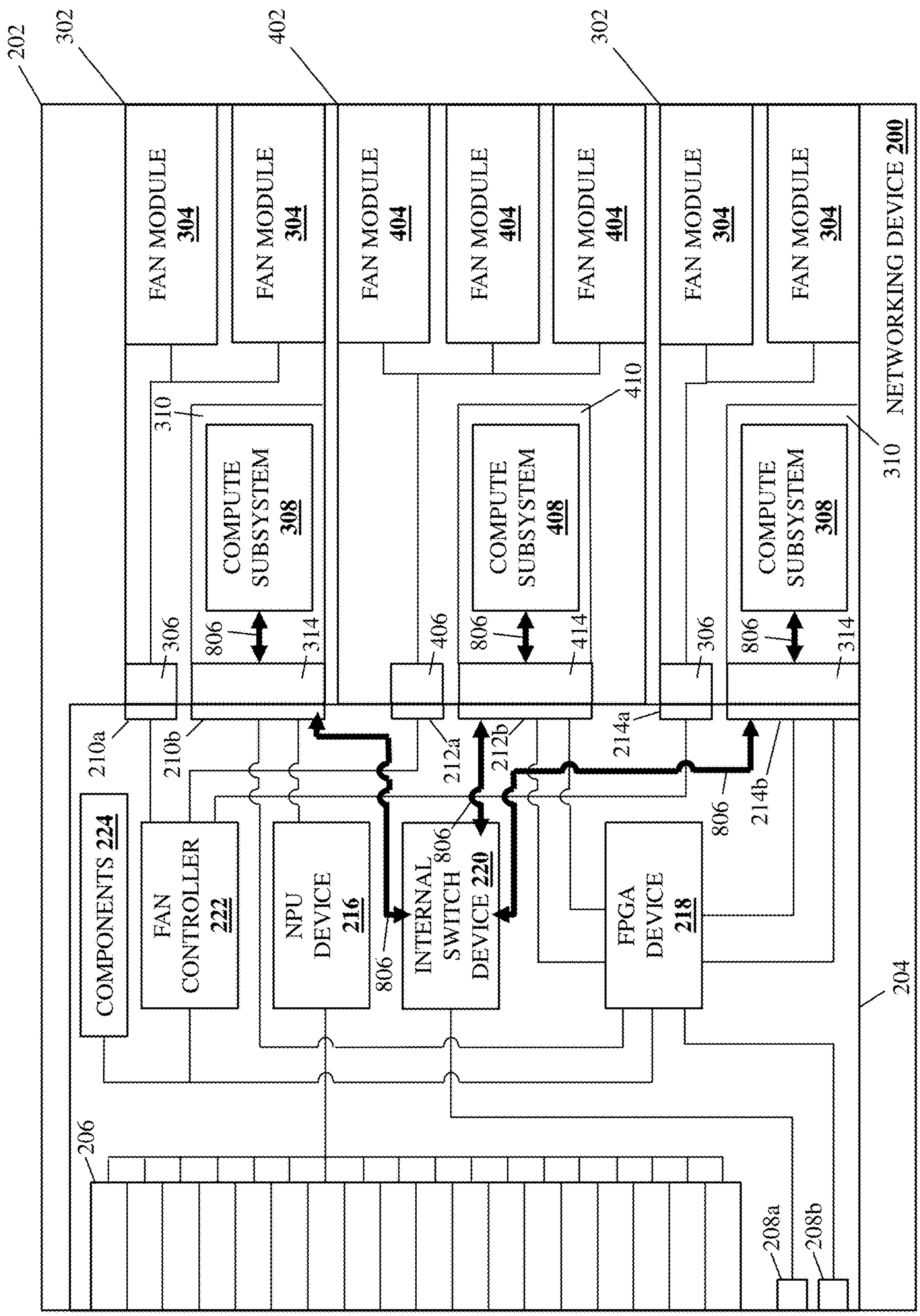
208b

FIG. 8E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COMPUTE SUBSYSTEM 308
COMPUTE SUBSYSTEM 408
COMPUTE SUBSYSTEM 308
NETWORKING DEVICE 200
306
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FPGA DEVICE 218
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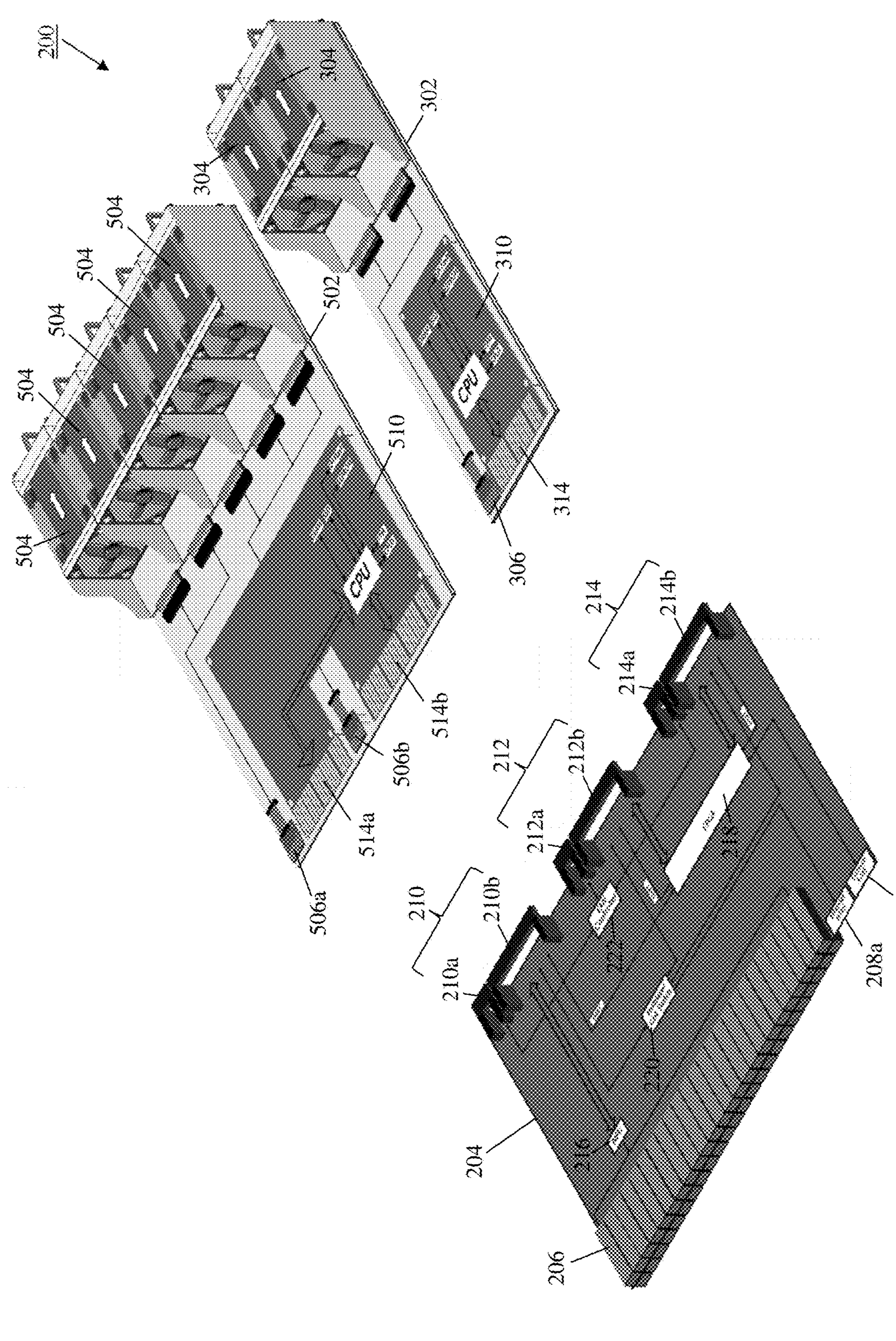
208a
208b

FIG. 8F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304
304
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CPU
CPU
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514b
214
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210b
222
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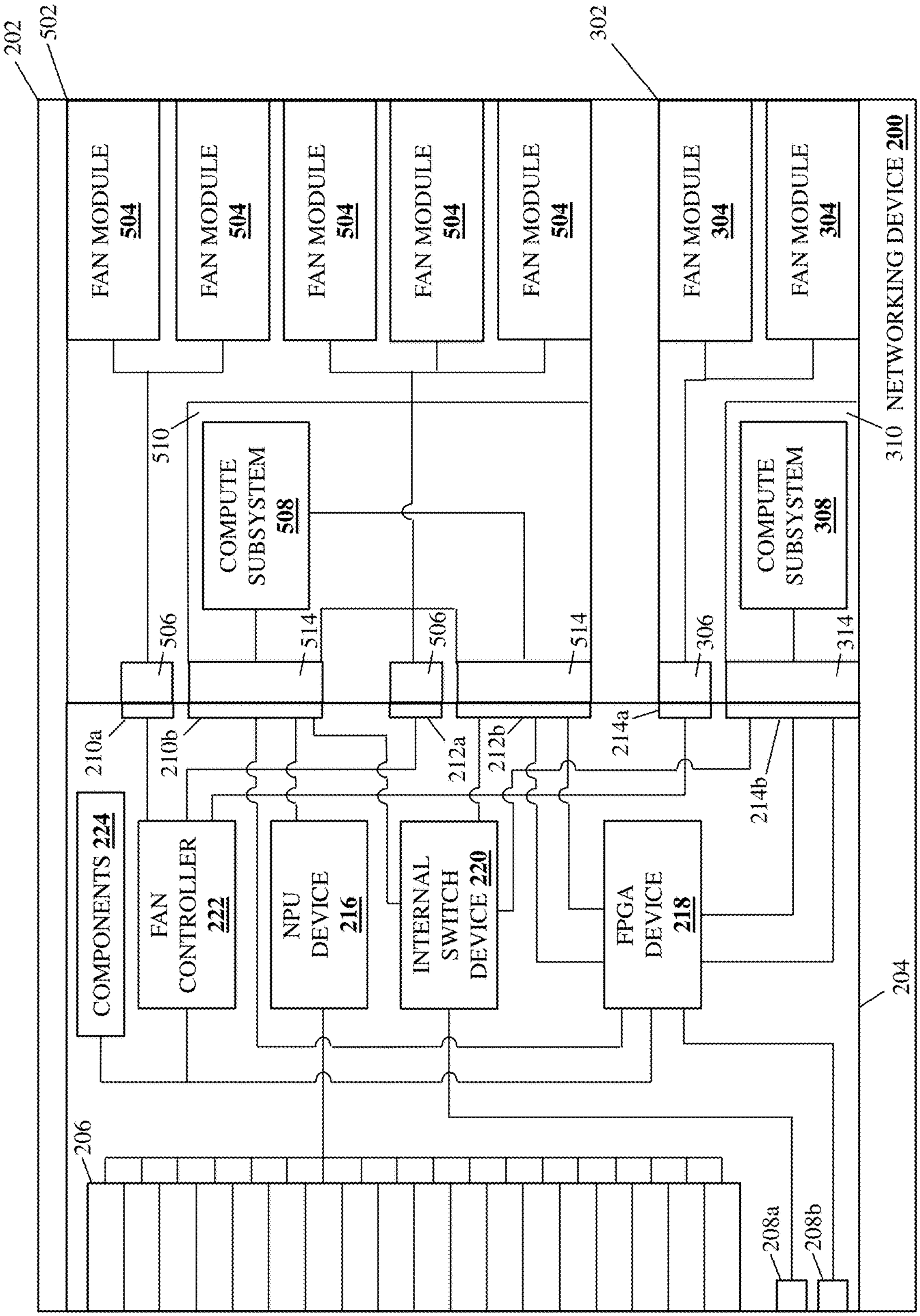
208a
208b

FIG. 9A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COMPUTE SUBSYSTEM 308
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INTERNAL SWITCH DEVICE 220
FPGA DEVICE 218
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206
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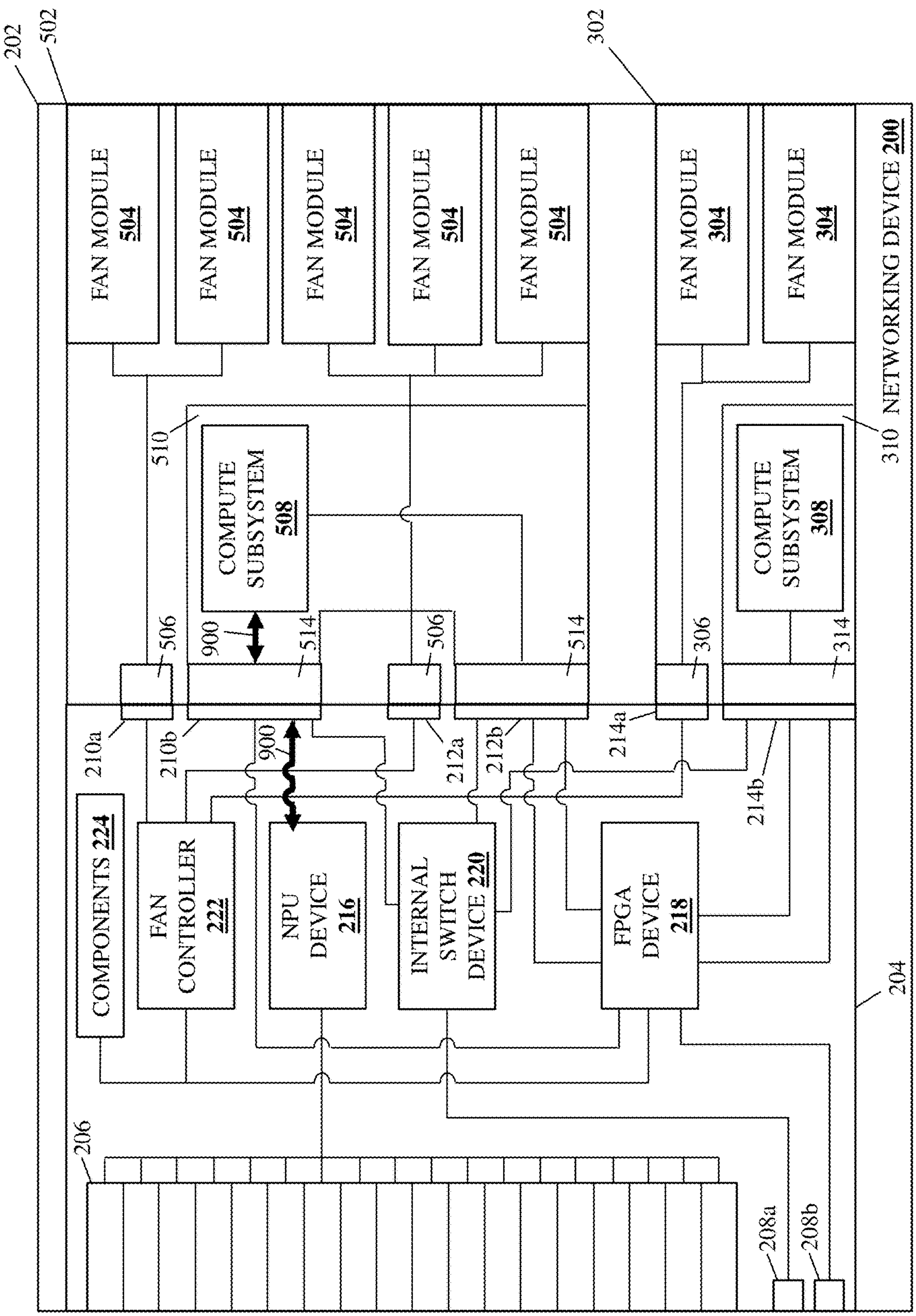
208b

FIG. 9B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212b
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INTERNAL SWITCH DEVICE 220
FPGA DEVICE 218
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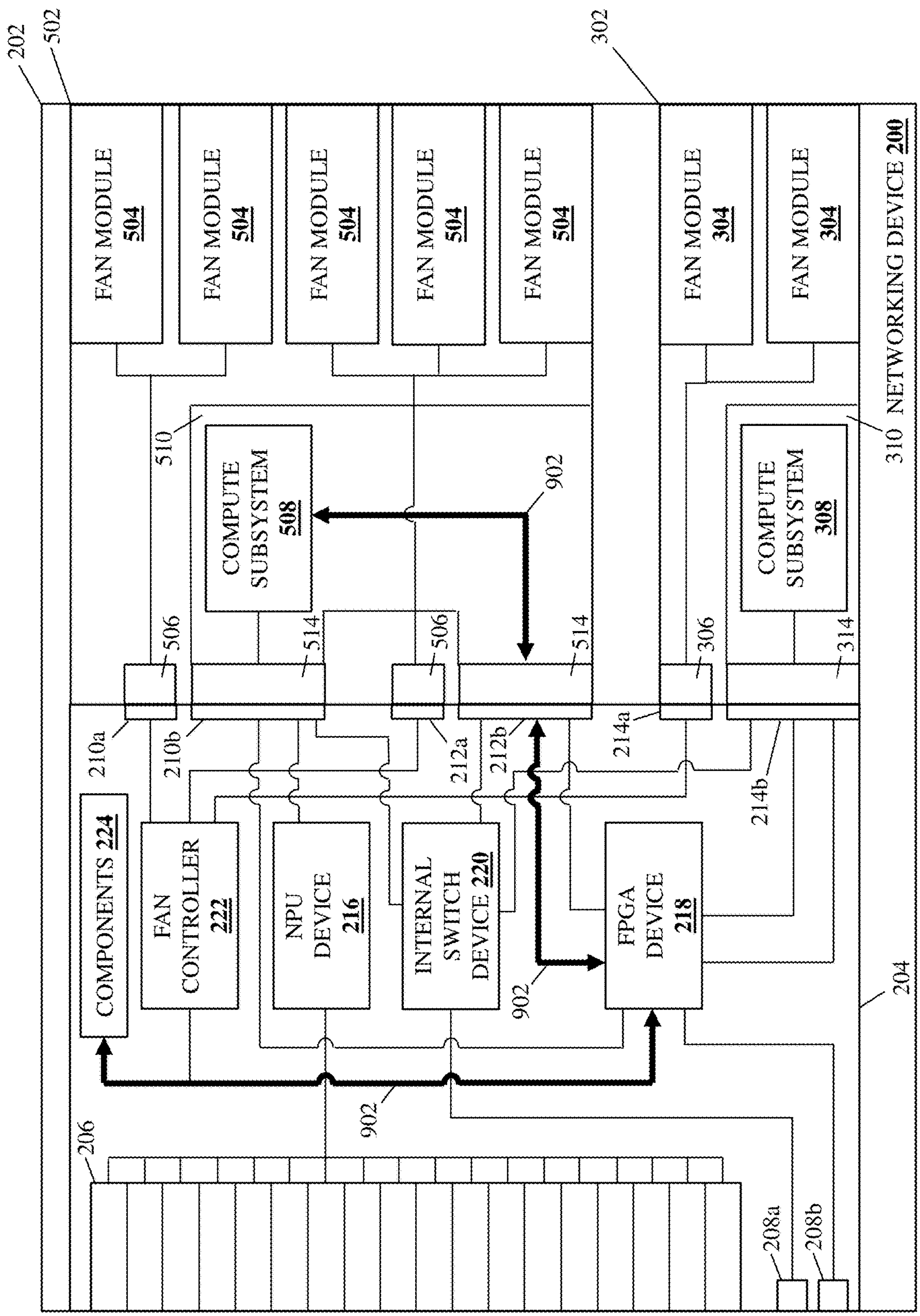
208a
208b

FIG. 9C



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COMPUTE SUBSYSTEM 308
NETWORKING DEVICE 200
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212b
214a
214b
COMPONENTS 224
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FPGA DEVICE 218
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208a
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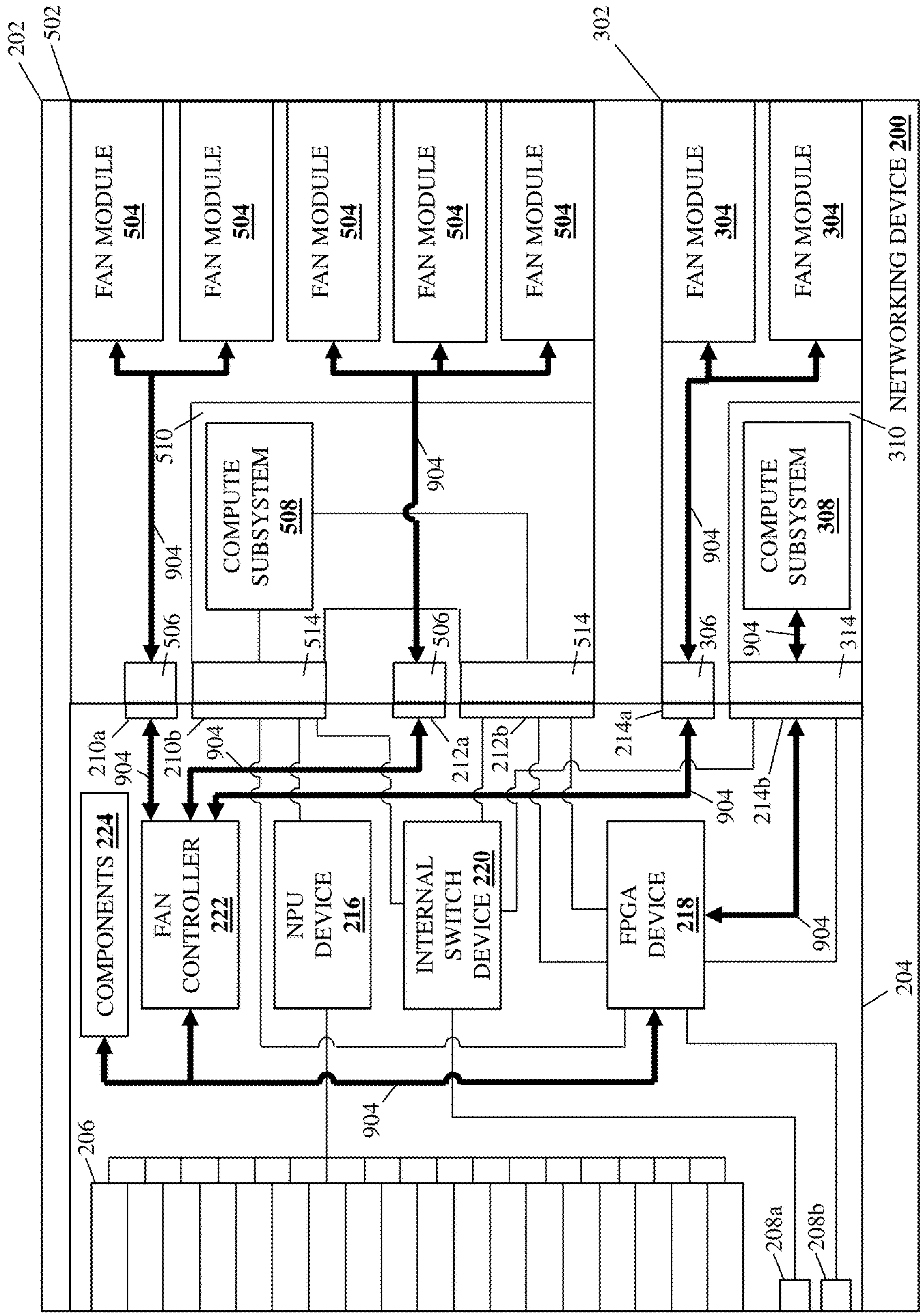
FIG. 9D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COMPUTE SUBSYSTEM 308
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214b
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FAN CONTROLLER 222
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FPGA DEVICE 218
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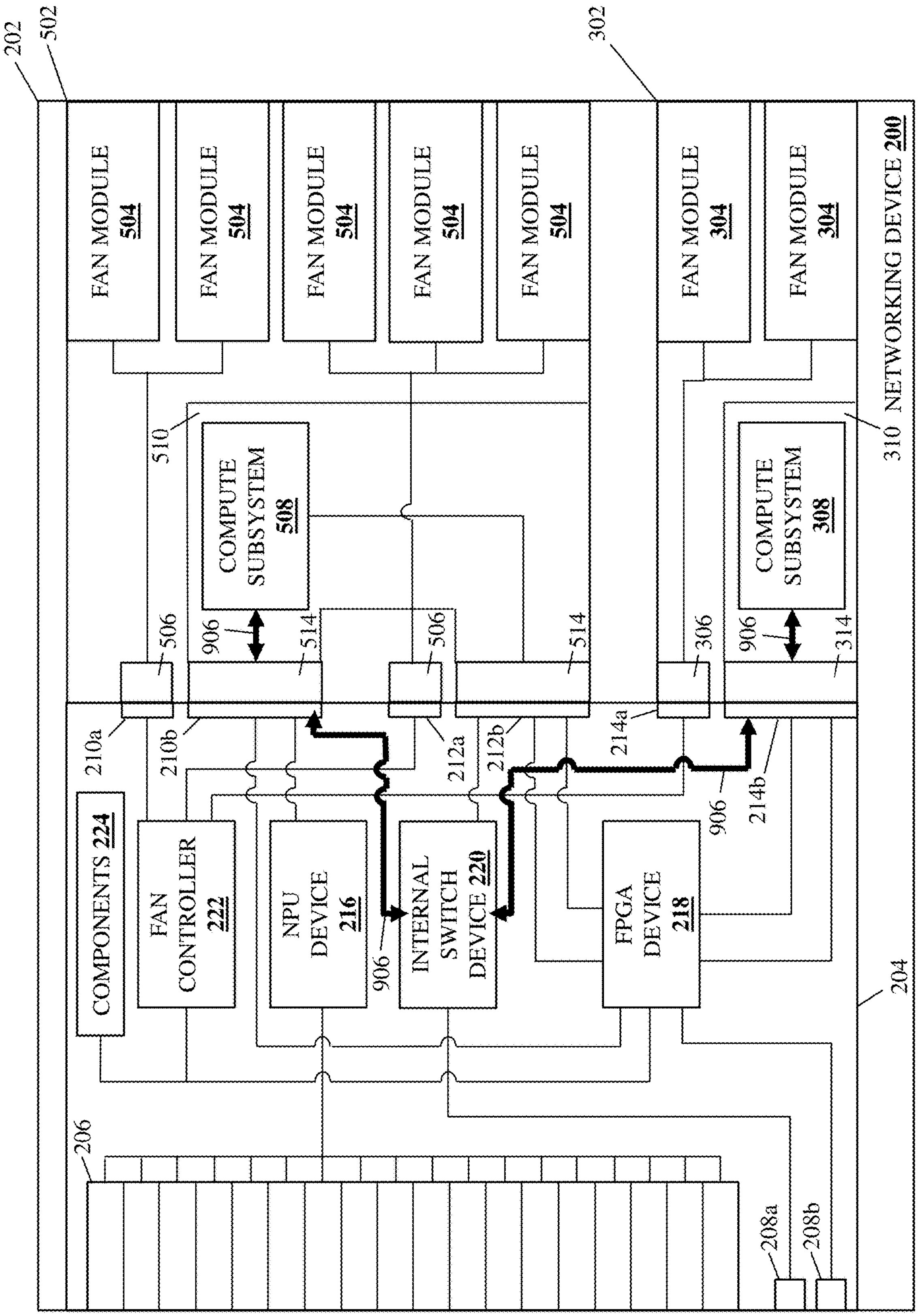
FIG. 9E



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FAN MODULE 504
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NETWORKING DEVICE 200
510
COMPUTE SUBSYSTEM 508
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COMPUTE SUBSYSTEM 308
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214b
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INTERNAL SWITCH DEVICE 220
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FPGA DEVICE 218
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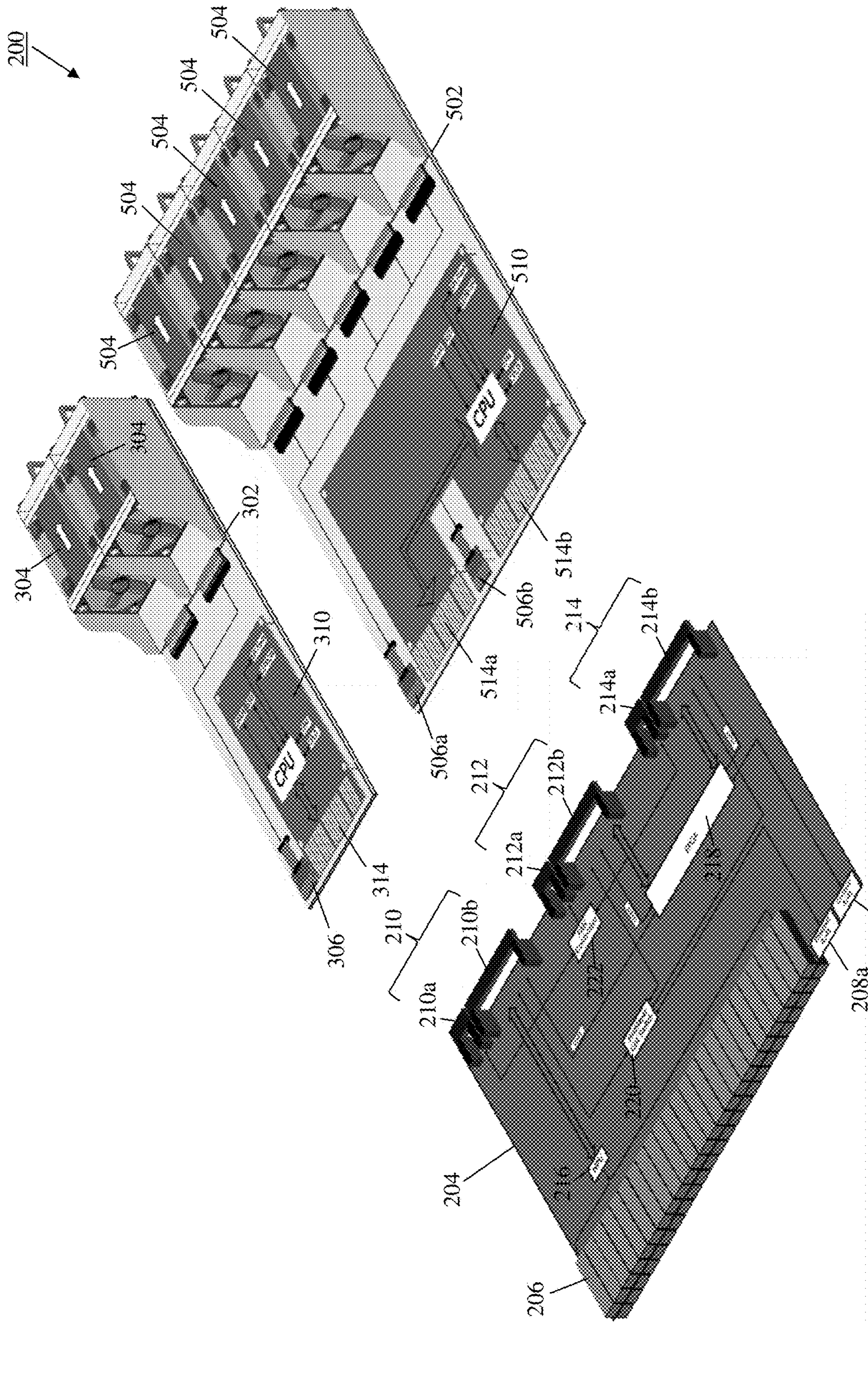
208b

FIG. 9F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CPU
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306
214
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208a
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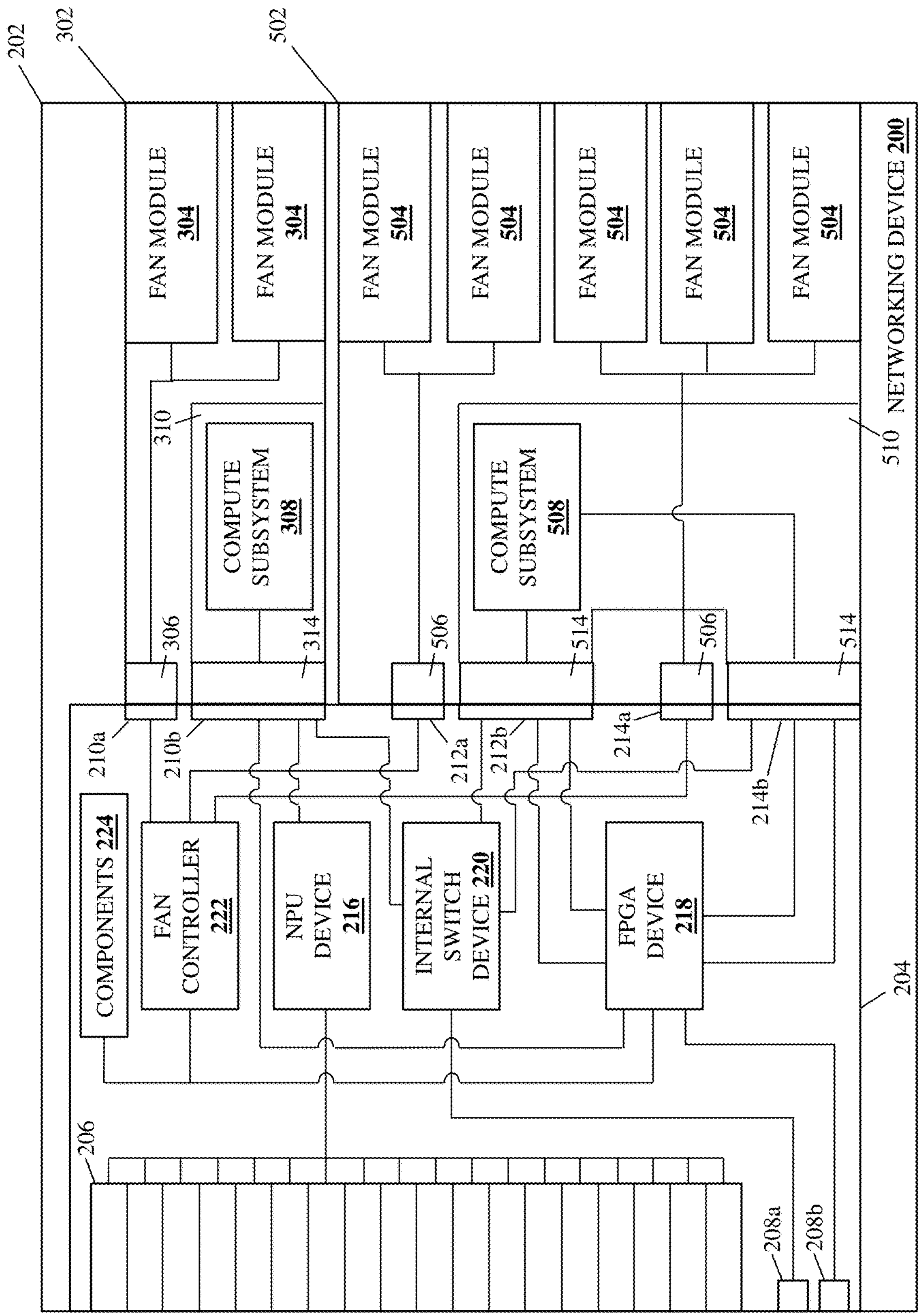
FIG. 10A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212b
214a
214b
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FPGA DEVICE 218
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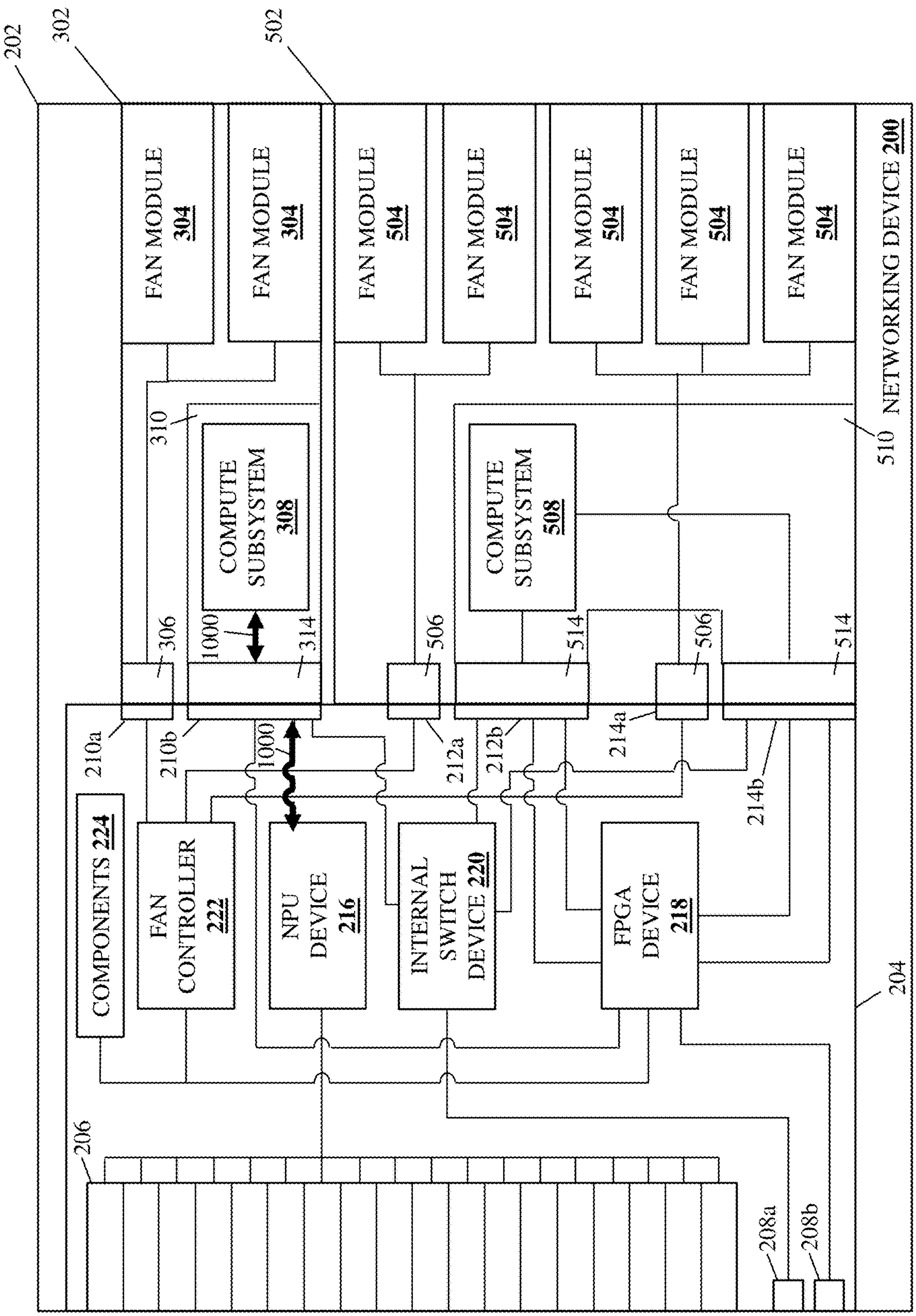
FIG. 10B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COMPUTE SUBSYSTEM 508
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212b
214a
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INTERNAL SWITCH DEVICE 220
FPGA DEVICE 218
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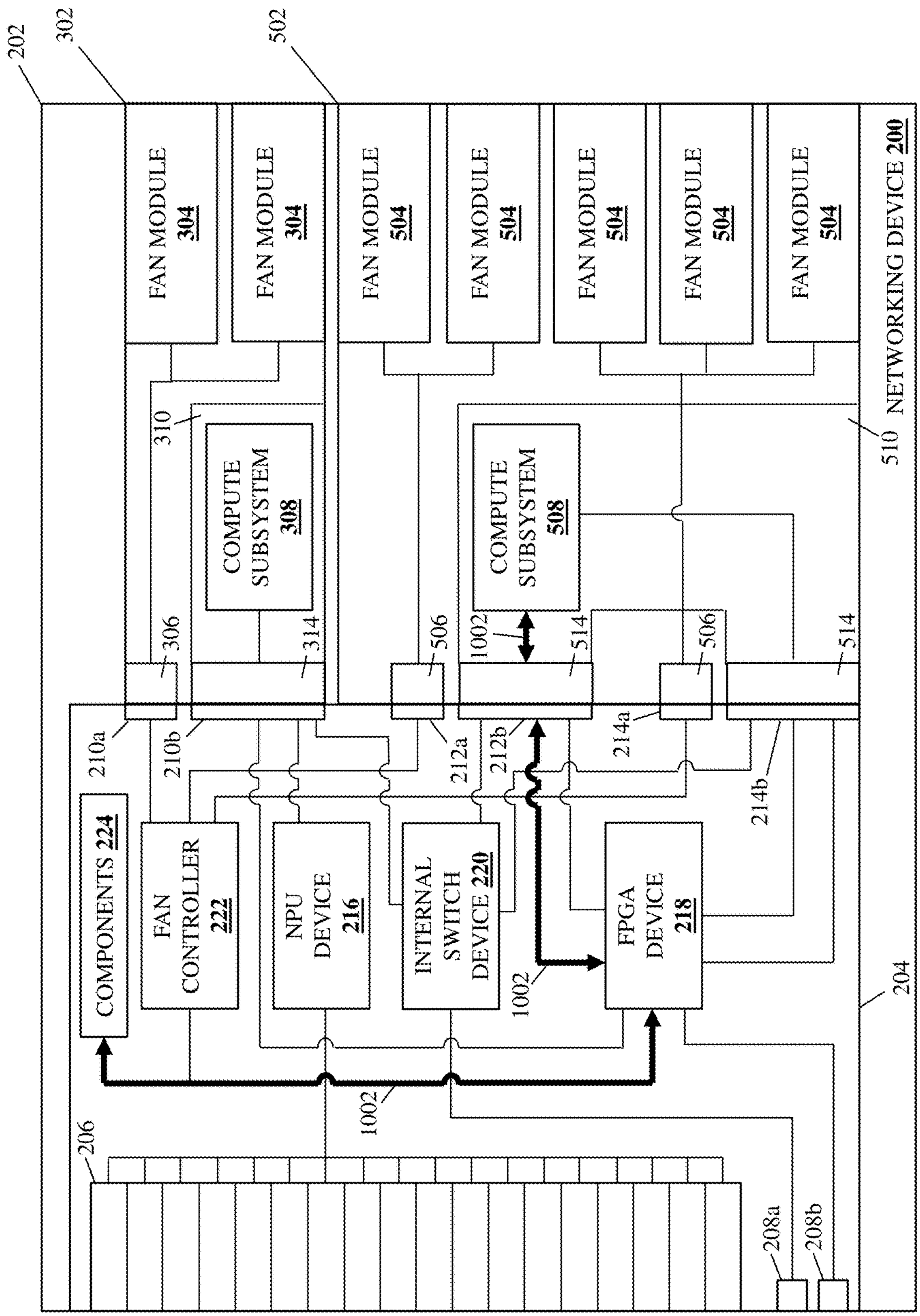
208a
208b

FIG. 10C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214b
COMPONENTS 224
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FPGA DEVICE 218
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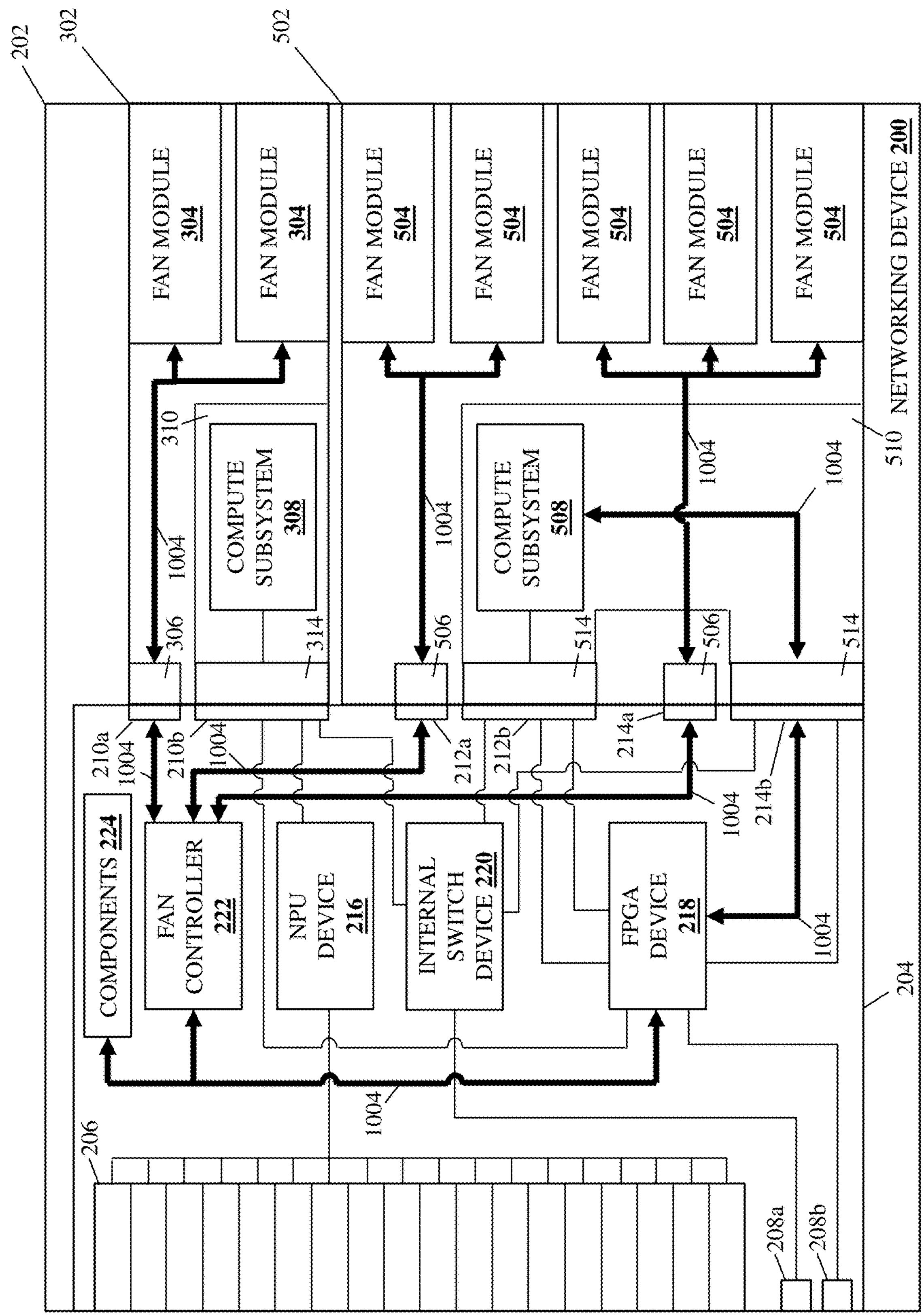
FIG. 10D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COMPUTE SUBSYSTEM 508
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FPGA DEVICE 218
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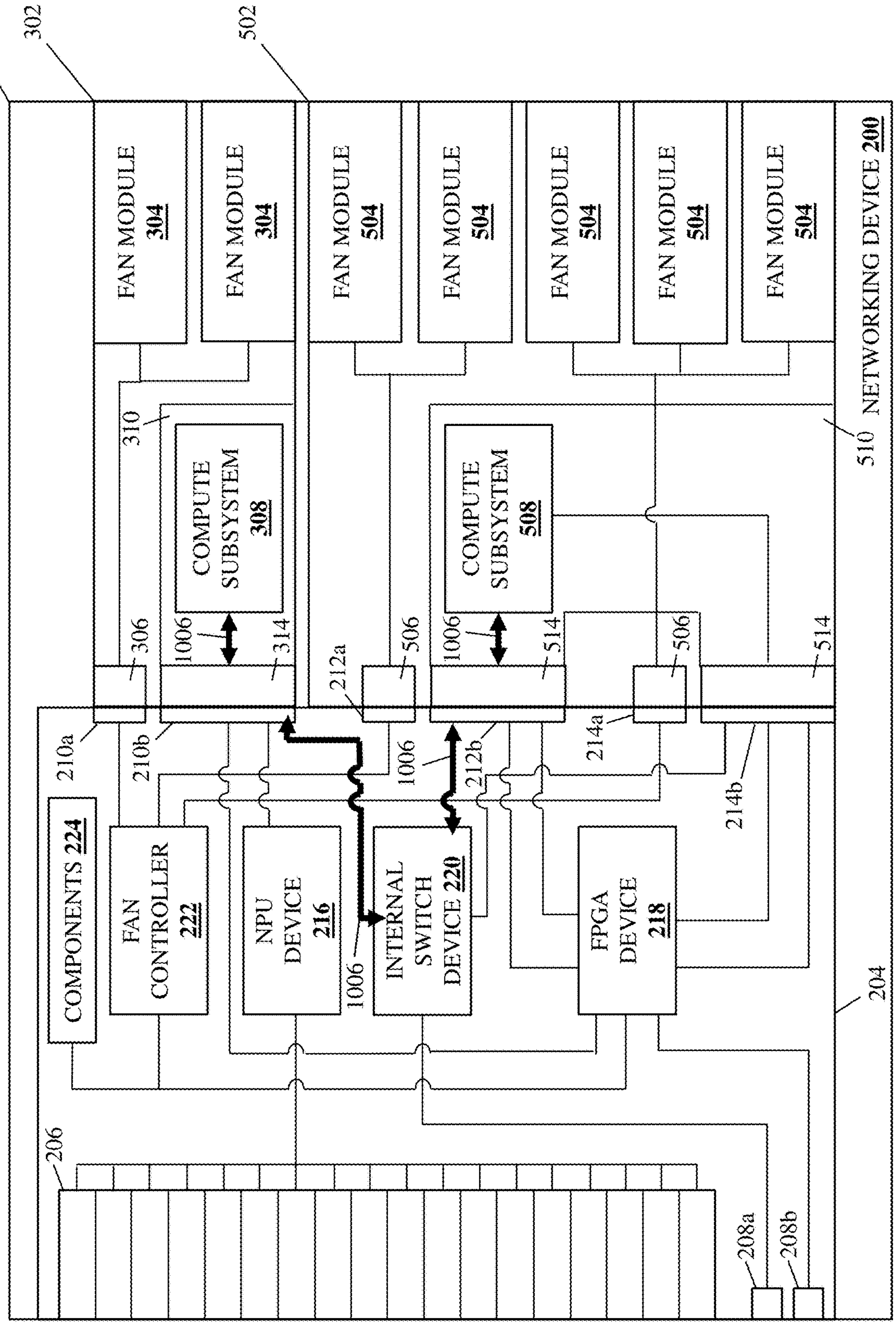
208a
208b

FIG. 10E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506
514
210a
210b
1006
212b
214a
214b
COMPONENTS 224
FAN CONTROLLER 222
NPU DEVICE 216
1006
INTERNAL SWITCH DEVICE 220
FPGA DEVICE 218
204
206
208a
208b

FIG. 10F

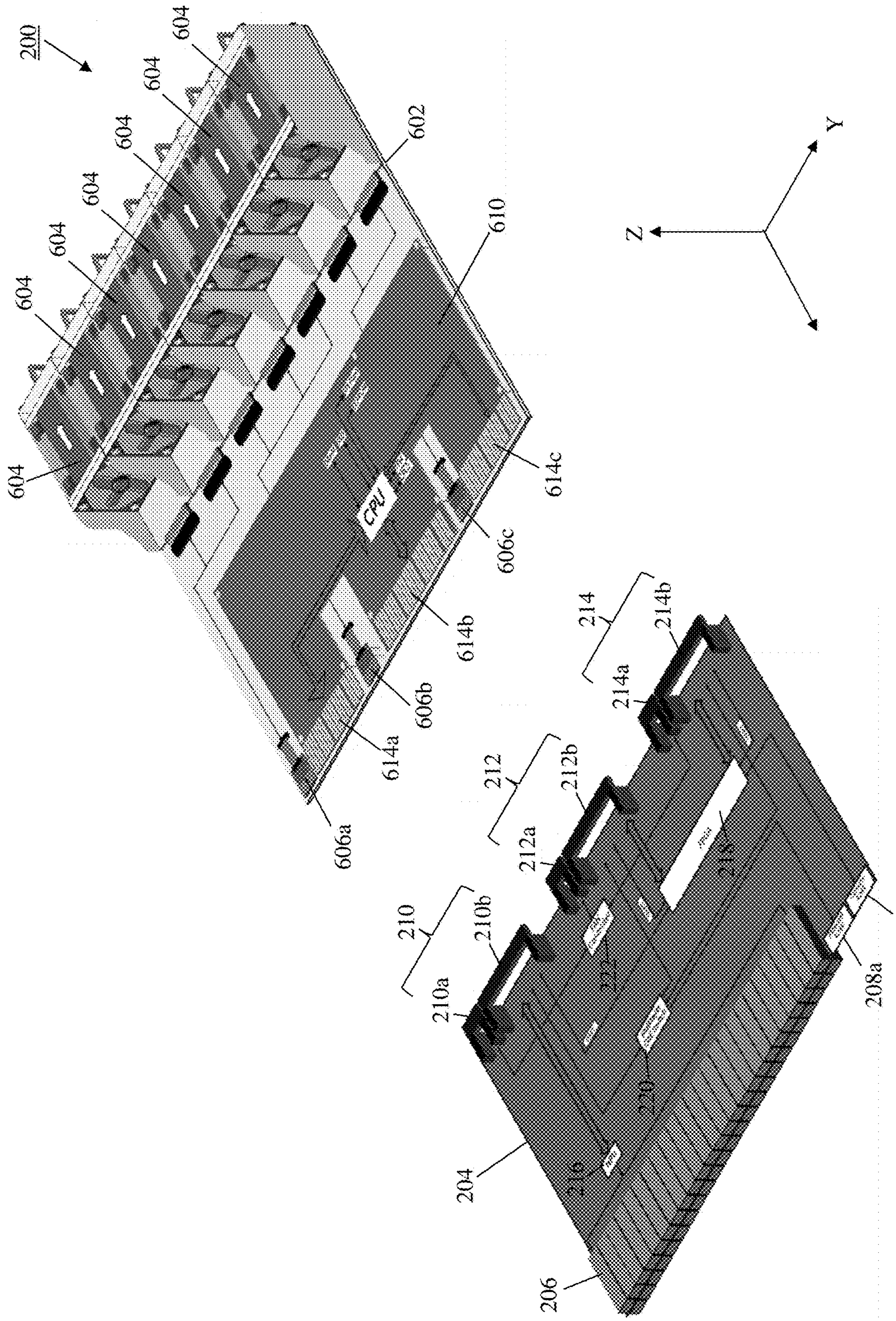
200
604
604
604
604
604
604
604
602
610
CPU
614a
614b
614c
606a
606b
606c
210
210a
210b
212
212a
212b
214
214a
214b
218
222
220
216
204
206
208a
208b
Z
Y

FIG. 11A

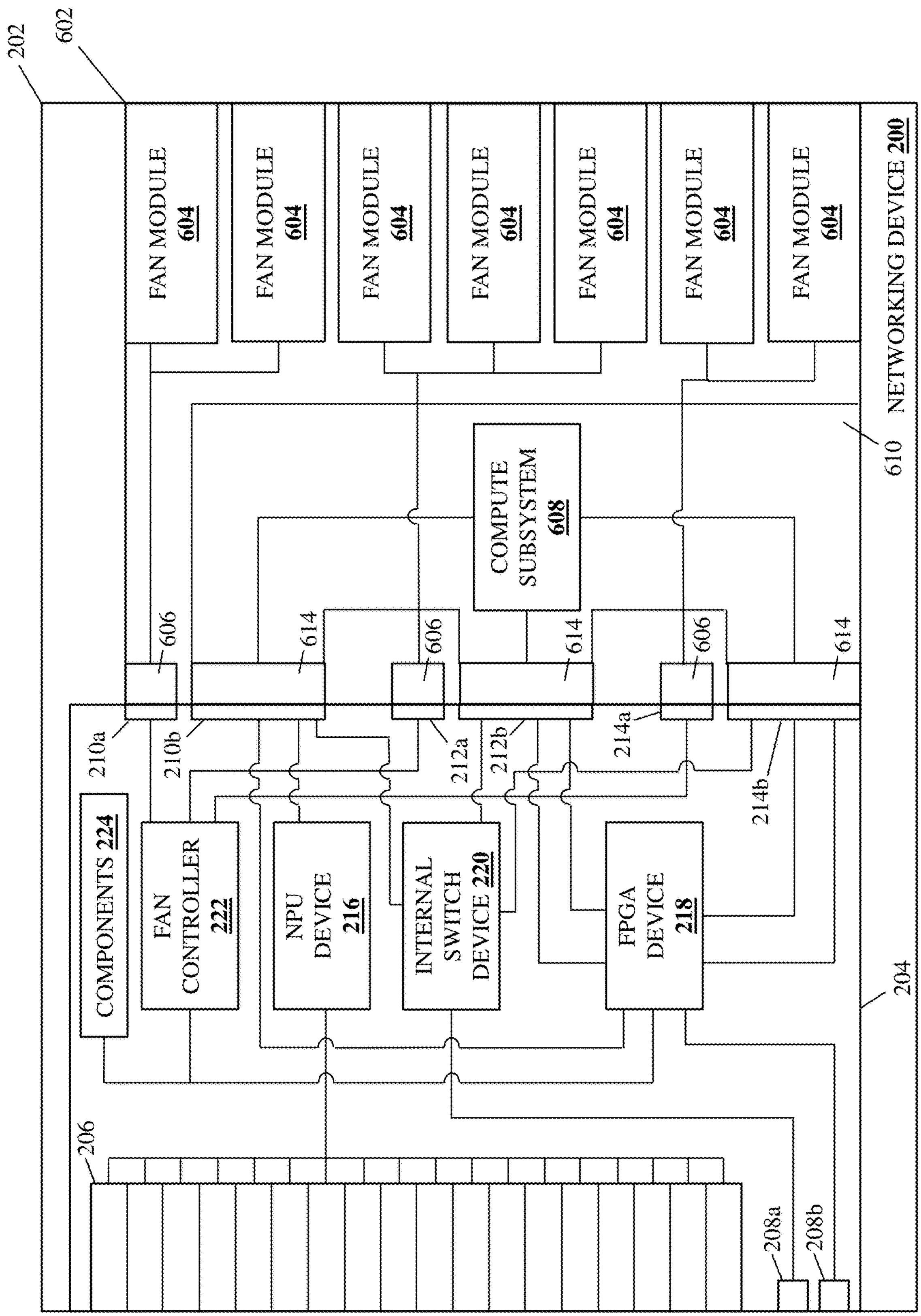
202
602
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
NETWORKING DEVICE 200
610
COMPUTE SUBSYSTEM 608
606
614
606
614
606
614
210a
210b
212a
212b
214a
214b
COMPONENTS 224
FAN CONTROLLER 222
NPU DEVICE 216
INTERNAL SWITCH DEVICE 220
FPGA DEVICE 218
204
206
208a
208b

FIG. 11B

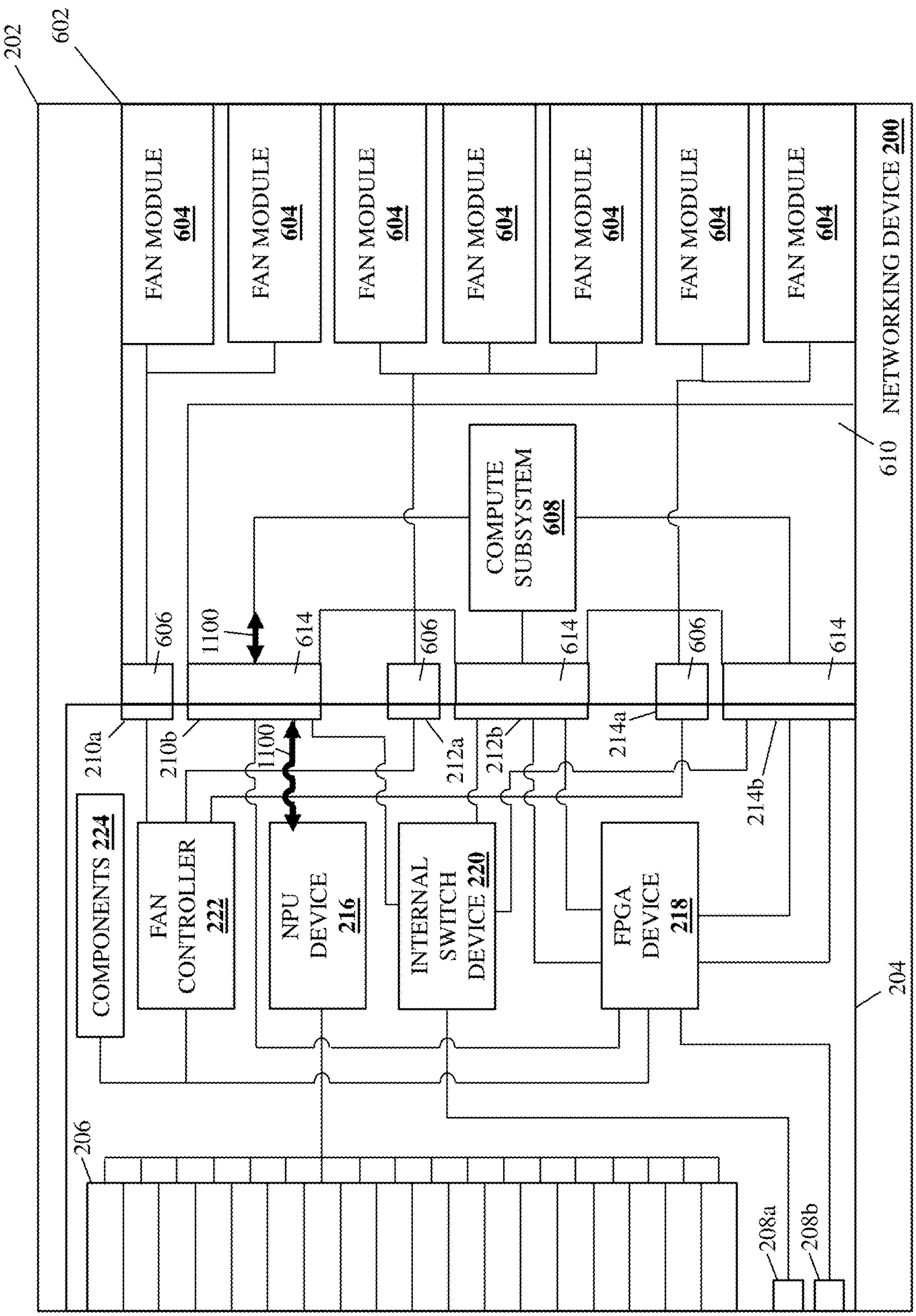
202
602
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
NETWORKING DEVICE 200
610
COMPUTE SUBSYSTEM 608
1100
606
614
606
614
606
614
210a
210b
1100
212a
212b
214a
214b
COMPONENTS 224
FAN CONTROLLER 222
NPU DEVICE 216
INTERNAL SWITCH DEVICE 220
FPGA DEVICE 218
204
206
208a
208b

FIG. 11C

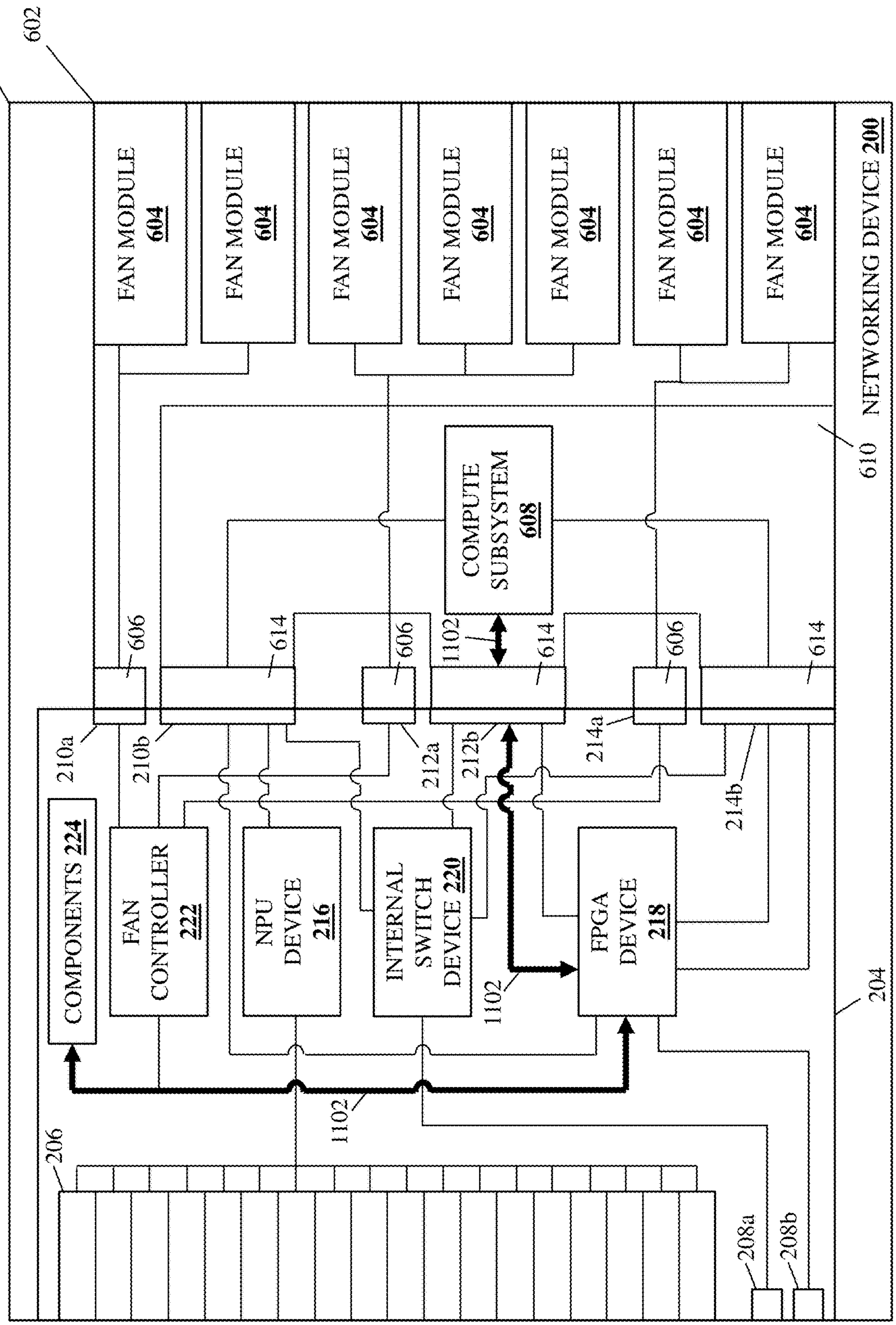
202
602
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
NETWORKING DEVICE 200
610
COMPUTE SUBSYSTEM 608
606
614
606
1102
614
606
614
210a
210b
212a
212b
214a
214b
COMPONENTS 224
FAN CONTROLLER 222
NPU DEVICE 216
INTERNAL SWITCH DEVICE 220
FPGA DEVICE 218
1102
204
1102
206
208a
208b

FIG. 11D

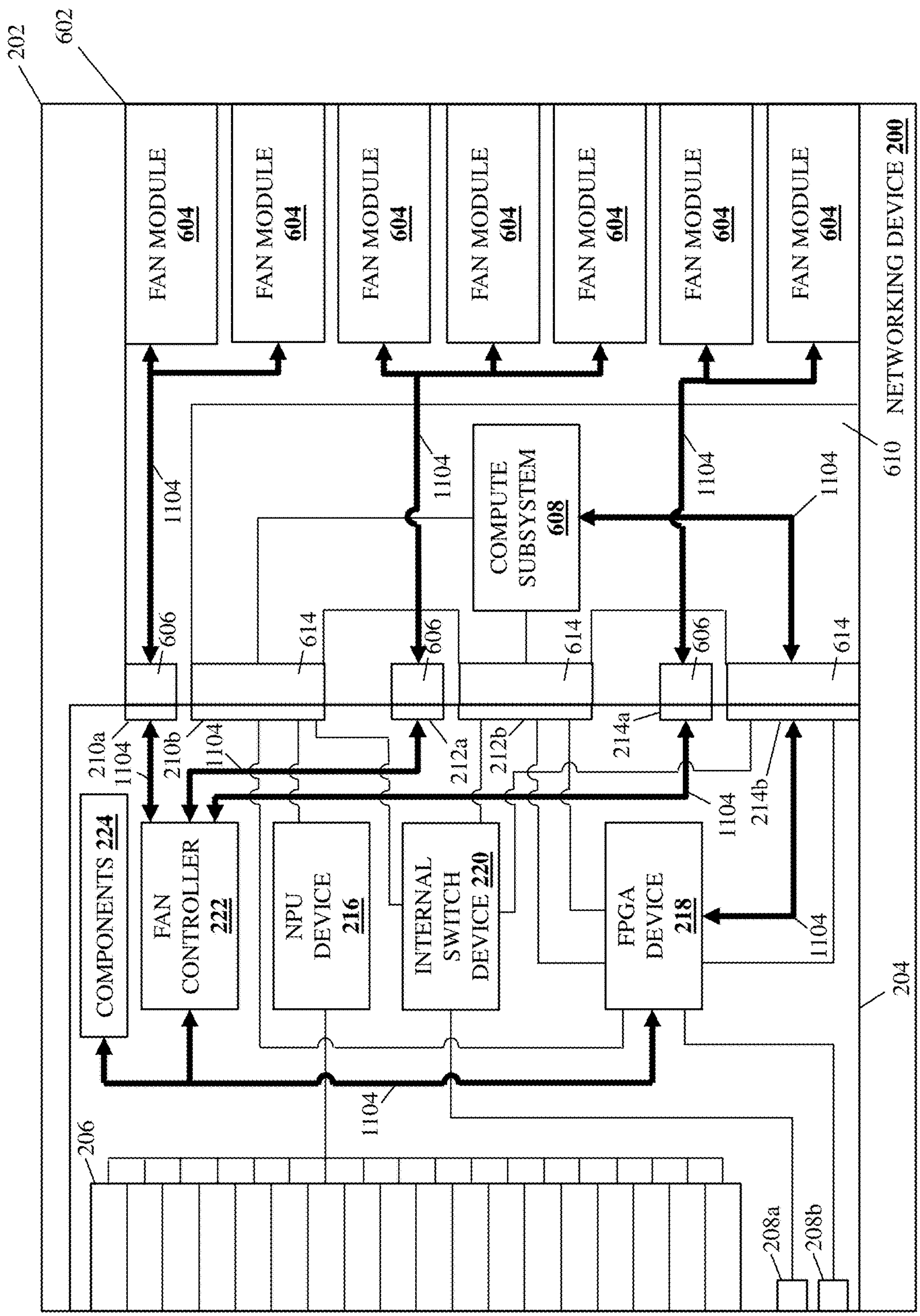
202
602
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
FAN MODULE 604
NETWORKING DEVICE 200
610
1104
1104
1104
1104
COMPUTE SUBSYSTEM 608
606
614
606
614
606
614
210a
1104
210b
1104
212a
212b
214a
1104
214b
COMPONENTS 224
FAN CONTROLLER 222
NPU DEVICE 216
INTERNAL SWITCH DEVICE 220
FPGA DEVICE 218
1104
204
1104
206
208a
208b

FIG. 11E

MODULAR COFTA-BASED NETWORKING DEVICE SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present disclosure is a continuation-in-part of U.S. patent application Ser. No. 18/600,978, attorney docket no. 137176.01, filed Mar. 11, 2024, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] The present disclosure relates generally to information handling systems, and more particularly to a modular networking information handling system provided using Computer-On-Fan-Tray-Assembly (COFTA) systems.

[0003] As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

[0004] Information handling systems such as, for example, switch devices and other networking devices known in the art, may be provided with a variety of switch management functionality that can vary based on different networking applications/use cases and that can affect their cost. However, such switch management functionality generally provides for management of the networking processing system (e.g., a Network Processing Unit (NPU), switch Application-Specific Integrated Circuit (ASIC), etc.) and associated networking components in the switch device, networking software utilized by the switch device, and non-networking hardware in the switch device. In conventional switch devices, a motherboard or other monolithic switch circuit board that includes the networking processing system is typically provided with a Central Processing Unit (CPU) that is integrated with the networking processing system and other components in the switch device and configured to provide a Networking Operating System (NOS) that operates to perform all of the switch management functionality described above.

[0005] However, the networking processing system and associated networking components in the switch device, the networking software utilized by the switch device, and the non-networking hardware in the switch device may each be provided by a different source, which can present issues with their management via the NOS discussed above. For example, capabilities of the networking drivers and Switch Abstraction Interface (SAI) used by the networking processing system may differ depending on the NOS being used, errors that occur in the switch device may present difficulties with regard to identifying their root cause (e.g., errors due to the operation of the networking processing system, networking software, or non-networking hardware), and/or other issues that would be apparent to one of skill in the art in possession of the present disclosure.

[0006] Accordingly, it would be desirable to provide a networking device that addresses the issues discussed above.

SUMMARY

[0007] According to one embodiment, a networking device includes a networking device chassis; a circuit board that is housed in the networking device chassis; a plurality of networking ports that are included on the circuit board and that are accessible on the networking device chassis; a networking processing system that is included on the circuit board and that is coupled to the plurality of ports; a plurality of non-networking components that are couped to the circuit board; and a networking device management connector system that is included on the circuit board and that includes: a networking processing system management connector that is coupled to the networking processing system; a non-networking component management connector that is coupled to the non-networking components; and a networking software management connector, wherein the networking device management connector system is configured to connect to at least two Computer-On-Fan-Tray-Assembly (COFTA) systems that are each configured to provide a respective operating system that is configured to manage a different subset of the networking processing system via networking processing system management connector, the non-networking components via the non-networking component management connector, and networking software via networking software management connector.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a schematic view illustrating an embodiment of an Information Handling System (IHS).

[0009] FIG. **2**A is a schematic view illustrating an embodiment of a networking device that may provide the modular COFTA-based networking device system of the present disclosure.

[0010] FIG. **2**B is a perspective view illustrating an embodiment of the networking device of FIG. **2**A.

[0011] FIG. **3**A is a schematic view illustrating an embodiment of a COFTA system that may provide the modular COFTA-based networking device system of the present disclosure.

[0012] FIG. **3**B is a perspective view illustrating an embodiment of the COFTA system of FIG. **3**A.

[0013] FIG. **4**A is a schematic view illustrating an embodiment of a COFTA system that may provide the modular COFTA-based networking device system of the present disclosure.

[0014] FIG. **4**B is a perspective view illustrating an embodiment of the COFTA system of FIG. **4**A.

[0015] FIG. **5**A is a schematic view illustrating an embodiment of a COFTA system that may provide the modular COFTA-based networking device system of the present disclosure.

[0016] FIG. **5**B is a perspective view illustrating an embodiment of the COFTA system of FIG. **5**A.

[0017] FIG. **6**A is a schematic view illustrating an embodiment of a COFTA system that may be used with the modular COFTA-based networking device system of the present disclosure.

[0018] FIG. **6**B is a perspective view illustrating an embodiment of the COFTA system of FIG. **6**A.

[0019] FIG. **7** is a flow chart illustrating an embodiment of a method for providing a modular COFTA-based networking device.

[0020] FIG. **8**A is a perspective view illustrating an embodiment of a pair of the COFTA systems of FIGS. **3**A and **3**B, and the COFTA system of FIGS. **4**A and **4**B, being coupled to the networking device of FIGS. **2**A and **2**B during the method of FIG. **7**.

[0021] FIG. **8**B is a schematic view illustrating an embodiment of the pair of the COFTA systems of FIGS. **3**A and **3**B, and the COFTA system of FIGS. **4**A and **4**B, coupled to the networking device of FIGS. **2**A and **2**B to provide a modular COFTA-based networking device system during the method of FIG. **7**.

[0022] FIG. **8**C is a schematic view illustrating an embodiment of the modular COFTA-based networking device system of FIG. **8**B operating during the method of FIG. **7**.

[0023] FIG. **8**D is a schematic view illustrating an embodiment of the modular COFTA-based networking device system of FIG. **8**B operating during the method of FIG. **7**

[0024] FIG. **8**E is a schematic view illustrating an embodiment of the modular COFTA-based networking device system of FIG. **8**B operating during the method of FIG. **7**

[0025] FIG. **8**F is a schematic view illustrating an embodiment of the modular COFTA-based networking device system of FIG. **8**B operating during the method of FIG. **7**

[0026] FIG. **9**A is a perspective view illustrating an embodiment of the COFTA systems of FIGS. **5**A and **5**B, and the COFTA system of FIGS. A and **4**B,**3** being coupled to the networking device of FIGS. **2**A and **2**B during the method of FIG. **7**.

[0027] FIG. **9**B is a schematic view illustrating an embodiment of the COFTA systems of FIGS. **5**A and **5**B, and the COFTA system of FIGS. **3**A and **3**B, coupled to the networking device of FIGS. **2**A and **2**B to provide a modular COFTA-based networking device system during the method of FIG. **7**.

[0028] FIG. **9**C is a schematic view illustrating an embodiment of the modular COFTA-based networking device system of FIG. **9**B operating during the method of FIG. **7**.

[0029] FIG. **9**D is a schematic view illustrating an embodiment of the modular COFTA-based networking device system of FIG. **9**B operating during the method of FIG. **7**.

[0030] FIG. **9**E is a schematic view illustrating an embodiment of the modular COFTA-based networking device system of FIG. **9**B operating during the method of FIG. **7**.

[0031] FIG. **9**F is a schematic view illustrating an embodiment of the modular COFTA-based networking device system of FIG. **9**B operating during the method of FIG. **7**.

[0032] FIG. **10**A is a perspective view illustrating an embodiment of the COFTA system of FIGS. **3**A and **3**B, and the COFTA system of FIGS. **5**A and **5**B, being coupled to the networking device of FIGS. **2**A and **2**B during the method of FIG. **7**.

[0033] FIG. **10**B is a schematic view illustrating an embodiment of the COFTA system of FIGS. **3**A and **3**B, and the COFTA system of FIGS. **5**A and **5**B, coupled to the networking device of FIGS. **2**A and **2**B during the method of FIG. **7**.

[0034] FIG. **10**C is a schematic view illustrating an embodiment of the modular COFTA-based networking device system of FIG. **10**B operating during the method of FIG. **7**.

[0035] FIG. **10**D is a schematic view illustrating an embodiment of the modular COFTA-based networking device system of FIG. **10**B operating during the method of FIG. **7**.

[0036] FIG. **10**E is a schematic view illustrating an embodiment of the modular COFTA-based networking device system of FIG. **10**B operating during the method of FIG. **7**.

[0037] FIG. **10**F is a schematic view illustrating an embodiment of the modular COFTA-based networking device system of FIG. **10**B operating during the method of FIG. **7**.

[0038] FIG. **11**A is a perspective view illustrating an embodiment of the COFTA system of FIGS. **6**A and **6**B being coupled to the networking device of FIGS. **2**A and **2**B.

[0039] FIG. **11**B is a schematic view illustrating an embodiment of the COFTA system of FIGS. **6**A and **6**B coupled to the networking device of FIGS. **2**A and **2**B.

[0040] FIG. **11**C is a schematic view illustrating the operation of the COFTA system of FIGS. **6**A and **6**B coupled to the networking device of FIGS. **2**A and **2**B.

[0041] FIG. **11**D is a schematic view illustrating the operation of the COFTA system of FIGS. **6**A and **6**B coupled to the networking device of FIGS. **2**A and **2**B.

[0042] FIG. **11**E is a schematic view illustrating the operation of the COFTA system of FIGS. **6**A and **6**B coupled to the networking device of FIGS. **2**A and **2**B.

DETAILED DESCRIPTION

[0043] For purposes of this disclosure, an information handling system may include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer (e.g., desktop or laptop), tablet computer, mobile device (e.g., personal digital assistant (PDA) or smart phone), server (e.g., blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, touchscreen and/or a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components.

[0044] In one embodiment, IHS **100**, FIG. **1**, includes a processor **102**, which is connected to a bus **104**. Bus **104** serves as a connection between processor **102** and other components of IHS **100**. An input device **106** is coupled to processor **102** to provide input to processor **102**. Examples of input devices may include keyboards, touchscreens, pointing devices such as mouses, trackballs, and trackpads, and/or a variety of other input devices known in the art. Programs and data are stored on a mass storage device **108**, which is coupled to processor **102**. Examples of mass storage devices may include hard discs, optical disks, magneto-optical discs, solid-state storage devices, and/or a variety of other mass storage devices known in the art. IHS **100** further includes a display **110**, which is coupled to processor **102** by a video controller **112**. A system memory **114** is coupled to processor **102** to provide the processor with fast storage to facilitate execution of computer programs by processor **102**. Examples of system memory may include random access memory (RAM) devices such as dynamic RAM (DRAM), synchronous DRAM (SDRAM), solid state memory devices, and/or a variety of other memory devices known in the art. In an embodiment, a chassis **116** houses some or all of the components of IHS **100**. It should be understood that other buses and intermediate circuits can be deployed between the components described above and processor **102** to facilitate interconnection between the components and the processor **102**.

[0045] Referring now to FIGS. **2**A and **2**B, an embodiment of a networking device **200** is illustrated that may provide the modular COFTA-based networking device system of the present disclosure. In an embodiment, the networking device **200** may be provided by the IHS **100** discussed above with reference to FIG. **1**, and/or may include some or all of the components of the IHS **100**, and in the specific examples provided below is provided by a switch device. However, while illustrated and discussed as being provided by a switch device, one of skill in the art in possession of the present disclosure will recognize that the modular COFTA-based networking device system of the present disclosure may be provided by a variety of networking devices while remaining within the scope of the present disclosure.

[0046] In the illustrated embodiment, the networking device **200** includes a networking device chassis **202** that houses the components of the networking device **200**, only some of which are illustrated and described below. In the examples illustrated and described below, the networking device chassis **202** houses a circuit board **204** that may be provided by a networking device motherboard and/or other networking device circuit boards that would be apparent to one of skill in the art in possession of the present disclosure. A plurality of networking ports **206** are included on the circuit board **204**, and one of skill in the art in possession of the present disclosure will appreciate how the networking ports **206** may be accessible on an outer surface of the networking device chassis **202**. In a specific example, the networking device ports **206** may be provided by transceiver systems that one of skill in the art in possession of the present disclosure will appreciate may include transceiver cages that are mounted to the circuit board **204** and that are configured to couple transceiver devices (e.g., Quad Small Form-factor Pluggable (QSFP) transceiver devices) to the circuit board **204**, although other networking device ports **206** will fall within the scope of the present disclosure as well.

[0047] A pair of management ports **208***a* and **208***b* are included on the circuit board **204**, and one of skill in the art in possession of the present disclosure will appreciate how the management ports **208***a* and **208***b* may be accessible on an outer surface of the networking device chassis **202**. In a specific example, the management port **208***a* may be provided by an Ethernet RJ45 port that one of skill in the art in possession of the present disclosure will appreciate may be utilized to perform remote management of the networking device **200** (e.g., via a networking cable coupled to the Ethernet RJ45 port and a network), and the management port **208***b* may be provided by a console RJ45 port that one of skill in the art in possession of the present disclosure will appreciate may be utilized to perform "in-person" management of the networking device **200** (e.g., via a networking cable coupled to the console RJ45 port and a laptop/notebook computing device located adjacent the networking device **200**), although other management ports **208***a* and **208***b* will fall within the scope of the present disclosure as well.

[0048] The circuit board **204** also includes a networking device management connector system that is provided by a plurality of COFTA system connectors, embodiments of which were described by the inventors of the present disclosure in U.S. patent application Ser. No. 18/600,978, attorney docket no. 137176.01, filed Mar. 11, 2024, the disclosure of which is incorporated by reference herein in its entirety. In the embodiments illustrated and described below, the networking device management connector system on the circuit board **204** includes three COFTA system connectors: a COFTA system connector **210** provided on the circuit board **204** by a COFTA fan subsystem connector **210***a* and a COFTA compute subsystem connector **210***b*, with the COFTA compute subsystem connector **210***b* operating as a "networking processing system management" connector in the examples described below; a COFTA system connector **212** provided on the circuit board **204** by a COFTA fan subsystem connector **212***a* and a COFTA compute subsystem connector **212***b*, with the COFTA compute subsystem connector **212***b* operating as a "networking software management" connector in the examples described below; and a COFTA system connector **214** provided on the circuit board **204** by a COFTA fan subsystem connector **214***a* and a COFTA compute subsystem connector **214***b*, with the COFTA compute subsystem connector **214***b* operating as a "non-networking component management" connector in the examples described below.

[0049] A networking processing system is included on the circuit board **204** and is illustrated and described below as including an Network Processing Unit (NPU) device **216**, but one of skill in the art in possession of the present disclosure will appreciate how the networking processing system may include a switch Application-Specific Integrated Circuit (ASIC) and/or other networking processors known in the art, as well as forwarding tables, telemetry hardware, and/or other networking hardware, firmware, and/or other networking processing system components known in the art. As can be seen in FIGS. **2**A and **2**B, the NPU device **216** is coupled to the plurality of networking ports **206**, as well as to the COFTA compute subsystem connector **210***b* on the

COFTA system connector **210** that operates as a "networking processing system management" connector in the examples described below.

[0050] A Field Programmable Gate Array (FPGA) device **218** is included on the circuit board **204**, but one of skill in the art in possession of the present disclosure will appreciate how the FPGA device **218** may be replaced by a Complex Programmable Logic Device (CPLD) and/or other processing devices that operate similarly to the FPGA device **218** described below. The FPGA device **218** includes a single connection to the COFTA compute subsystem connector **210***b* on the COFTA system connector **210** that operates as a "networking processing system management" connector in the examples described below, and as illustrated in FIG. **2**B that connection may be provided by a Universal Asynchronous Receiver-Transmitter (UART) connection and/or other similar connection known in the art.

[0051] The FPGA device **218** also includes a pair of connections to the COFTA compute subsystem connector **212***b* on the COFTA system connector **212** that operates as a "networking software management" connector in the examples described below, and as illustrated in FIG. **2**B one of those connections may be provided by a UART connection and/or other similar connection known in the art. The FPGA device **218** also includes a pair of connections to the COFTA compute subsystem connector **214***b* on the COFTA system connector **214** that operates as a "non-networking component management" connector in the examples described below, and as illustrated in FIG. **2**B one of those connections may be provided by a UART connection and/or other similar connection known in the art. The FPGA device **218** also includes a connection to the management port **208***b* that one of skill in the art in possession of the present disclosure will appreciate may be used for the "in-person" management of the networking device **200** described above and in further detail below.

[0052] An internal switch device **220** is included on the circuit board **204** and is illustrated in FIG. **2**B as being provided by a multilayer Gigabit Ethernet (GbE) switch, but one of skill in the art in possession of the present disclosure will appreciate how the internal switch device **220** may be provided by other switch devices or switching components that operate similarly to the internal switch device **220** described below. The internal switch device **220** includes a connection to the COFTA compute subsystem connector **210***b* on the COFTA system connector **210** that operates as a "networking processing system management" connector in the examples described below, a connection to the COFTA compute subsystem connector **212***b* on the COFTA system connector **212** that operates as a "networking software management" connector in the examples described below, and a connection to the COFTA compute subsystem connector **214***b* on the COFTA system connector **214** that operates as a "non-networking component management" connector in the examples described below. The FPGA device **218** also includes a connection to the management port **208***a* that one of skill in the art in possession of the present disclosure will appreciate may be used for the remote management of the networking device **200** described above and in further detail below.

[0053] A fan controller **222** is included on the circuit board **204** and includes a connection to the COFTA fan subsystem connector **210***a* on the COFTA system connector **210**, a connection to the COFTA fan subsystem connector **212***a* on the COFTA system connector **212**, and a connection to the COFTA fan subsystem connector **214***a* on the COFTA system connector **214**. The fan controller **222** also includes a connection to the FPGA device **218**. A plurality of components **224** are also illustrated as being included on the circuit board **204** and including connections to the FPGA device **218**, but one of skill in the art in possession of the present disclosure will appreciate how the components may be coupled to the circuit board **204** via connectors, cables, and/or other couplings while remaining within the scope of the present disclosure as well. In specific examples, the components **224** may include power systems (e.g., power adapter components, power supply components, and/or other power system components known in the art), Light Emitting Device (LED) indicator system (e.g., LED components such as port LEDs for the ports **206**, LED control components, and/or other LED indicator system components known in the art), thermal control systems (e.g., thermal sensor components used with the fan controller **222**), networking software components that interact with networking software used by the networking device **200** (e.g., firmware that interacts with the Networking Operating System (NOS) and/or that is otherwise included in a networking software management system, a Trusted Platform Module (TPM), a Power over Ethernet (POE) manager or controller, a cryptographic accelerator, a console buffer, etc.), and/or any other components that would be apparent to one of skill in the art in possession of the present disclosure.

[0054] As described below, the embodiment of the portion of the modular COFTA-based networking device system provided by the networking device **200** illustrated and described below includes the "networking processing system management" connector provided by the COFTA compute subsystem connector **210***b* in the COFTA system connector **210** that has exclusive access the NPU device **216** that is coupled to the ports **206** (as well as other networking processing system components) in order to enable management of the networking processing system in the networking device **200**.

[0055] the embodiment of the portion of the modular COFTA-based networking device system provided by the networking device **200** illustrated and described below also includes the "networking software management" connector provided by the COFTA compute subsystem connector **212***b* in the COFTA system connector **212** that has access to the FPGA device **218** that is connected to a subset of the components **224** that may interact with networking software provided by the Network Operating System (NOS) used with the networking device **200** as described below (e.g., firmware that interacts with the NOS and/or that is otherwise included in a networking software management system, a TPM, a PoE manager or controller, a cryptographic accelerator, a console buffer, etc.) in order to enable management of the networking software used with the networking device **200**.

[0056] The embodiment of the portion of the modular COFTA-based networking device system provided by the networking device **200** illustrated and described below also includes the "non-networking component management" connector provided by the COFTA compute subsystem connector **214***b* in the COFTA system connector **214** that has access to the FPGA device **218** that is connected to the fan controller **222** and the subset of the components **224** that provide "non-networking" hardware and other components (e.g., the power system, the LED indicator system, and other non-networking components described above) in the networking device **200** as described below in order to enable management of the non-networking components in the networking device **200**.

[0057] Furthermore, the coupling of the FPGA device **218** to the management port **208*b*** also allows a user to access the COFTA systems that may be connected to the COFTA compute subsystem connectors **210*b***, **212*b***, and **214*b*** discussed below via, for example, the UART connections between the FPGA device **218** and those COFTA compute subsystem connectors **210*b***, **212*b***, and **214*b***. The coupling of the internal switch device **220** the COFTA compute subsystem connectors **210*b***, **212*b***, and **214*b*** also allows respective operating systems provided by COFTA systems (which may be connected to those COFTA compute subsystem connectors **210*b***, **212*b***, and **214*b*** as discussed below) to communicate with each other, while the coupling of the internal switch device **220** to the management port **208*a*** also allows a user to access the COFTA systems that may be connected to the COFTA compute subsystem connectors **210*b***, **212*b***, and **214*b*** discussed below.

[0058] The coupling of the fan controller **222** to the FPGA device **218** and the COFTA fan subsystem connectors **210*a***, **212*a***, and **214*a***, respectively, allows for the use of the fan controller **222** (e.g., by a COFTA system connected to one of those COFTA fan subsystem connectors **210*a***, **212*a***, or **214*a*** as described below) to control fan subsystems on the COFTA systems that may be connected to the COFTA fan subsystem connectors **210*a***, **212*a***, and **214*a*** discussed below. However, while a specific networking device **200** has been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that networking devices (or other devices operating according to the teachings of the present disclosure in a manner similar to that described below for the networking device **200**) may include a variety of components and/or component configurations for providing conventional networking device functionality, as well as the modular COFTA-based networking device functionality discussed below, while remaining within the scope of the present disclosure as well.

[0059] Referring now to FIGS. **3**A and **3**B, an embodiment of a COFTA system **300** is illustrated that may provide the modular COFTA-based networking device system of the present disclosure. Embodiments of COFTA systems were described by the inventors of the present disclosure in U.S. patent application Ser. No. 18/600,978, attorney docket no. 137176.01, filed Mar. 11, 2024, the disclosure of which is incorporated by reference herein in its entirety. The COFTA system **300** includes a fan chassis **302** that supports the components of the COFTA system **300**. As will be appreciated by one of skill in the art in possession of the present disclosure, the fan chassis **302** of the COFTA system **300** may be considered a "fan tray assembly" that is configured support one or more fan modules, as well as configured to be positioned in the networking device **200** to couple those fan module(s) to the networking device **200** in order to allow those fan modules to be used to provide cooling for the networking device **200**. As such, the fan chassis **302** may include any of a variety of device coupling features that one of skill in the art in possession of the present disclosure would appreciate allow the coupling of the fan chassis **302** to the networking device **200** as described below.

[0060] In the illustrated embodiment, the fan chassis **302** is configured to support two fan modules **304** that may be provided by fan modules like those described by the inventors of the present disclosure in U.S. patent application Ser. No. 18/479,235, attorney docket no. 133643.01, filed Oct. 2, 2023, the disclosure of which is incorporated by reference herein in its entirety, as well as any other fan modules that would be apparent to one of skill in the art in possession of the present disclosure. As described in U.S. patent application Ser. No. 18/600,978, attorney docket no. 137176.01, filed Mar. 11, 2024, the fan chassis **302** may include a fan coupling subsystem that includes respective fan slots that are located adjacent a rear edge of the fan chassis **302** and that may include any of a variety of fan module coupling features that are configured to couple the fan modules **304** to the fan chassis **302** as illustrated in FIGS. **3**A and **3**B below, respective fan module connectors for each fan module **304** (illustrated in FIG. **3**B) it is configured to support, and/or any other fan module couplings that would be apparent to one of skill in the art in possession of the present disclosure. The fan chassis **302** is also provided with a fan controller connector **306** that is coupled to each of the fan modules **304** (e.g., via the fan module connectors illustrated in FIG. **3**B), and that may be configured to aggregate signals received from fan modules **304** (i.e., via the fan module connectors), split signals received for fan modules **304** (i.e., via the fan module connectors), and/or perform any other signal transmission operations that would be apparent to one of skill in the art in possession of the present disclosure.

[0061] As illustrated, a compute subsystem **308** may be mounted to the fan chassis **302**. For example, the compute subsystem **308** may include a circuit board **310** that, as can be seen in FIG. **3**B, may be mounted to the fan chassis **302** via a plurality of stand-offs **312** (i.e., with the two stand-offs **312** visible in FIG. **3**B engaging the circuit board **310** adjacent its corners along a common edge, as well as two stand-offs that are not visible in FIG. **3**B but that may engage the circuit board **310** opposite the circuit board **310** from the visible stand-offs **312** and adjacent its corners along a common opposing edge) in order to provide the circuit board **310** in a spaced-apart orientation along a Z-axis from the adjacent surface of the fan chassis **302** in order to, for example, provide additional thermal cooling for the compute subsystem **308** by allowing airflow from fan modules **304** to move past opposing surfaces (e.g., the top and bottom surfaces in FIG. **3**B) of the circuit board **310**.

[0062] As will be appreciated by one of skill in the art in possession of the present disclosure, the compute subsystem **308** may include any of a variety of compute components. For example, in the embodiment illustrated in FIG. **3**B and discussed below, the compute subsystem **308** includes a primary processing system provided by a Central Processing Unit (CPU) **308*a*** that is coupled to storage devices provided by a Serial Peripheral Interface (SPI) flash storage device **308*b*** and a Solid State Drive (SSD) storage device **308*c***, memory devices provided by Dynamic Random Access Memory (DRAM) device(s) **308*d*** and an Electronically Erasable Programmable Read-Only Memory (EEPROM) device **308*e***, a security device provided by a Trusted Platform Module (TPM) device **308*f***, and a secondary processing device provided by a Complex Programmable Logic Device (CPLD) **308*g***.

[0063] However, while specific processing devices, storage devices, memory devices, and security devices are illustrated and described herein, one of skill in the art in possession of the present disclosure will appreciate how the compute subsystem **308** may include any of a variety of processing devices, storage devices, memory devices, security devices, and/or other devices based on the compute functionality required from the compute subsystem **308** for any particular COFTA application. As described below, the compute subsystem **308** may be configured to provide either a networking processing system management operating system or a non-networking component management operating system when used in different embodiments of the modular COFTA-based networking device system of the present disclosure.

[0064] As can be seen in FIGS. **3**A and **3**B, the compute subsystem **308** includes a networking device connector **314** that, in the illustrated example, is coupled to the CPU **308***a*. As described in U.S. patent application Ser. No. 18/600,978, attorney docket no. 137176.01, filed Mar. 11, 2024, the compute subsystem **308** may be an adaptation of a Compute-On-Module (COM) Express device that includes a modified connector, and thus may be provided with a COM Express form-factor with the exception of that modified connector. For example, the networking device connector **314** may be configured similarly to a COM Express device connector, with the exception that the networking device connector **314** is oriented, and configured to connect along, an X-axis (i.e., the axis parallel to the circuit board **310** and the fan chassis **302**) rather than an Z-axis (i.e., the axis perpendicular to the circuit board **310** and the fan chassis **302**) that conventional COM Express device connectors are oriented and configured to connect along. However, while a specific compute subsystem and networking device connector are illustrated and described, one of skill in the art in possession of the present disclosure will appreciate how other compute subsystems and networking device connectors will fall within the scope of the present disclosure as well.

[0065] Referring now to FIGS. **4**A and **4**B, an embodiment of a COFTA system **400** is illustrated that may provide the modular COFTA-based networking device system of the present disclosure. As discussed above, embodiments of COFTA systems were described by the inventors of the present disclosure in U.S. patent application Ser. No. 18/600,978, attorney docket no. 137176.01, filed Mar. 11, 2024, the disclosure of which is incorporated by reference herein in its entirety. The COFTA system **400** includes a fan chassis **402** that supports the components of the COFTA system **400**. As will be appreciated by one of skill in the art in possession of the present disclosure, the fan chassis **402** of the COFTA system **400** may be considered a "fan tray assembly" that is configured support one or more fan modules, as well as configured to be positioned in the networking device **200** to couple those fan module(s) to the networking device **200** in order to allow those fan modules to be used to provide cooling for the networking device **200**. As such, the fan chassis **302** may include any of a variety of device coupling features that one of skill in the art in possession of the present disclosure would appreciate allow the coupling of the fan chassis **302** to the networking device **200** as described below.

[0066] In the illustrated embodiment, the fan chassis **402** is configured to support three fan modules **404** that may be provided by fan modules like those described by the inventors of the present disclosure in U.S. patent application Ser. No. 18/479,235, attorney docket no. 133643.01, filed Oct. 2, 2023, the disclosure of which is incorporated by reference herein in its entirety, as well as any other fan modules that would be apparent to one of skill in the art in possession of the present disclosure. As described in U.S. patent application Ser. No. 18/600,978, attorney docket no. 137176.01, filed Mar. 11, 2024, the fan chassis **402** may include a fan coupling subsystem that includes respective fan slots that are located adjacent a rear edge of the fan chassis **402** and that may include any of a variety of fan module coupling features that are configured to couple the fan modules **404** to the fan chassis **402** as illustrated in FIGS. **4**A and **4**B below, respective fan module connectors for each fan module **404** (illustrated in FIG. **4**B) it is configured to support, and/or any other fan module couplings that would be apparent to one of skill in the art in possession of the present disclosure. The fan chassis **402** is also provided with a fan controller connector **406** that is coupled to each of the fan modules **404** (e.g., via the fan module connectors illustrated in FIG. **4**B), and that may be configured to aggregate signals received from fan modules **404** (i.e., via the fan module connectors), split signals received for fan modules **404** (i.e., via the fan module connectors), and/or perform any other signal transmission operations that would be apparent to one of skill in the art in possession of the present disclosure.

[0067] As illustrated, a compute subsystem **408** may be mounted to the fan chassis **402**. For example, the compute subsystem **408** may include a circuit board **410** that, as can be seen in FIG. **4**B, may be mounted to the fan chassis **402** via a plurality of stand-offs **412** (i.e., with the two stand-offs **412** visible in FIG. **4**B engaging the circuit board **410** adjacent its corners along a common edge, as well as two stand-offs that are not visible in FIG. **4**B but that may engage the circuit board **410** opposite the circuit board **410** from the visible stand-offs **412** and adjacent its corners along a common opposing edge) in order to provide the circuit board **410** in a spaced-apart orientation along a Z-axis from the adjacent surface of the fan chassis **402** in order to, for example, provide additional thermal cooling for the compute subsystem **408** by allowing airflow from fan modules **404** to move past opposing surfaces (e.g., the top and bottom surfaces in FIG. **4**B) of the circuit board **410**.

[0068] As will be appreciated by one of skill in the art in possession of the present disclosure, the compute subsystem **408** may include any of a variety of compute components. For example, in the embodiment illustrated in FIG. **4**B and discussed below, the compute subsystem **408** includes a primary processing system provided by a Central Processing Unit (CPU) **408***a* that is coupled to storage devices provided by a Serial Peripheral Interface (SPI) flash storage device **408***b* and a Solid State Drive (SSD) storage device **408***c*, memory devices provided by Dynamic Random Access Memory (DRAM) device(s) **408***d* and an Electronically Erasable Programmable Read-Only Memory (EEPROM) device **408***e*, a security device provided by a Trusted Platform Module (TPM) device **408***f*, and a secondary processing device provided by a Complex Programmable Logic Device (CPLD) **408***g*.

[0069] However, while specific processing devices, storage devices, memory devices, and security devices are illustrated and described herein, one of skill in the art in possession of the present disclosure will appreciate how the compute subsystem **408** may include any of a variety of processing devices, storage devices, memory devices, security devices, and/or other devices based on the compute functionality required from the compute subsystem **408** for any particular COFTA application. As described below, the compute subsystem **408** may be configured to provide a networking software management operating system when used in an embodiment of the modular COFTA-based networking device system of the present disclosure.

[0070] As can be seen in FIGS. **4**A and **4**B, the compute subsystem **408** includes a networking device connector **414** that, in the illustrated example, is coupled to the CPU **408***a*. As described in U.S. patent application Ser. No. 18/600,978, attorney docket no. 137176.01, filed Mar. 11, 2024, the compute subsystem **408** may be an adaptation of a Compute-On-Module (COM) Express device that includes a modified connector, and thus may be provided with a COM Express form-factor with the exception of that modified connector. For example, the networking device connector **414** may be configured similarly to a COM Express device connector, with the exception that the networking device connector **414** is oriented, and configured to connect along, an X-axis (i.e., the axis parallel to the circuit board **410** and the fan chassis **402**) rather than an Z-axis (i.e., the axis perpendicular to the circuit board **410** and the fan chassis **402**) that conventional COM Express device connectors are oriented and configured to connect along. However, while a specific compute subsystem and networking device connector are illustrated and described, one of skill in the art in possession of the present disclosure will appreciate how other compute subsystems and networking device connectors will fall within the scope of the present disclosure as well.

[0071] Referring now to FIGS. **5**A and **5**B, an embodiment of a COFTA system **500** is illustrated that may provide the modular COFTA-based networking device system of the present disclosure. As discussed above, embodiments of COFTA systems were described by the inventors of the present disclosure in U.S. patent application Ser. No. 18/600,978, attorney docket no. 137176.01, filed Mar. 11, 2024, the disclosure of which is incorporated by reference herein in its entirety. The COFTA system **500** includes a fan chassis **502** that supports the components of the COFTA system **500**. As will be appreciated by one of skill in the art in possession of the present disclosure, the fan chassis **402** of the COFTA system **4500** may be considered a "fan tray assembly" that is configured support one or more fan modules, as well as configured to be positioned in the networking device **200** to couple those fan module(s) to the networking device **200** in order to allow those fan modules to be used to provide cooling for the networking device **200**. As such, the fan chassis **502** may include any of a variety of device coupling features that one of skill in the art in possession of the present disclosure would appreciate allow the coupling of the fan chassis **502** to the networking device **200** as described below.

[0072] In the illustrated embodiment, the fan chassis **502** is configured to support five fan modules **504** that may be provided by fan modules like those described by the inventors of the present disclosure in U.S. patent application Ser. No. 18/479,235, attorney docket no. 133643.01, filed Oct. 2, 2023, the disclosure of which is incorporated by reference herein in its entirety, as well as any other fan modules that would be apparent to one of skill in the art in possession of the present disclosure. As described in U.S. patent application Ser. No. 18/600,978, attorney docket no. 137176.01, filed Mar. 11, 2024, the fan chassis **502** may include a fan coupling subsystem that includes respective fan slots that are located adjacent a rear edge of the fan chassis **502** and that may include any of a variety of fan module coupling features that are configured to couple the fan modules **504** to the fan chassis **502** as illustrated in FIGS. **5**A and **5**B below, respective fan module connectors for each fan module **504** (illustrated in FIG. **5**B) it is configured to support, and/or any other fan module couplings that would be apparent to one of skill in the art in possession of the present disclosure. The fan chassis **502** is also provided with fan controller connectors **506***a* and **506***b* that are each coupled to a subset of the fan modules **504** (e.g., with the fan controller connector **506***a* coupled to two of the fan modules **504** and the fan controller connector **506***b* coupled to three of the fan modules **504** via the fan module connectors illustrated in FIG. **5**B), and that may each be configured to aggregate signals received from fan modules **504** (i.e., via the fan module connectors), split signals received for fan modules **504** (i.e., via the fan module connectors), and/or perform any other signal transmission operations that would be apparent to one of skill in the art in possession of the present disclosure.

[0073] As illustrated, a compute subsystem **508** may be mounted to the fan chassis **502**. For example, the compute subsystem **508** may include a circuit board **510** that, as can be seen in FIG. **5**B, may be mounted to the fan chassis **502** via a plurality of stand-offs **512** (i.e., with the two stand-offs **512** visible in FIG. **5**B engaging the circuit board **510** adjacent its corners along a common edge, as well as two stand-offs that are not visible in FIG. **5**B but that may engage the circuit board **510** opposite the circuit board **510** from the visible stand-offs **512** and adjacent its corners along a common opposing edge) in order to provide the circuit board **510** in a spaced-apart orientation along a Z-axis from the adjacent surface of the fan chassis **502** in order to, for example, provide additional thermal cooling for the compute subsystem **508** by allowing airflow from fan modules **504** to move past opposing surfaces (e.g., the top and bottom surfaces in FIG. **5**B) of the circuit board **510**.

[0074] As will be appreciated by one of skill in the art in possession of the present disclosure, the compute subsystem **508** may include any of a variety of compute components. For example, in the embodiment illustrated in FIG. **5**B and discussed below, the compute subsystem **508** includes a primary processing system provided by a Central Processing Unit (CPU) **508***a* that is coupled to storage devices provided by a Serial Peripheral Interface (SPI) flash storage device **508***b* and a Solid State Drive (SSD) storage device **508***c*, memory devices provided by Dynamic Random Access Memory (DRAM) device(s) **508***d* and an Electronically Erasable Programmable Read-Only Memory (EEPROM) device **508***e*, a security device provided by a Trusted Platform Module (TPM) device **508***f*, and a secondary processing device provided by a Complex Programmable Logic Device (CPLD) **508***g*.

[0075] However, while specific processing devices, storage devices, memory devices, and security devices are illustrated and described herein, one of skill in the art in possession of the present disclosure will appreciate how the compute subsystem **508** may include any of a variety of processing devices, storage devices, memory devices, security devices, and/or other devices based on the compute functionality required from the compute subsystem **508** for any particular COFTA application. As described below, the compute subsystem **508** may be configured to provide either a networking processing system/networking software management operating system or a networking software/non-networking component management operating system when used in different embodiments of the modular COFTA-based networking device system of the present disclosure.

[0076] As can be seen in FIGS. **5**A and **5**B, the compute subsystem **508** includes networking device connectors **514***a* and **514***b* that, in the illustrated example, are coupled to the CPU **508***a*. As described in U.S. patent application Ser. No. 18/600,978, attorney docket no. 137176.01, filed Mar. 11, 2024, the compute subsystem **508** may be an adaptation of a Compute-On-Module (COM) Express device that includes a modified connector, and thus may be provided with a COM Express form-factor with the exception of that modified connector. For example, each of the networking device connectors **514***a* and **514***b* may be configured similarly to a COM Express device connector, with the exception that the networking device connectors **514***a* and **514***b* are oriented, and configured to connect along, an X-axis (i.e., the axis parallel to the circuit board **510** and the fan chassis **502**) rather than an Z-axis (i.e., the axis perpendicular to the circuit board **510** and the fan chassis **502**) that conventional COM Express device connectors are oriented and configured to connect along. However, while a specific compute subsystem and networking device connector are illustrated and described, one of skill in the art in possession of the present disclosure will appreciate how other compute subsystems and networking device connectors will fall within the scope of the present disclosure as well.

[0077] Referring now to FIGS. **6**A and **6**B, an embodiment of a COFTA system **600** is illustrated that may be used with the networking device **200**. As discussed above, embodiments of COFTA systems were described by the inventors of the present disclosure in U.S. patent application Ser. No. 18/600,978, attorney docket no. 137176.01, filed Mar. 11, 2024, the disclosure of which is incorporated by reference herein in its entirety. The COFTA system **600** includes a fan chassis **602** that supports the components of the COFTA system **600**. As will be appreciated by one of skill in the art in possession of the present disclosure, the fan chassis **602** of the COFTA system **4500** may be considered a "fan tray assembly" that is configured support one or more fan modules, as well as configured to be positioned in the networking device **200** to couple those fan module(s) to the networking device **200** in order to allow those fan modules to be used to provide cooling for the networking device **200**. As such, the fan chassis **602** may include any of a variety of device coupling features that one of skill in the art in possession of the present disclosure would appreciate allow the coupling of the fan chassis **602** to the networking device **200** as described below.

[0078] In the illustrated embodiment, the fan chassis **602** is configured to support seven fan modules **604** that may be provided by fan modules like those described by the inventors of the present disclosure in U.S. patent application Ser. No. 18/479,235, attorney docket no. 133643.01, filed Oct. 2, 2023, the disclosure of which is incorporated by reference herein in its entirety, as well as any other fan modules that would be apparent to one of skill in the art in possession of the present disclosure. As described in U.S. patent application Ser. No. 18/600,978, attorney docket no. 137176.01, filed Mar. 11, 2024, the fan chassis **602** may include a fan coupling subsystem that includes respective fan slots that are located adjacent a rear edge of the fan chassis **602** and that may include any of a variety of fan module coupling features that are configured to couple the fan modules **604** to the fan chassis **602** as illustrated in FIGS. **6**A and **6**B below, respective fan module connectors for each fan module **604** (illustrated in FIG. **6**B) it is configured to support, and/or any other fan module couplings that would be apparent to one of skill in the art in possession of the present disclosure. The fan chassis **602** is also provided with fan controller connectors **606***a*, **606***b*, and **606***c* that are each coupled to a subset of the fan modules **604** (e.g., with the fan controller connector **606***a* coupled to two of the fan modules **604**, the fan controller connector **606***b* coupled to three of the fan modules **604**, and the fan controller connector **606***c* coupled to two of the fan modules **604** via the fan module connectors illustrated in FIG. **6**B), and that may each be configured to aggregate signals received from fan modules **604** (i.e., via the fan module connectors), split signals received for fan modules **604** (i.e., via the fan module connectors), and/or perform any other signal transmission operations that would be apparent to one of skill in the art in possession of the present disclosure.

[0079] As illustrated, a compute subsystem **608** may be mounted to the fan chassis **602**. For example, the compute subsystem **608** may include a circuit board **610** that, as can be seen in FIG. **6**B, may be mounted to the fan chassis **602** via a plurality of stand-offs **612** (i.e., with the two stand-offs **612** visible in FIG. **6**B engaging the circuit board **610** adjacent its corners along a common edge, as well as two stand-offs that are not visible in FIG. **6**B but that may engage the circuit board **610** opposite the circuit board **610** from the visible stand-offs **612** and adjacent its corners along a common opposing edge) in order to provide the circuit board **610** in a spaced-apart orientation along a Z-axis from the adjacent surface of the fan chassis **602** in order to, for example, provide additional thermal cooling for the compute subsystem **608** by allowing airflow from fan modules **604** to move past opposing surfaces (e.g., the top and bottom surfaces in FIG. **6**B) of the circuit board **610**.

[0080] As will be appreciated by one of skill in the art in possession of the present disclosure, the compute subsystem **608** may include any of a variety of compute components. For example, in the embodiment illustrated in FIG. **6**B and discussed below, the compute subsystem **608** includes a primary processing system provided by a Central Processing Unit (CPU) **608***a* that is coupled to storage devices provided by a Serial Peripheral Interface (SPI) flash storage device **608***b* and a Solid State Drive (SSD) storage device **608***c*, memory devices provided by Dynamic Random Access Memory (DRAM) device(s) **608***d* and an Electronically Erasable Programmable Read-Only Memory (EEPROM) device **608***e*, a security device provided by a Trusted Platform Module (TPM) device **608***f*, and a secondary processing device provided by a Complex Programmable Logic Device (CPLD) **608***g*.

[0081] However, while specific processing devices, storage devices, memory devices, and security devices are illustrated and described herein, one of skill in the art in possession of the present disclosure will appreciate how the compute subsystem **608** may include any of a variety of processing devices, storage devices, memory devices, security devices, and/or other devices based on the compute functionality required from the compute subsystem **508** for any particular COFTA application. As described below, the compute subsystem **608** may be configured to provide the networking processing system/networking software/non-networking component management operating system when used in an embodiment of the modular COFTA-based networking device system of the present disclosure.

[0082] As can be seen in FIGS. **6**A and **6**B, the compute subsystem **608** includes networking device connectors **614**a, **614**b, and **614**c that, in the illustrated example, are coupled to the CPU **608**a. As described in U.S. patent application Ser. No. 18/600,978, attorney docket no. 137176.01, filed Mar. 11, 2024, the compute subsystem **608** may be an adaptation of a Compute-On-Module (COM) Express device that includes a modified connector, and thus may be provided with a COM Express form-factor with the exception of that modified connector. For example, each of the networking device connectors **614**a, **614**b, and **614**c may be configured similarly to a COM Express device connector, with the exception that the networking device connectors **614**a. **614**b. and **614**c are oriented, and configured to connect along, an X-axis (i.e., the axis parallel to the circuit board **610** and the fan chassis **602**) rather than an Z-axis (i.e., the axis perpendicular to the circuit board **610** and the fan chassis **602**) that conventional COM Express device connectors are oriented and configured to connect along. However, while a specific compute subsystem and networking device connector are illustrated and described, one of skill in the art in possession of the present disclosure will appreciate how other compute subsystems and networking device connectors will fall within the scope of the present disclosure as well.

[0083] As described below, the COFTA systems **300**, **400**, and **500** discussed above may be utilized with the networking device **200** in different combinations to configure the modular COFTA-based networking device of the present disclosure with different management operating systems, while the COFTA system **600** may be utilized with the networking device **200** when a single management operating system is desired. However, while specific COFTA systems are illustrated and described herein as being used with a specific networking device, one of skill in the art in possession of the present disclosure will appreciate how the teachings of the present disclosure may be applied to other networking device configurations using other COFTA system configurations while remaining within the scope of the present disclosure as well.

[0084] Referring now to FIG. **7**, an embodiment of a method **700** for providing a modular Computer-On-Fan-Tray-Assembly (COFTA)-based networking device is illustrated. As discussed below, the systems and methods of the present disclosure utilize COFTA systems as modular components in a networking device that are each configured to provide a respective operating system to manage different subsets of a networking processing system in the networking device, non-networking components in the networking device, and networking software used by the networking device. For example, the modular COFTA networking device system of the present disclosure may include a networking device chassis housing a circuit board coupled to a plurality of non-networking components and including: a networking processing system coupled to networking ports that are accessible on the networking device chassis, a networking processing system management connector coupled to the networking processing system, a non-networking component management connector coupled to the non-networking components, and a networking software management connector. At least two COFTA systems are each connected to a different subset of the networking processing system management connector, the non-networking component management connector, and the networking software management connector. Each COFTA system provides a respective operating system that manages a different subset of the networking processing system via the networking processing system management connector, the plurality of non-networking components via the non-networking component management connector, and networking software via the networking software management connector. As such, different networking device management functionality for the networking device may be performed by dedicated operating systems, reducing or eliminating the issues with managing conventional networking devices discussed above.

[0085] The method **700** begins at block **702** where COFTA systems are connected to different subsets of COFTA system connectors in a networking device management connector system on a networking device. In a first embodiment of the method **700**, at block **702**, a first of the COFTA system **300** discussed above with reference to FIGS. **3**A and **3**B provides a networking processing system management COFTA system, the COFTA system **400** discussed above with reference to FIGS. **4**A and **4**B provides a networking software management COFTA system, and a second of the COFTA system **300** discussed above with reference to FIGS. **3**A and **3**B provides a non-networking component management COFTA system, each of which are coupled to the networking device management connector system on the circuit board **204** of the networking device **200**. While not illustrated or described in detail, one of skill in the art in possession of the present disclosure will appreciate how the COFTA systems **300** and **400** and the networking device management connector system on the circuit board **204** of the networking device **200** may be keyed to only allow the first COFTA system **300** to be connected to the COFTA system connector **210**, to only allow the COFTA system **400** to be connected to the COFTA system connector **212**, and to only all the second COFTA system **300** to be connected to the COFTA system connector **214**.

[0086] As illustrated in FIGS. **8**A and **8**B, the first COFTA system **300** may be positioned adjacent the COFTA system connector **210** on the circuit board **204** of the networking device **200** (e.g., by positioned the first COFTA system **300** in a COFTA system housing defined by the networking device chassis **200** adjacent the COFTA system connector **210**), and the first COFTA system **300** may then be moved towards the circuit board **204** such that the fan controller connector **306** on the first COFTA system **300** engages the COFTA fan subsystem connector **210**a on the COFTA system connector **210**, and the networking device connector **314** on the first COFTA system **300** engages the COFTA compute subsystem connector **210**b on the COFTA system connector **210**.

[0087] As also illustrated in FIGS. **8**A and **8**B, the COFTA system **400** may be positioned adjacent the COFTA system connector **212** on the circuit board **204** of the networking device **200** (e.g., by positioned the first COFTA system **400** in a COFTA system housing defined by the networking device chassis **200** adjacent the COFTA system connector **212**), and the COFTA system **400** may then be moved towards the circuit board **204** such that the fan controller connector **406** on the COFTA system **400** engages the

COFTA fan subsystem connector **212**_a_ on the COFTA system connector **212**, and the networking device connector **414** on the COFTA system **400** engages the COFTA compute subsystem connector **212**_b_ on the COFTA system connector **212**.

[0088] As also illustrated in FIGS. **8**A and **8**B, the second COFTA system **300** may be positioned adjacent the COFTA system connector **214** on the circuit board **204** of the networking device **200** (e.g., by positioned the second COFTA system **300** in a COFTA system housing defined by the networking device chassis **200** adjacent the COFTA system connector **214**), and the second COFTA system **300** may then be moved towards the circuit board **204** such that the fan controller connector **306** on the second COFTA system **300** engages the COFTA fan subsystem connector **214**_a_ on the COFTA system connector **214**, and the networking device connector **314** on the second COFTA system **300** engages the COFTA compute subsystem connector **214**_b_ on the COFTA system connector **214**.

[0089] In a second embodiment of the method **700**, at block **702** the COFTA system **500** discussed above with reference to FIGS. **5**A and **5**B provides a networking processing system/networking software management COFTA system, and the COFTA system **300** discussed above with reference to FIGS. **3**A and **3**B provides a non-networking component management COFTA system, each of which are coupled to the networking device management connector system on the circuit board **204** of the networking device **200**. While not illustrated or described in detail, one of skill in the art in possession of the present disclosure will appreciate how the COFTA systems **500** and **300** and the networking device management connector system on the circuit board **204** of the networking device **200** may be keyed to only allow the COFTA system **500** to be connected to the COFTA system connectors **210** and **212**, and to only allow the COFTA system **300** to be connected to the COFTA system connector **214**.

[0090] As illustrated in FIGS. **9**A and **9**B, the COFTA system **500** may be positioned adjacent the COFTA system connectors **210** and **212** on the circuit board **204** of the networking device **200** (e.g., by positioned the COFTA system **500** in a COFTA system housing defined by the networking device chassis **200** adjacent the COFTA system connectors **210** and **212**), and the COFTA system **500** may then be moved towards the circuit board **204** such that the fan controller connectors **506**_a_ and **506**_b_ on the COFTA system **500** engage the COFTA fan subsystem connectors **210**_a_ and **212**_a_, respectively, on the COFTA system connectors **210** and **212**, respectively, and the networking device connectors **514**_a_ and **514**_b_ on the COFTA system **500** engage the COFTA compute subsystem connectors **210**_b_ and **212**_b_, respectively, on the COFTA system connectors **210** and **212**, respectively.

[0091] As also illustrated in FIGS. **9**A and **9**B, the COFTA system **300** may be positioned adjacent the COFTA system connector **214** on the circuit board **204** of the networking device **200** (e.g., by positioning the COFTA system **300** in a COFTA system housing defined by the networking device chassis **200** adjacent the COFTA system connector **214**), and the COFTA system **300** may then be moved towards the circuit board **204** such that the fan controller connector **306** on the COFTA system **300** engages the COFTA fan subsystem connector **214**_a_ on the COFTA system connector **214**, and the networking device connector **314** on the COFTA system **300** engages the COFTA compute subsystem connector **214**_b_ on the COFTA system connector **214**.

[0092] In a third embodiment of the method **700**, at block **702** the COFTA system **300** discussed above with reference to FIGS. **3**A and **3**B provides a networking processing system management COFTA system, and the COFTA system **500** discussed above with reference to FIGS. **5**A and **5**B provides a networking software/non-networking component management COFTA system, each of which are coupled to the networking device management connector system on the circuit board **204** of the networking device **200**. While not illustrated or described in detail, one of skill in the art in possession of the present disclosure will appreciate how the COFTA systems **300** and **500** and the networking device management connector system on the circuit board **204** of the networking device **200** may be keyed to only allow the COFTA system **300** to be connected to the COFTA system connector **210**, and to only allow the COFTA system **500** to be connected to the COFTA system connectors **212** and **214**.

[0093] As illustrated in FIGS. **10**A and **10**B, the COFTA system **300** may be positioned adjacent the COFTA system connector **210** on the circuit board **204** of the networking device **200** (e.g., by positioning the COFTA system **300** in a COFTA system housing defined by the networking device chassis **200** adjacent the COFTA system connector **210**), and the COFTA system **300** may then be moved towards the circuit board **204** such that the fan controller connector **306** on the COFTA system **300** engages the COFTA fan subsystem connector **210**_a_ on the COFTA system connector **210**, and the networking device connector **314** on the COFTA system **300** engages the COFTA compute subsystem connector **210**_b_ on the COFTA system connector **210**.

[0094] As also illustrated in FIGS. **10**A and **10**B, the COFTA system **500** may be positioned adjacent the COFTA system connectors **212** and **214** on the circuit board **204** of the networking device **200** (e.g., by positioned the COFTA system **500** in a COFTA system housing defined by the networking device chassis **200** adjacent the COFTA system connectors **212** and **214**), and the COFTA system **500** may then be moved towards the circuit board **204** such that the fan controller connectors **506**_a_ and **506**_b_ on the COFTA system **500** engage the COFTA fan subsystem connectors **212**_a_ and **214**_a_, respectively, on the COFTA system connectors **212** and **214**, respectively, and the networking device connectors **514**_a_ and **514**_b_ on the COFTA system **500** engage the COFTA compute subsystem connectors **212**_b_ and **214**_b_, respectively, on the COFTA system connectors **212** and **214**, respectively.

[0095] The method **700** then proceeds to block **704** where the COFTA systems provide respective operating systems. Continuing with the first embodiment of the method **700** introduced with reference to FIGS. **8**A and **8**B above, at block **704** the first of the COFTA system **300** discussed above with reference to FIGS. **3**A and **3**B that provides the networking processing system management COFTA system in this embodiment may operate to provide a networking processing system management operating system, the COFTA system **400** discussed above with reference to FIGS. **4**A and **4**B that provides the networking software management COFTA system in this embodiment may operate to provide a networking software management operating system, and the second of the COFTA system **300** discussed above with reference to FIGS. **3**A and **3**B that provides the non-networking component management COFTA system in this embodiment may operate to provide a non-networking component management operating system.

[0096] For example, the compute subsystem **308** in the first COFTA system **300** that is connected to the COFTA system connector **210** may have previously been configured to provide the networking processing system management operating system, and one of skill in the art in possession of the present disclosure will appreciate how that compute subsystem **308** may power on, boot, reset, reboot, and or otherwise initialize in order to enter a runtime state in which it provides the networking processing system management operating system. Similarly, the compute subsystem **408** in the COFTA system **400** that is connected to the COFTA system connector **212** may have previously been configured to provide the networking software management operating system, and one of skill in the art in possession of the present disclosure will appreciate how that compute subsystem **408** may power on, boot, reset, reboot, and or otherwise initialize in order to enter a runtime state in which it provides the networking software management operating system. Similarly as well, the compute subsystem **308** in the second COFTA system **300** that is connected to the COFTA system connector **214** may have previously been configured to provide the non-networking component management operating system, and one of skill in the art in possession of the present disclosure will appreciate how that compute subsystem **308** may power on, boot, reset, reboot, and or otherwise initialize in order to enter a runtime state in which it provides the non-networking component management operating system.

[0097] Continuing with the second embodiment of the method **700** introduced with reference to FIGS. **9**A and **9**B above, at block **704** the COFTA system **500** discussed above with reference to FIGS. **5**A and **5**B that provides the networking processing system/networking software management COFTA system in this embodiment may operate to provide a networking processing system/networking software management operating system, and the COFTA system **300** discussed above with reference to FIGS. **3**A and **3**B that provides the non-networking component management COFTA system in this embodiment may operate to provide a non-networking component management operating system.

[0098] For example, the compute subsystem **508** in the COFTA system **500** that is connected to the COFTA system connectors **210** and **212** may have previously been configured to provide the networking processing system/networking software management operating system, and one of skill in the art in possession of the present disclosure will appreciate how that compute subsystem **508** may power on, boot, reset, reboot, and or otherwise initialize in order to enter a runtime state in which it provides the networking processing system/networking software management operating system. Similarly, the compute subsystem **308** in the COFTA system **300** that is connected to the COFTA system connector **214** may have previously been configured to provide the non-networking component management operating system, and one of skill in the art in possession of the present disclosure will appreciate how that compute subsystem **308** may power on, boot, reset, reboot, and or otherwise initialize in order to enter a runtime state in which it provides the non-networking component management operating system.

[0099] Continuing with the third embodiment of the method **700** introduced with reference to FIGS. **10**A and **10**B above, at block **704** the COFTA system **300** discussed above with reference to FIGS. **3**A and **3**B that provides the networking processing system management COFTA system in this embodiment may operate to provide a networking processing system management operating system, and the COFTA system **500** discussed above with reference to FIGS. **5**A and **5**B that provides the networking software/non-networking component management COFTA system in this embodiment may operate to provide a networking software/non-networking component management operating system.

[0100] For example, the compute subsystem **308** in the COFTA system **300** that is connected to the COFTA system connector **210** may have previously been configured to provide the networking processing system management operating system, and one of skill in the art in possession of the present disclosure will appreciate how that compute subsystem **308** may power on, boot, reset, reboot, and or otherwise initialize in order to enter a runtime state in which it provides the networking processing system management operating system. Similarly, the compute subsystem **508** in the COFTA system **500** that is connected to the COFTA system connectors **212** and **214** may have previously been configured to provide the networking software/non-networking component management operating system, and one of skill in the art in possession of the present disclosure will appreciate how that compute subsystem **508** may power on, boot, reset, reboot, and or otherwise initialize in order to enter a runtime state in which it provides the networking software/non-networking component management operating system.

[0101] The method **700** then proceeds to block **706** where the respective operating systems provided by the COFTA systems manage different subsets of a networking processing system in the networking device, non-networking components in the networking device, and networking software used by the networking device. Continuing with the first embodiment of the method **700** discussed above with reference to FIGS. **8**A and **8**B, at block **706**, the networking processing system management operating system provided by the networking processing system management COFTA system (i.e., the first of the COFTA system **300** discussed above with reference to FIGS. **3**A and **3**B that is connected to the COFTA system connector **210**) may operate to manage the networking processing system (e.g., the NPU device **216** and related component) in the networking device **200**, the networking software management operating system provided by the networking software management COFTA system (i.e., the COFTA system **400** discussed above with reference to FIGS. **4**A and **4**B that is connected to the COFTA system connector **212**) may operate to manage the networking software used by the networking device **200** (as well as a subset of the components **224** that interact with that networking software such as, for example, firmware that interacts with the NOS and/or that is otherwise included in a networking software management system, a TPM, a PoE manager or controller, a cryptographic accelerator, a console buffer, etc.)), and the non-networking component management operating system provided by the non-networking component management COFTA system (i.e., the second of the COFTA system **300** discussed above with reference to FIGS. **3**A and **3**B that is connected to the COFTA system connector **214**) may operate to manage the fan controller

222, a subset of the components **224**, and/or other non-networking components in the networking device **200**.

[0102] For example, with reference to FIG. **8**C, the networking processing system management operating system provided by the compute subsystem **308** in the first COFTA system **300** discussed above may perform networking processing system management operations **800** that include managing the NPU device **216** (and related components) via the networking device connector **314** and the COFTA compute subsystem connector **210***b*. As will be appreciated by one of skill in the art in possession of the present disclosure, the networking processing system management operations **800** may include any of a variety of management operations related to the NPU device **216** (or switch ASIC), related networking hardware, forwarding tables, telemetry hardware (e.g., NPU device/silicon telemetry hardware), NPU-specific drivers, a Software Development Kit (SDK) accessed via a Switch Abstraction Interface (SAI), firmware and/or other functionality that is relatively tightly coupled to the NPU device **216** (e.g., Precision Time Protocol (PTP) functionality, Synchronous Ethernet (SyncE) functionality, etc.), and/or other networking processing system components that would be apparent to one of skill in the art in possession of the present disclosure.

[0103] In another example, with reference to FIG. **8**D, the networking software management operating system provided by the compute subsystem **408** in the COFTA system **400** discussed above may perform networking software management operations **802** that include managing the networking software used by the networking device **200**, which in some examples may include managing the subset of the components **224** that interact with that networking software (e.g., firmware that interacts with the NOS and/or that is otherwise included in a networking software management system, a TPM, a PoE manager or controller, a cryptographic accelerator, a console buffer, etc.) via the networking device connector **414**, the COFTA compute subsystem connector **212***b*, and the FPGA device **218**. As will be appreciated by one of skill in the art in possession of the present disclosure, the networking software management operations **802** may include any of a variety of management operations related to a Network Operating System (NOS) provided by the compute subsystem **408** for use by the networking device **200**, a management framework provided for a user of the networking device **200** (e.g., a Command Line Interface (CLI), programmatic NorthBound Interface (NBI), REpresentational State Transfer (REST), Google Remote Procedure Call (gRPC) Network Management Interface (GNMI), gRPC Network Operations Interface (GNOI), etc.), Layer 2 (L2)/Layer 3 (L3) protocols, operating system images, Zero-Touch Provisioning (ZTP), Authentication Authorization and Accounting (AAA), Network Time Protocol (NTP), Domain Name Service (DNS), Role-Based Access Control (RBAC), Simple Network Management Protocol (SNMP), Dynamic Host Configuration Protocol (DHCP), and/or other networking software components that would be apparent to one of skill in the art in possession of the present disclosure.

[0104] In another example, with reference to FIG. **8**E, the non-networking component management operating system provided by the compute subsystem **308** in the second COFTA system **300** discussed above may perform non-networking component management operations **804** that include managing the fan controller **222**, the subset of the components **224**, and/or other non-networking components in the networking device **200** via the networking device connector **314**, the COFTA compute subsystem connector **214***b*, and the FPGA device **218**. As will be appreciated by one of skill in the art in possession of the present disclosure, the non-networking component management operations **804** may include any of a variety of management operations related to power monitoring, control/reset mechanisms, sensor monitoring (e.g., thermal sensor monitoring, voltage sensor monitoring, etc.), power supply system status monitoring and control, fan status monitoring and control, LED control, firmware updates, and/or other non-networking components that would be apparent to one of skill in the art in possession of the present disclosure.

[0105] As will be appreciated by one of skill in the art in possession of the present disclosure, the non-networking component management operations **804** described above are often conventionally performed by a Baseboard Management Controller (BMC) device, Platform Management Controller (PMC) device, and/or other management controller known in the art, and thus the second COFTA system **300** may operate as a dedicated management controller (with a dedicated management operating system) in the embodiment illustrated in FIGS. **8**A-**8**E. Furthermore, FIG. **8**E illustrates a specific example in which the non-networking component management operations **804** include using the fan controller **222** to control any of the fan modules **304** on the first COFTA system **300**, the fan modules **404** on the COFTA system **400**, and the fan modules **304** on the second COFTA system **300**.

[0106] With reference to FIG. **8**F, any of the management operations discussed above may include any of the networking processing system management operating system provided by the compute subsystem **308** in the first COFTA system **300**, the networking software management operating system provided by the compute subsystem **408** in the COFTA system **400**, and the non-networking component management operating system provided by the compute subsystem **308** in the second COFTA system **300** performing operating system communication operations **806** via the networking device connector **314**/COFTA compute subsystem connector **210***b* connection, the networking device connector **314**/COFTA compute subsystem connector **212***b* connection, and the networking device connector **314**/COFTA compute subsystem connector **214***b* connection, respectively, and using the internal switch device **220**.

[0107] While not illustrated or described in detail, as discussed above, a user may access any of the compute subsystem **308** in the first of the COFTA system **300**, the compute subsystem **408** in the COFTA system **400**, and the compute subsystem **308** in the second of the COFTA system **300** (i.e., via the networking device connector **314**/COFTA compute subsystem connector **210***b* connection, the networking device connector **314**/COFTA compute subsystem connector **212***b* connection, and the networking device connector **314**/COFTA compute subsystem connector **214***b* connection, respectively) using the management port **208***a* and the internal switch device **220**, or using the management port **208***a* and the FPGA device **218** (via its UART connections discussed above).

[0108] Continuing with the second embodiment of the method **700** discussed above with reference to FIGS. **9**A and **9**B, at block **706**, the networking processing system/networking software management operating system provided by the networking processing system/networking software management COFTA system (i.e., the COFTA system **500** discussed above with reference to FIGS. **5**A and **5**B that is connected to the COFTA system connectors **210** and **212**) may operate to manage the networking processing system (e.g., the NPU device **216** and related component) and the networking software used by the networking device **200** (as well as a subset of the components **224** that interact with that networking software (such as, for example, firmware that interacts with the NOS and/or that is otherwise included in a networking software management system, a TPM, a PoE manager or controller, a cryptographic accelerator, a console buffer, etc.)), and the non-networking component management operating system provided by the non-networking component management COFTA system (i.e., the COFTA system **300** discussed above with reference to FIGS. **3**A and **3**B that is connected to the COFTA system connector **214**) may operate to manage the fan controller, a subset of the components **224**, and/or other non-networking components in the networking device **200**.

[0109] For example, with reference to FIG. **9**C, the networking processing system/networking software management operating system provided by the compute subsystem **508** in the COFTA system **500** discussed above may perform networking processing system management operations **900** that include managing the NPU device **216** (and related components) via the networking device connector **314** and the COFTA compute subsystem connector **210***b*. Similarly as discussed above, the networking processing system management operations **900** may include any of a variety of management operations related to the NPU device **216** (or switch ASIC), related networking hardware, forwarding tables, telemetry hardware (e.g., NPU device/silicon telemetry hardware), NPU-specific drivers, a Software Development Kit (SDK) accessed via a Switch Abstraction Interface (SAI), firmware and/or other functionality that is relatively tightly coupled to the NPU device **216** (e.g., Precision Time Protocol (PTP) functionality, Synchronous Ethernet (SyncE) functionality, etc.), and/or other networking processing system components that would be apparent to one of skill in the art in possession of the present disclosure.

[0110] In another example, with reference to FIG. **9**D, the networking processing system/networking software management operating system provided by the compute subsystem **508** in the COFTA system **500** discussed above may perform networking software management operations **902** that include managing the networking software used by the networking device **200**, which in some examples may include managing the subset of the components **224** that interact with that networking software (e.g., firmware that interacts with the NOS and/or that is otherwise included in a networking software management system, a TPM, a PoE manager or controller, a cryptographic accelerator, a console buffer, etc.) via the networking device connector **414**, the COFTA compute subsystem connector **212***b*, and the FPGA device **218**. Similarly as discussed above, the networking software management operations **902** may include any of a variety of management operations related to a Network Operating System (NOS) provided by the compute subsystem **508** for use by the networking device **200**, a management framework provided for a user of the networking device **200** (e.g., a Command Line Interface (CLI), programmatic NorthBound Interface (NBI), REpresentational State Transfer (REST), Google Remote Procedure Call (gRPC) Network Management Interface (GNMI), gRPC Network Operations Interface (GNOI), etc.), Layer 2 (L2)/Layer 3 (L3) protocols, operating system images, Zero-Touch Provisioning (ZTP), Authentication Authorization and Accounting (AAA), Network Time Protocol (NTP), Domain Name Service (DNS), Role-Based Access Control (RBAC), Simple Network Management Protocol (SNMP), Dynamic Host Configuration Protocol (DHCP), and/or other networking software components that would be apparent to one of skill in the art in possession of the present disclosure.

[0111] In another example, with reference to FIG. **9**E, the non-networking component management operating system provided by the compute subsystem **308** in the COFTA system **300** discussed above may perform non-networking component management operations **904** that include managing the fan controller **222**, the subset of the components **224**, and/or other non-networking components in the networking device **200** via the networking device connector **314**, the COFTA compute subsystem connector **214***b*, and the FPGA device **218**. As will be appreciated by one of skill in the art in possession of the present disclosure, the non-networking component management operations **904** may include any of a variety of management operations related to power monitoring, control/reset mechanisms, sensor monitoring (e.g., thermal sensor monitoring, voltage sensor monitoring, etc.), power supply system status monitoring and control, fan status monitoring and control, LED control, firmware updates, and/or other non-networking components that would be apparent to one of skill in the art in possession of the present disclosure.

[0112] As will be appreciated by one of skill in the art in possession of the present disclosure, the non-networking component management operations **904** described above are often conventionally performed by a Baseboard Management Controller (BMC) device, Platform Management Controller (PMC) device, and/or other management controller known in the art, and thus the COFTA system **300** may operate as a dedicated management controller (with a dedicated management operating system) in the embodiment illustrated in FIGS. **9**A-**9**E. Furthermore, FIG. **9**E illustrates a specific example in which the non-networking component management operations **904** include using the fan controller **222** to control any of the fan modules **504** on the COFTA system **500**, and the fan modules **304** on the COFTA system **300**.

[0113] With reference to FIG. **9**F, any of the management operations discussed above may include the networking processing system/networking software management operating system provided by the compute subsystem **508** in the COFTA system **500** and the non-networking component management operating system provided by the compute subsystem **308** in the COFTA system **300** performing operating system communication operations **906** via the networking device connector **314**/COFTA compute subsystem connector **210***b* connection and the networking device connector **314**/COFTA compute subsystem connector **214***b* connection, respectively, and using the internal switch device **220**.

[0114] While not illustrated or described in detail, as discussed above, a user may access either of the compute subsystem **508** in the COFTA system **500** and the compute subsystem **308** in the COFTA system **300** (i.e., via the networking device connector **314**/COFTA compute subsystem connector **210***b* connection or the networking device connector **314**/COFTA compute subsystem connector **212***b* connection, and the networking device connector **314**/COFTA compute subsystem connector **214***b* connection, respectively) using the management port **208***a* and the internal switch device **220**, or using the management port **208***a* and the FPGA device **218** (via its UART connections discussed above).

[0115] Continuing with the third embodiment of the method **700** discussed above with reference to FIGS. **10**A and **10**B, at block **706**, the networking processing system management operating system provided by the networking processing system management COFTA system (i.e., the COFTA system **300** discussed above with reference to FIGS. **3**A and **3**B that is connected to the COFTA system connector **210**) may operate to manage the networking processing system (e.g., the NPU device **216** and related component), and the networking software/non-networking component management operating system provided by the networking software/non-networking component management COFTA system (i.e., the COFTA system **500** discussed above with reference to FIGS. **5**A and **5**B that is connected to the COFTA system connectors **212** and **214**) may operate to manage networking software used by the networking device **200**, as well as the fan controller, a subset of the components **224**, and/or other non-networking components in the networking device **200**.

[0116] For example, with reference to FIG. **10**C, the networking processing system management operating system provided by the compute subsystem **308** in the COFTA system **300** discussed above may perform networking processing system management operations **1000** that include managing the NPU device **216** (and related components) via the networking device connector **314** and the COFTA compute subsystem connector **210***b*. Similarly as discussed above, the networking processing system management operations **1000** may include any of a variety of management operations related to the NPU device **216** (or switch ASIC), related networking hardware, forwarding tables, telemetry hardware (e.g., NPU device/silicon telemetry hardware), NPU-specific drivers, a Software Development Kit (SDK) accessed via a Switch Abstraction Interface (SAI), firmware and/or other functionality that is relatively tightly coupled to the NPU device **216** (e.g., Precision Time Protocol (PTP) functionality, Synchronous Ethernet (SyncE) functionality, etc.), and/or other networking processing system components that would be apparent to one of skill in the art in possession of the present disclosure.

[0117] In another example, with reference to FIG. **10**D, the networking software/non-networking component management operating system provided by the compute subsystem **508** in the COFTA system **500** discussed above may perform networking software management operations **1002** that include managing the networking software used by the networking device **200**, which in some examples may include managing the subset of the components **224** that interact with that networking software (e.g., firmware that interacts with the NOS and/or that is otherwise included in a networking software management system, a TPM, a PoE manager or controller, a cryptographic accelerator, a console buffer, etc.) via the networking device connector **414**, the COFTA compute subsystem connector **212***b*, and the FPGA device **218**. Similarly as discussed above, the networking software management operations **1002** may include any of a variety of management operations related to a Network Operating System (NOS) provided by the compute subsystem **508** for use by the networking device **200**, a management framework provided for a user of the networking device **200** (e.g., a Command Line Interface (CLI), programmatic NorthBound Interface (NBI), REpresentational State Transfer (REST), Google Remote Procedure Call (gRPC) Network Management Interface (GNMI), gRPC Network Operations Interface (GNOI), etc.), Layer 2 (L2)/Layer 3 (L3) protocols, operating system images, Zero-Touch Provisioning (ZTP), Authentication Authorization and Accounting (AAA), Network Time Protocol (NTP), Domain Name Service (DNS), Role-Based Access Control (RBAC), Simple Network Management Protocol (SNMP), Dynamic Host Configuration Protocol (DHCP), and/or other networking software components that would be apparent to one of skill in the art in possession of the present disclosure.

[0118] In another example, with reference to FIG. **10**E, the networking software/non-networking component management operating system provided by the compute subsystem **508** in the COFTA system **500** discussed above may perform non-networking component management operations **1004** that include managing the fan controller **222**, the subset of the components **224**, and/or other non-networking components in the networking device **200** via the networking device connector **314**, the COFTA compute subsystem connector **214***b*, and the FPGA device **218**. As will be appreciated by one of skill in the art in possession of the present disclosure, the non-networking component management operations **1004** may include any of a variety of management operations related to power monitoring, control/reset mechanisms, sensor monitoring (e.g., thermal sensor monitoring, voltage sensor monitoring, etc.), power supply system status monitoring and control, fan status monitoring and control, LED control, firmware updates, and/or other non-networking components that would be apparent to one of skill in the art in possession of the present disclosure.

[0119] As will be appreciated by one of skill in the art in possession of the present disclosure, the non-networking component management operations **1004** described above are often conventionally performed by a Baseboard Management Controller (BMC) device, Platform Management Controller (PMC) device, and/or other management controller known in the art, and thus the COFTA system **500** may operate as a management controller (i.e., in addition to managing the networking software as described above) in the embodiment illustrated in FIGS. **10**A-**10**E. Furthermore, FIG. **10**E illustrates a specific example in which the non-networking component management operations **1004** include using the fan controller **222** to control any of the fan modules **304** on the COFTA system **300**, and the fan modules **504** on the COFTA system **500**.

[0120] With reference to FIG. **10**F, any of the management operations discussed above may include the networking processing system management operating system provided by the compute subsystem **308** in the COFTA system **300** and the networking software/non-networking component management operating system provided by the compute subsystem **508** in the COFTA system **500** performing operating system communication operations **1006** via the networking device connector **314**/COFTA compute subsystem connector **210***b* connection and the networking device connector **314**/COFTA compute subsystem connector **212***b* connection, respectively, and using the internal switch device **220**.

[0121] While not illustrated or described in detail, as discussed above, a user may access either of the compute subsystem **308** in the COFTA system **300** and the compute subsystem **508** in the COFTA system **500** (i.e., via the networking device connector **314**/COFTA compute subsystem connector **210***b* connection, and the networking device connector **314**/COFTA compute subsystem connector **212***b* connection or the networking device connector **314**/COFTA compute subsystem connector **214***b* connection, respectively) using the management port **208***a* and the internal switch device **220**, or using the management port **208***a* and the FPGA device **218** (via its UART connections discussed above).

[0122] Thus, the networking device **200** discussed above with reference to FIGS. **2**A and **2**B may be used with different combinations of the COFTA system **300** of FIGS. **3**A and **3**B, the COFTA system **400** of FIGS. **4**A and **4**B, and the COFTA system **500** of FIGS. **5**A and **5**B to provide different modular networking configurations that may be selected depending on how the networking device **200** will be used. For example, providing a dedicated networking processing system management operating system to perform networking processing system management (e.g., as illustrated and described above with reference to the COFTA system **300** in FIGS. **8**A-**8**F and **10**A-**10**F) allows a single vendor to control the networking processing system management operating system provided by the networking processing system management COFTA system to perform that networking processing system management, as well as associated drivers, applications, and other networking processing system management components, and allows a user to select the networking processing system management operating system that is best suited for their use of the networking device **200**.

[0123] Furthermore, such solutions that provide a dedicated networking processing system management operating system to perform networking processing system management allow the networking software management and non-networking component management for the networking device **200** to be provided by differentiated management solutions that may evolve independently of the networking processing system management. For example, a networking processing system management COFTA system provided according to the teachings of the present disclosure may provide an operating system kernel that is optimally configured (or provides the best available configuration) to run NPU drivers, firmware functionality, and/or other networking processing subsystems in terms of security, stability, and/or other characteristics of a networking processing system management operating system that would be apparent to one of skill in the art in possession of the present disclosure. As such, networking processing issues with the networking device **200** (e.g., "field" issues) to be triaged independently of the networking software management and/or non-networking component management, providing the ability to resolve such issues relatively quicker than conventional networking devices due to the management functionality demarcation provided by the teachings of the present disclosure.

[0124] Furthermore, while particular benefits associated with providing a dedicated networking processing system management operating system to perform networking processing system management are described above, one of skill in the art in possession of the present disclosure will appreciate how similar benefits may be realized by providing a dedicated networking software management operating system to perform networking software management (e.g., as illustrated and described above with reference to the COFTA system **400** in FIGS. **8**A-**8**F), and/or providing a dedicated non-networking component management operating system to perform non-networking component management (e.g., as illustrated and described above with reference to the COFTA system **300** in FIGS. **8**A-**8**F and **9**A-**9**F).

[0125] Furthermore, one of skill in the art in possession of the present disclosure will recognize a variety of networking device use cases may call for the embodiment illustrated in FIGS. **9**A-**9**F in which the networking processing system management and networking software management are performed by the COFTA system **500** while the non-networking component management is performed by the COFTA system **300**, as well as a variety of networking device use cases may call for the embodiment illustrated in FIGS. **10**A-**10**F in which the networking processing system management is performed by the COFTA system **300** and the networking software management and non-networking component management is performed by the COFTA system **500**. As such, one of skill in the art in possession of the present disclosure will appreciate how the modularity provided by the networking device **200** and different combinations of the COFTA systems **300**, **400**, and **500** allow a user to configure (and provide different levels of management functionality disaggregation for the device networking **200** based on desired features/functionalities/capabilities of the networking device **200**, workloads that will be performed using the networking device **200**, security requirements of the networking device **200**, and/or other characteristics of the networking device **200**.

[0126] As will be appreciated by one of skill in the art in possession of the present disclosure, the networking device **200** is also capable of being configured without the management functionality disaggregation described above in order to support users that desire the conventional integrated management functionality discussed above, and/or networking device use cases that may benefit from such integrated management functionality. For example, the COFTA system **600** discussed above with reference to FIGS. **6**A and **6**B may provide a networking processing system/networking software/non-networking component management COFTA system, and may be coupled to the networking device management connector system on the circuit board **204** of the networking device **200**. As illustrated in FIGS. **11**A and **11**B, the COFTA system **600** may be positioned adjacent the COFTA system connectors **210**, **212**, and **214** on the circuit board **204** of the networking device **200** (e.g., by positioning the COFTA system **600** in a COFTA system housing defined by the networking device chassis **200** adjacent the COFTA system connectors **210**, **212**, and **214**), and the COFTA system **600** may then be moved towards the circuit board **204** such that the fan controller connectors **606***a*, **606***b*, and **606***c* on the COFTA system **600** engage the COFTA fan subsystem connectors **210***a*, **212***a*, and **214***a*, respectively, on the COFTA system connectors **210**, **212**, and **214**, respectively, and the networking device connectors **614***a*, **614***b*, and **614***c* on the COFTA system **600** engage the COFTA compute subsystem connectors **210***b*, **212***b*, and **214***b*, respectively, on the COFTA system connectors **210**, **212**, and **214**.

[0127] The COFTA system **600** that provides the networking processing system/networking software/non-networking component management COFTA system may operate to provide a networking processing system/networking software/non-networking component management operating system. With reference back to FIGS. **11**A and **11**B, the compute subsystem **608** in the COFTA system **600** that is connected to the COFTA system connectors **210**, **212**, and **214** may have previously been configured to provide the networking processing system/networking software/non-networking component management operating system, and one of skill in the art in possession of the present disclosure will appreciate how that compute subsystem **608** may power on, boot, reset, reboot, and or otherwise initialize in order to enter a runtime state in which it provides the networking processing system/networking software/non-networking component management operating system.

[0128] The networking processing system/networking software/non-networking component management operating system provided by the networking processing system/networking software/non-networking component COFTA system (i.e., the COFTA system **600** discussed above with reference to FIGS. **6**A and **6**B that is connected to the COFTA system connectors **210**, **212**, and **214**) may operate to manage the networking processing system (e.g., the NPU device **216** and related component), the networking software used by the networking device **200** (as well as a subset of the components **224** that interact with that networking software such as, for example, firmware that interacts with the NOS and/or that is otherwise included in a networking software management system, a TPM, a PoE manager or controller, a cryptographic accelerator, a console buffer, etc.), and the fan controller, a subset of the components **224**, and/or other non-networking components in the networking device **200**.

[0129] For example, with reference to FIG. **11**C, the networking processing system/networking software/non-networking component management operating system provided by the compute subsystem **608** in the COFTA system **600** discussed above may perform networking processing system management operations **1100** that include managing the NPU device **216** (and related components) via the networking device connector **314** and the COFTA compute subsystem connector **210***b*. Similarly as discussed above, the networking processing system management operations **1100** may include any of a variety of management operations related to the NPU device **216** (or switch ASIC), related networking hardware, forwarding tables, telemetry hardware (e.g., NPU device/silicon telemetry hardware), NPU-specific drivers, a Software Development Kit (SDK) accessed via a Switch Abstraction Interface (SAI), firmware and/or other functionality that is relatively tightly coupled to the NPU device **216** (e.g., Precision Time Protocol (PTP) functionality, Synchronous Ethernet (SyncE) functionality, etc.), and/or other networking processing system components that would be apparent to one of skill in the art in possession of the present disclosure.

[0130] In another example, with reference to FIG. **11**D, the networking processing system/networking software/non-networking component management operating system provided by the compute subsystem **608** in the COFTA system **600** discussed above may perform networking software management operations **1102** that include managing the networking software used by the networking device **200**, which in some examples may include managing the subset of the components **224** that interact with that networking software (e.g., firmware that interacts with the NOS and/or that is otherwise included in a networking software management system, a TPM, a PoE manager or controller, a cryptographic accelerator, a console buffer, etc.) via the networking device connector **414**, the COFTA compute subsystem connector **212***b*, and the FPGA device **218**. Similarly as discussed above, the networking software management operations **1102** may include any of a variety of management operations related to a Network Operating System (NOS) provided by the compute subsystem **508** for use by the networking device **200**, a management framework provided for a user of the networking device **200** (e.g., a Command Line Interface (CLI), programmatic North-Bound Interface (NBI), REpresentational State Transfer (REST), Google Remote Procedure Call (gRPC) Network Management Interface (GNMI), gRPC Network Operations Interface (GNOI), etc.), Layer 2 (L2)/Layer 3 (L3) protocols, operating system images, Zero-Touch Provisioning (ZTP), Authentication Authorization and Accounting (AAA), Network Time Protocol (NTP), Domain Name Service (DNS), Role-Based Access Control (RBAC), Simple Network Management Protocol (SNMP), Dynamic Host Configuration Protocol (DHCP), and/or other networking software components that would be apparent to one of skill in the art in possession of the present disclosure.

[0131] In another example, with reference to FIG. **11**E, the networking processing system/networking software/non-networking component management operating system provided by the compute subsystem **608** in the COFTA system **600** discussed above may perform non-networking component management operations **1104** that include managing the fan controller **222**, the subset of the components **224**, and/or other non-networking components in the networking device **200** via the networking device connector **314**, the COFTA compute subsystem connector **214***b*, and the FPGA device **218**. As will be appreciated by one of skill in the art in possession of the present disclosure, the non-networking component management operations **1104** may include any of a variety of management operations related to power monitoring, control/reset mechanisms, sensor monitoring (e.g., thermal sensor monitoring, voltage sensor monitoring, etc.), power supply system status monitoring and control, fan status monitoring and control, LED control, firmware updates, and/or other non-networking components that would be apparent to one of skill in the art in possession of the present disclosure.

[0132] As will be appreciated by one of skill in the art in possession of the present disclosure, the non-networking component management operations **1104** described above are often conventionally performed by a Baseboard Management Controller (BMC) device, Platform Management Controller (PMC) device, and/or other management controller known in the art, and thus the COFTA system **600** may operate as a management controller (i.e., in addition to managing the networking processing system and the networking software as described above) in the embodiment illustrated in FIGS. **11**A-**11**E. Furthermore, FIG. **11**E illustrates a specific example in which the non-networking component management operations **1104** include using the fan controller **222** to control any of the fan modules **604** on the COFTA system **600**.

[0133] While not illustrated or described in detail, as discussed above, a user may access either of the compute subsystem **608** in the COFTA system **600** (i.e., via the networking device connector **314**/COFTA compute subsystem connector **210***b* connection, the networking device connector **314**/COFTA compute subsystem connector **212***b* connection, or the networking device connector **314**/COFTA compute subsystem connector **214***b* connection, respectively) using the management port **208***a* and the internal switch device **220**, or using the management port **208***a* and the FPGA device **218** (via its UART connections discussed above).

[0134] As such, the networking device **200** may be configured without the management functionality disaggregation described above in order to provide conventional integrated management functionality that, while being subject to the issues discussed above, may be desired for some networking device use cases. As will be appreciated by one of skill in the art in possession of the present disclosure, each of the COFTA systems **300**, **400**, **500**, and **600** may be associated with a respective Stock Keeping Unit (SKU), and any potential networking device user may configure a networking device with desired management functionality by selecting different combinations of those SKUs. Furthermore, one of skill in the art in possession of the present disclosure will appreciate how the COFTA systems **300**, **400**, **500**, and **600** are field replaceable, allowing the management functionality provided for the networking device **200** to be changed when desired.

[0135] Thus, systems and methods have been described that utilize COFTA systems as modular components in a networking device that are each configured to provide a respective operating system to manage different subsets of a networking processing system in the networking device, non-networking components in the networking device, and networking software used by the networking device. For example, the modular COFTA networking device system of the present disclosure may include a networking device chassis housing a circuit board coupled to a plurality of non-networking components and including: a networking processing system coupled to networking ports that are accessible on the networking device chassis, a networking processing system management connector coupled to the networking processing system, a non-networking component management connector coupled to the non-networking components, and a networking software management connector. At least two COFTA systems are each connected to a different subset of the networking processing system management connector, the non-networking component management connector, and the networking software management connector. Each COFTA system provides a respective operating system that manages a different subset of the networking processing system via the networking processing system management connector, the plurality of non-networking components via the non-networking component management connector, and networking software via the networking software management connector. As such, different switch management functionality for the networking device may be performed by dedicated operating systems, reducing or eliminating the issues with managing conventional networking devices discussed above.

[0136] Although illustrative embodiments have been shown and described, a wide range of modification, change and substitution is contemplated in the foregoing disclosure and in some instances, some features of the embodiments may be employed without a corresponding use of other features. Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the scope of the embodiments disclosed herein.

What is claimed is:

1. A modular Computer-On-Fan-Tray-Assembly (COFTA) networking device system, comprising:
- a networking device chassis;
- a circuit board that is housed in the networking device chassis;
- a plurality of networking ports that are included on the circuit board and that are accessible on the networking device chassis;
- a networking processing system that is included on the circuit board and that is coupled to the plurality of ports;
- a plurality of non-networking components that are couped to the circuit board;
- a networking processing system management connector that is included on the circuit board and that is coupled to the networking processing system;
- a non-networking component management connector that is included on the circuit board and that is coupled to the non-networking components;
- a networking software management connector that is included on the circuit board; and
- at least two Computer-On-Fan-Tray-Assembly (COFTA) systems that are each connected to a different subset of the networking processing system management connector, the non-networking component management connector, and the networking software management connector, wherein each COFTA system is configured to provide a respective operating system that is configured to manage a different subset of the networking processing system via the networking processing system management connector, the plurality of non-networking components via the non-networking component management connector, and networking software via the networking software management connector.

2. The system of claim **1**, further comprising:
- an internal switch device that is included on the circuit board, that is coupled to each of the networking processing system management connector, the non-networking component management connector, and the networking software management connector, and that is configured to transmit communications between the respective operating systems provided by the at least two COFTA systems.

3. The system of claim **1**, further comprising:
- a respective fan connector that is included on the circuit board and located adjacent each of the networking processing system management connector, the non-networking component management connector, and the networking software management connector, wherein each respective fan connector is configured to connect to one of the at least two COFTA systems to couple to a fan subsystem on that COFTA system when that COFTA system is connected to the networking processing system management connector, the non-networking component management connector, or the networking software management connector that is located adjacent that fan connector.

4. The system of claim **3**, wherein the plurality of non-networking components include a fan controller that is coupled to each of the respective fan connectors and that is configured for use in controlling the fan subsystem coupled to that respective fan connector.

5. The system of claim **1**, wherein the at least two COFTA systems include:

a networking processing system management COFTA system that is connected to the networking processing system management connector and that is configured to provide a networking processing system management operating system that is configured to manage the networking processing system via the networking processing system management connector;

a non-networking component management COFTA system that is connected to the non-networking component management connector and that is configured to provide a non-networking component management operating system that is configured to manage the plurality of non-networking components via the non-networking component management connector; and

a networking software management COFTA system that is connected to the networking software management connector and that is configured to provide a networking software management operating system that is configured to manage the networking software via the networking software management connector.

6. The system of claim **1**, wherein the at least two COFTA systems include:

a networking processing system/networking software management COFTA system that is connected to the networking processing system management connector and the networking software management connector, and that is configured to provide a networking processing system/networking software management operating system that is configured to manage the networking processing system via the networking processing system management connector, and manage the networking software via the networking software management connector; and

a non-networking component management COFTA system that is connected to the non-networking component management connector and that is configured to provide a non-networking component management operating system that is configured to manage the plurality of non-networking components via the non-networking component management connector.

7. The system of claim **1**, wherein the at least two COFTA systems include:

a networking processing system management COFTA system that is connected to the networking processing system management connector and that is configured to provide a networking processing system management operating system that is configured to manage the networking processing system via the networking processing system management connector; and

a networking software/non-networking component management COFTA system that is connected to the networking software management connector and the non-networking component management connector, and that is configured to provide a networking software/non-networking component management operating system that is configured to manage the networking software via the networking software management connector, and manage the plurality of non-networking components via the non-networking component management connector.

8. A networking device, comprising:

a networking device chassis;

a circuit board that is housed in the networking device chassis;

a plurality of networking ports that are included on the circuit board and that are accessible on the networking device chassis;

a networking processing system that is included on the circuit board and that is coupled to the plurality of ports;

a plurality of non-networking components that are couped to the circuit board; and

a networking device management connector system that is included on the circuit board and that includes:

a networking processing system management connector that is coupled to the networking processing system;

a non-networking component management connector that is coupled to the non-networking components; and

a networking software management connector, wherein the networking device management connector system is configured to connect to at least two Computer-On-Fan-Tray-Assembly (COFTA) systems that are each configured to provide a respective operating system that is configured to manage a different subset of the networking processing system via networking processing system management connector, the non-networking components via the non-networking component management connector, and networking software via networking software management connector.

9. The networking device of claim **8**, further comprising:

an internal switch device that is included on the circuit board, that is coupled to each of the networking processing system management connector, the non-networking component management connector, and the networking software management connector, and that is configured to transmit communications between the respective operating systems provided by the at least two COFTA systems when the at least two COFTA systems are connected to networking device management connector system.

10. The networking device of claim **8**, further comprising:

a respective fan connector that is included on the circuit board and located adjacent each of the networking processing system management connector, the non-networking component management connector, and the networking software management connector, wherein each respective fan connector is configured to connect to one of the at least two COFTA systems, when the at least two COFTA systems are connected to the networking device management connector system, to couple to a fan subsystem on that COFTA system when that COFTA system is connected to the networking processing system management connector, the non-networking component management connector, or the networking software management connector that is located adjacent that fan connector.

11. The networking device of claim **10**, wherein the plurality of non-networking components include a fan controller that is coupled to each of the respective fan connectors and that is configured for use in controlling the fan subsystem coupled to that respective fan connector when the at least two COFTA systems are connected to the networking device management connector system.

12. The networking device of claim **8**, wherein the plurality of non-networking components include a power system.

13. The networking device of claim **8**, wherein the plurality of non-networking components include a Light-Emitting Device (LED) indicator system.

14. A method for providing a modular Computer-On-Fan-Tray-Assembly (COFTA)-based networking device, comprising:

connecting, by each of at least two Computer-On-Fan-Tray-Assembly (COFTA) systems, to a different subset of:

a networking processing system management connector in a networking device management connector system that is included on a circuit board housed in a networking device chassis;

a non-networking component management connector in the networking device management connector system; and

a networking software management connector in the networking device management connector system;

providing, by each of the least two COFTA systems, a respective operating system; and

managing, by each of the respective operating systems provided by the at least two COFTA systems, a different subset of:

a networking processing system via the networking processing system management connector, wherein the networking processing system is included on the circuit board and coupled to a plurality of networking ports that are included on the circuit board and that are accessible via the networking device chassis;

a plurality of non-networking components via the non-networking component management connector, wherein the plurality of non-networking components are coupled to the circuit board; and

networking software via the networking software management connector.

15. The method of claim **14**, further comprising:

transmitting, by an internal switch device that is included on the circuit board and coupled to each of the networking processing system management connector, the non-networking component management connector, and the networking software management connector, communications between the respective operating systems provided by the at least two COFTA systems.

16. The method of claim **14**, further comprising:

connecting, by a respective fan connector that is included on the circuit board and located adjacent each of the networking processing system management connector, the non-networking component management connector, and the networking software management connector, to one of the at least two COFTA systems to couple that respective fan connector to a fan subsystem on that COFTA system.

17. The method of claim **16**, further comprising:

controlling, by a fan controller that is included in the plurality of non-networking components and coupled to each of the respective fan connectors, the fan subsystem coupled to that respective fan connector.

18. The method of claim **14**, wherein the at least two COFTA systems include:

a networking processing system management COFTA system that is connected to the networking processing system management connector and that provides a networking processing system management operating system that manages the networking processing system via the networking processing system management connector;

a non-networking component management COFTA system that is connected to the non-networking component management connector and that provides a non-networking component management operating system that manages the plurality of non-networking components via the non-networking component management connector; and

a networking software management COFTA system that is connected to the networking software management connector and that provides a networking software management operating system that manages the networking software via the networking software management connector.

19. The method of claim **14**, wherein the at least two COFTA systems include:

a networking processing system/networking software management COFTA system that is connected to the networking processing system management connector and the networking software management connector, and that provides a networking processing system/networking software management operating system that manages the networking processing system via the networking processing system management connector, and manages the networking software via the networking software management connector; and

a non-networking component management COFTA system that is connected to the non-networking component management connector and that provides a non-networking component management operating system that manages the plurality of non-networking components via the non-networking component management connector.

20. The method of claim **14**, wherein the at least two COFTA systems include:

a networking processing system management COFTA system that is connected to the networking processing system management connector and that provides a networking processing system management operating system that manages the networking processing system via the networking processing system management connector; and

a networking software/non-networking component management COFTA system that is connected to the networking software management connector and the non-networking component management connector, and that provides a networking software/non-networking component management operating system that manages the networking software via the networking software management connector, and manages the plurality of non-networking components via the non-networking component management connector.

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